

Work, energy and power

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5.1 Introduction

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We use the term ‘work’ in everyday conversation to mean many different things. We talk about going to work, doing homework, working in class. Physicists mean something very specific when they talk about work.

In Physics we use the term work to describe the process of transferring energy from object or system to another or converting energy from one form to another.

You will learn that work and energy are closely related to Newton’s laws of motion. You shall see that the energy of an object is its capacity to do work and doing work is the process of transferring energy from one object or form to another by means of a force. In other words,

- an object with lots of energy can do lots of work.
- when object A transfers energy to object B, the energy of object A decreases by the same amount as the energy of object B increases, we say that object A does work on object B.

Lifting objects or throwing them requires that you do work on them. Even making an electrical current flow requires that something do work. Objects or systems must have energy to be able to do work on other objects or systems by transferring some of their energy.

Key linked concepts

- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Techniques of vector addition — Physical Sciences, Grade 10, Vectors and scalars
- Newton’s laws — Physical Sciences, Grade 11, Forces
- Force diagrams — Physical Sciences, Grade 11, Forces

▶ See video: 27RG at www.everythingscience.co.za

5.2 Work

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We cover different topics in different chapters in different grades but that doesn’t mean that they are not related. In fact, it is very important to note that all of the different topics related to mechanics (forces, mechanical energy, momentum, rectilinear motion) actually form a **consistent** picture of the same physical system. There have been examples where we’ve shown the same results using two methods, for example determining speed or velocity using equations of motion or conservation of mechanical energy. Learning about work will help us tie everything we’ve learnt about previously together.

Work will allow us to connect energy transfer to forces, which we have already linked

to momentum and the equations of motion. When a force tends to act in or against the direction of motion of an object we say that the force is doing work on the object. Specifically, work is defined mathematically in terms of the force and the displacement of the object.

DEFINITION: *Work*

When a force acts in or against the direction of motion of an object, work is done on the object.

$$W = F\Delta x \cos \theta$$

This means that in order for work to be done, an object must have its position changed by an amount $\Delta \vec{x}$ while a force, $\vec{F}$, is acting on it such that there is some non-zero component of the force in the direction of the displacement.

Work is calculated as:

$$W = F\Delta x \cos \theta$$

Where:

- W = work in joules (J)
- F = magnitude of force for which we are calculating work in newtons (N)
- Δx = magnitude of displacement in metres (m)
- θ = angle between the force and the direction of displacement

Note that:

- $F \cos \theta$ is the magnitude of the component of $\vec{F}$ in the direction of $\Delta \vec{x}$. If $\theta > 90^\circ$ then the component is parallel to the direction of displacement but point in the opposite direction and the force is opposing the motion.
- We will deal with constant forces not forces that are continuously changing in this chapter. All of the principles we have covered are applicable when forces are not constant.

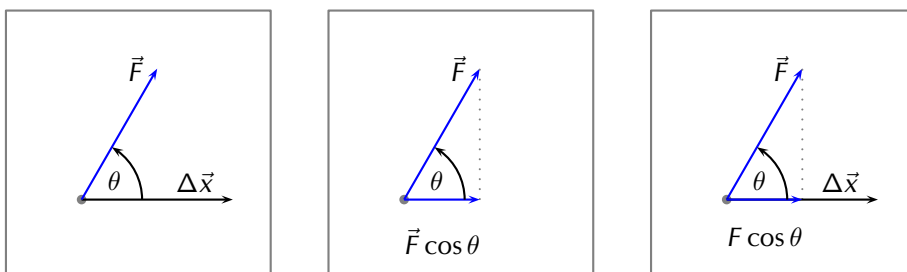
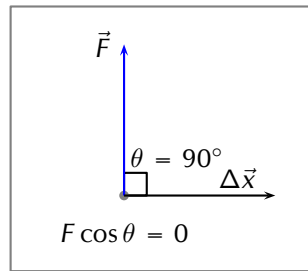


Figure 5.1: The force $\vec{F}$ causes the object to be displaced by $\Delta \vec{x}$ at angle θ .

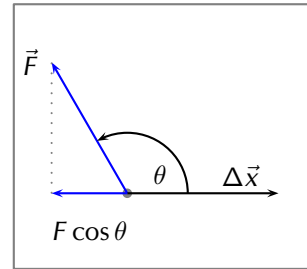
Important: $\cos \theta$ tells you the relative direction of the force and the displacement which is important. If the component of the force along the direction of the displacement is opposite in direction to the displacement then the sign of the displacement vector and force vector will be different. This is regardless of which direction was chosen as a positive direction.

Let us look at some examples to understand this properly. In the images below the grey dot represents an object. A force, $\vec{F}$, acts on the object. The object moves through a displacement, $\Delta \vec{x}$. What is the sign of the work done in each case?

At $\theta = 90^\circ$ the magnitude of the component of the force in the direction of the displacement is zero so **no work is done** as $\cos(90) = 0$.



At $\theta > 90^\circ$ the magnitude of the component of the force along the line of the displacement is opposite in direction to the displacement so **negative work is done**, $\cos \theta < 0$.



It is very important to note that for work to be done there must be a component of the applied force in the direction of motion. Forces perpendicular to the direction of motion do no work.

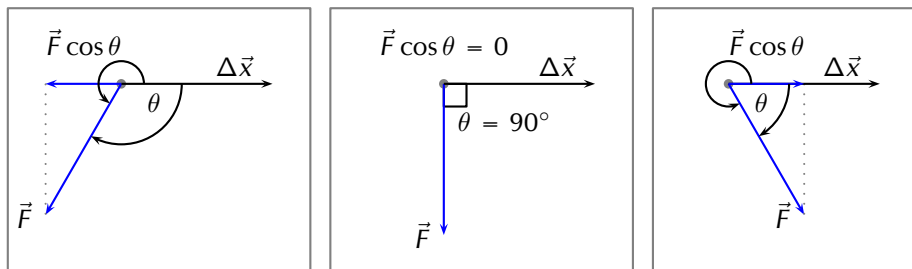


Figure 5.2: The force $\vec{F}$ causes the object to be displaced by $\Delta\vec{x}$ at angle θ .

It is only the direction of the force on the object that matters and not the direction from the source of the force to the object. In Figure 5.3 both powerlifters are exerting an upwards force on the weights. On the left the weight is being pulled upwards and on the right it is being pushed upwards.



Figure 5.3: Left: A powerlifter deadlifts a weight. Right: A powerlifter benchpresses a weight.

Weight lifting is a good context in which to think about work because it helps to identify misconceptions introduced by everyday use of the term ‘work’. In the two cases in Figure 5.3 everyone would describe moving the weights upwards as very hard work. From a physics perspective, as the powerlifters lift the weight they are exerting a force in the direction of the displacement so positive work is done on the weights.

Consider the strongman walking in Figure 5.4. He carries two very heavy sleds as far as he can in a competition. What work is the man doing on the sleds and why?

Most people would say he is working very hard because the sleds are heavy to carry but from a physics perspective he is doing no work on the sleds. The reason that he does no work is because the force he exerts is directly upwards to balance the force of gravity and the displacement is in the horizontal direction. Therefore there is no component of the force in the direction of displacement ($\theta = 90^\circ$) and no work done.

His muscles do need to use their energy reserves to maintain the force to balance gravity. That does not result in energy transfer to the sleds.

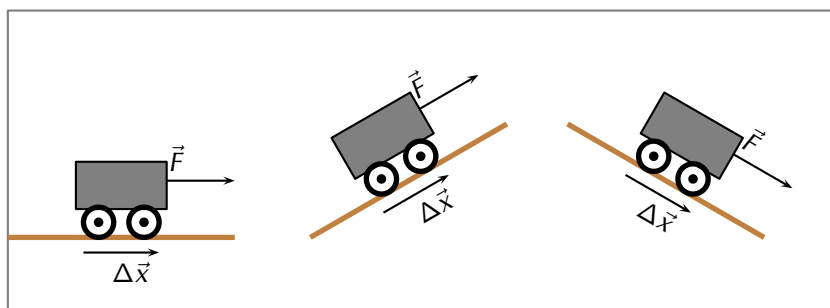


Figure 5.4: A strongman carries heavy sleds as far as possible in a competition.

Investigation: Is work done?

Decide whether or not work is done in the following situations. Remember that for work to be done a force must be applied in the direction of motion and there must be a displacement. Identify which two objects are interacting, what the action-reaction pairs of forces are and why the force described is or isn't doing work.

1. Max pushes against a wall and becomes tired.
2. A book falls off a table and free falls to the ground.
3. A rocket accelerates through space.
4. A waiter holds a tray full of plates above his head with one arm and carries it straight across the room at constant speed.



For each of the above pictures, the force vector is acting in the same direction as the displacement vector. As a result, the angle $\theta = 0^\circ$ because there is no difference in angle between the direction of applied force and the direction of displacement.

TIP

The everyday use of the word “work” differs from the physics use. In physics, only the component of the applied force that is parallel to the motion does work on an object. So, for example, a person holding up a heavy book does no work on the book.

The work done by a force can then be positive or negative. This sign tells us about the direction of the energy transfer. Work is a scalar so the sign should not be misinterpreted to mean that work is a vector. Work is defined as energy transfer, energy is a scalar quantity and the sign indicates whether energy was increased or decreased.

- If $\vec{F}_{\text{applied}}$ acts or has a component acting in the same direction as the motion, then positive work is being done. In this case the object on which the force is applied gains energy.
- If the direction of motion and $\vec{F}_{\text{applied}}$ are opposite, then negative work is being done. This means that energy is lost and the object exerting the force gains energy. For example, if you try to push a car uphill by applying a force up the slope and instead the car rolls down the hill you are doing negative work on the car. Alternatively, the car is doing positive work on you!

As with all physical quantities, work must have units. Following from the definition, work is measured in N·m. The name given to this combination of S.I. units is the joule (symbol J).

Worked example 1: Calculating work on a car when speeding up.

QUESTION

A car is travelling along a straight horizontal road. A force of 500 N is applied to the car in the direction that it is travelling, speeding it up. While it is speeding up it covers a distance of 20 m. Calculate the work done on the car.



SOLUTION

Step 1: Analyse the question to determine what information is provided

- The magnitude of the force applied is $F = 500 \text{ N}$.
- The distance moved is $\Delta x = 20 \text{ m}$.
- The applied force and distance moved are in the same direction. Therefore, the angle between the force and displacement is $\theta = 0^\circ$.

These quantities are all in SI units, so no unit conversions are required.

Step 2: Analyse the question to determine what is being asked

- We are asked to find the work done on the car. We know from the definition that work done is $W = F\Delta x \cos \theta$.

Step 3: Next we substitute the values and calculate the work done

$$\begin{aligned} W &= F\Delta x \cos \theta \\ &= (500)(20)(\cos 0) \\ &= (500)(20)(1) \\ &= 10\,000 \text{ J} \end{aligned}$$

Remember that the answer must be *positive*, as the applied force and the displacement are in the same direction. In this case, the car gains kinetic energy.

Worked example 2: Calculating work on the car while braking

QUESTION

The same car now slows down when a force of 300 N is applied opposite to the direction of motion while it travels 25 m forward. Calculate the work done on the car.

SOLUTION

Step 1: Analyse the question to determine what information is provided

- The magnitude of the force applied is $F = 300 \text{ N}$.
- The distance moved is $\Delta x = 25 \text{ m}$.
- The applied force and distance moved are in the opposite direction. Therefore, the angle between the force and displacement is $\theta = 180^\circ$.

These quantities are all in the correct units, so no unit conversions are required.

Step 2: Analyse the question to determine what is being asked

- We are asked to find the work done on the car. We know from the definition that work done is $W = F\Delta x \cos \theta$

Step 3: Next we substitute the values and calculate the work done

$$\begin{aligned} W &= F\Delta x \cos \theta \\ &= (300)(25)(\cos 180) \\ &= (300)(25)(-1) \\ &= -7500 \text{ J} \end{aligned}$$

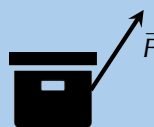
Note that the answer must be *negative* as the applied force and the displacement are in opposite directions. This means that the energy is being lost by the car. This may be energy lost as heat to the environment. Energy conservation still holds, the energy has just been transferred to a larger system that contains the car.

What happens when the applied force and the motion are not parallel? By using the formula $W = F\Delta x \cos \theta$, we are actually calculating the *component* of the applied force in the direction of motion. Note that the component of the force perpendicular to the direction of motion does no work.

Worked example 3: Calculating work done on a box pulled at an angle.

QUESTION

Calculate the work done on a box, if it is pulled 5 m along the ground by applying a force of $F = 20 \text{ N}$ at an angle of 60° to the horizontal.



SOLUTION

Step 1: Analyse the question to determine what information is provided

- The force applied is $F = 20 \text{ N}$
- The distance moved is $\Delta x = 5 \text{ m}$ along the ground
- The angle between the applied force and the motion is $\theta = 60^\circ$

These quantities are in the correct units so we do not need to perform any unit conversions.

Step 2: Analyse the question to determine what is being asked

- We are asked to find the work done on the box.

Step 3: Substitute and calculate the work done

Now we can calculate the work done on the box:

$$\begin{aligned} W &= F \Delta x \cos \theta \\ &= (20)(5)(\cos 60) \\ &= 50 \text{ J} \end{aligned}$$

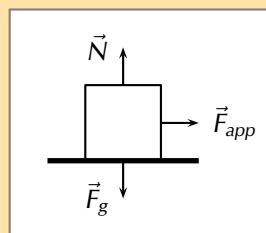
Note that the answer is positive as the component of the force parallel to the direction of motion is in the same direction as the motion.

The work done on the box is 50 J.

Exercise 5 – 1: Work

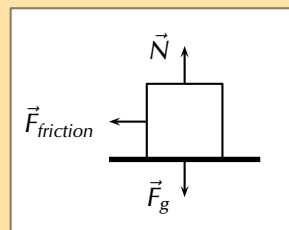
A 10 N force is applied to push a block across a frictionless surface for a displacement of 5,0 m to the right. The

1. block has a weight $\vec{F}_g$ of 20 N. Determine the work done by the following forces: normal force, weight $\vec{F}_g$, applied force.



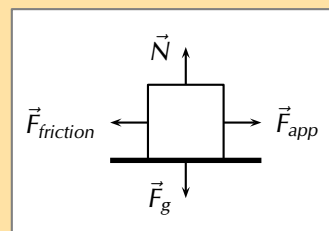
A 10 N frictional force slows a moving block to a stop after a displacement of 5,0 m to the right. The block has

2. a weight of 20 N. Determine the work done by the following forces: normal force, weight, frictional force.



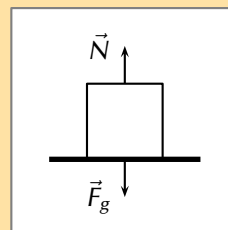
A 10 N force is applied to push a block across a frictional surface at constant speed for a displacement of 5,0 m to the right. The block has a weight of 20 N and the frictional force is 10 N. Determine the work done by the following forces: normal force, weight, applied force and frictional force.

- 3.

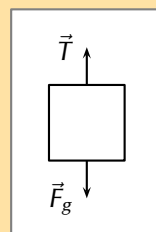


An object with a weight of 20 N is sliding at constant speed across a frictionless surface for a displacement of 5 m to the right. Determine if there is any work done.

- 4.



- An object with a weight of 20 N is pulled upward at constant speed by a
5. 20 N force for a vertical displacement of 5 m. Determine if there is any work done.

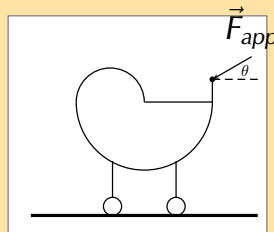


6. Before beginning its descent, a roller coaster is always pulled up the first hill to a high initial height. Work is done on the roller coaster to achieve this initial height. A coaster designer is considering three different incline angles **of the hill** at which to drag the 2000 kg car train to the top of the 60 m high hill. In each case, the force applied to the car will be applied parallel to the hill. Her critical question is: which angle would require the least work? Analyse the data, determine the work done in each case, and answer this critical question.

Angle of Incline	Applied Force	Distance	Work
35°	$1,1 \times 10^4$ N	100 m	
45°	$1,3 \times 10^4$ N	90 m	
55°	$1,5 \times 10^4$ N	80 m	

7. A traveller carries a 150 N suitcase up four flights of stairs (a total height of 12 m) and then pushes it with a horizontal force of 60 N at a constant speed of $0,25 \text{ m}\cdot\text{s}^{-1}$ for a horizontal distance of 50 m on a frictionless surface. How much work does the traveller do on the suitcase during this entire trip?

8. A parent pushes down on a pram with a force of 50 N at an angle of 30° to the horizontal. The pram is moving on a frictionless surface. If the parent pushes the pram for a horizontal distance of 30 m, how work is done on the pram?



9. How much work is done by the force required to raise a 2000 N lift 5 floors vertically at a constant speed? The vertical distance between floors is 5 m high.
10. A student with a mass of 60 kg runs up three flights of stairs in 15 s, covering a vertical distance of 10 m. Determine the amount of work done by the student to elevate her body to this height.
11. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27RH 2. 27RJ 3. 27RK 4. 27RM 5. 27RN 6. 27RP
7. 27RQ 8. 27RR 9. 27RS 10. 27RT



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We have only looked at a single force acting on an object. Sometimes more than one force acts at the same time (we dealt with this in Grade 11). We call the work done after taking all the forces into account the **net work done**. If there is only one force

acting then the work it does, if any, is the net work done. In this case there are two equivalent approaches we can adopt to finding the net work done on the object. We can:

- **Approach 1:** calculate the work done by **each force individually** and then sum them **taking the signs into account**. If one force does positive work and another does the same amount of work but it is negative then they cancel out.
- **Approach 2:** calculate the resultant force from all the forces acting and calculate the work done using the resultant force. This will be equivalent to Approach 1. If the resultant force parallel to the direction of motion is zero, no net work will be done.

Remember that work done tells you about the energy transfer to or from an object by means of a force. That is why we can have zero net work done even if multiple large forces are acting on an object. Forces that result in positive work increase the energy of the object, forces that result in negative work reduce the energy of an object. If as much energy is transferred to an object as is transferred away then the final result is that the object gains no energy overall.

Worked example 4: Approach 1, calculating the net work on a car

QUESTION

The same car is now accelerating forward, but friction is working against the motion of the car. A force of 300 N is applied forward on the car while it is travelling 20 m forward. A frictional force of 100 N acts to oppose the motion. Calculate the net work done on the car.



Only forces with a component in the plane of motion are shown on the diagram. No work is done by F_g or F_{Normal} as they act perpendicular to the direction of motion.

SOLUTION

Step 1: Analyse the question to determine what information is provided

- The force applied is $F_{applied} = 300\text{ N}$ forwards.
- The force of friction is $F_{friction} = 100\text{ N}$ opposite to the direction of motion.
- The distance moved is $\Delta x = 20\text{ m}$.
- The applied force and distance moved are in the same plane so we can calculate the work done by the applied forward force and the work done by the force of friction backwards.

These quantities are all in the correct units, so no unit conversions are required.

Step 2: Analyse the question to determine what is being asked

- We are asked to find the net work done on the car. We know from the definition that work done is $W = F\Delta x \cos \theta$

Step 3: Next we calculate the work done by each force.

$$\begin{aligned}W_{\text{applied}} &= F_{\text{applied}} \Delta x \cos \theta \\&= (300)(20)(\cos 0) \\&= (300)(20)(1) \\&= 6000 \text{ J}\end{aligned}$$

$$\begin{aligned}W_{\text{net}} &= W_{\text{applied}} + W_{\text{friction}} \\&= (6000) + (-2000) \\&= 4000 \text{ J}\end{aligned}$$

$$\begin{aligned}W_{\text{friction}} &= F_{\text{friction}} \Delta x \cos \theta \\&= (100)(20)(\cos 180) \\&= (100)(20)(-1) \\&= -2000 \text{ J}\end{aligned}$$

The answer shown in this worked example shows that although energy has been lost by the car to friction, the total work done on the car has resulted in a net energy gain. This can be seen by the positive answer.

As mentioned before, there is an alternative method to solving the same problem, which is to determine the net force acting on the car and to use this to calculate the work. This means that the vector forces acting in the plane of motion must be added to get the net force $\vec{F}_{\text{net}}$. The net force is then applied over the displacement to get the net work W_{net} .

Worked example 5: Approach 2, calculating the net force

QUESTION

The same car is now accelerating forward, but friction is working against the motion of the car. A force of 300 N is applied forward on the car while it is travelling 20 m forward. A frictional force of 100 N acts to oppose the motion. Calculate the net work done on the car.

SOLUTION

Step 1: Analyse the question to determine what information is provided

- The force applied is $\vec{F}_{\text{applied}} = 300 \text{ N}$ forwards.
- The force of friction is $\vec{F}_{\text{friction}} = 100 \text{ N}$ backwards.
- The distance moved is $\Delta x = 20 \text{ m}$.
- The applied forces $\vec{F}_{\text{applied}} = 300 \text{ N}$ and the force of friction $\vec{F}_{\text{friction}} = 100 \text{ N}$ are in the same plane as the distance moved. Therefore, we can add the vectors. As vectors require direction, we will say that forward is positive and therefore backward is negative. Note, the force of friction is acting at 180° i.e. backwards and so is acting in the opposite vector direction i.e. negative.

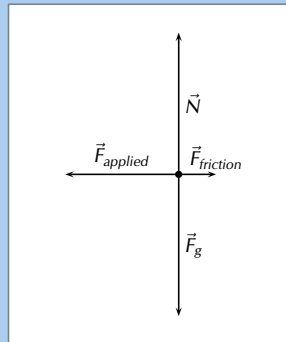
These quantities are all in the correct units, so no unit conversions are required.

Step 2: Analyse the question to determine what is being asked

- We are asked to find the net work done on the car. We know from the definition that work done is $W_{\text{net}} = F_{\text{net}} \Delta x \cos \theta$

Step 3: We calculate the net force acting on the car, and we convert this into net work.

First we draw the force diagram:



Let forwards (to the left in the picture) be positive. We know that the motion of the car is in the horizontal direction so we can neglect the force due to gravity, $\vec{F}_g$, and the normal force, $\vec{N}$. **Note:** if the car were on a slope we would need to calculate the component of gravity parallel to the slope.

$$\begin{aligned}\vec{F}_{net} &= \vec{F}_{applied} + \vec{F}_{friction} \\ &= (+300) + (-100) \\ \vec{F}_{net} &= 200 \text{ N forwards}\end{aligned}$$

$\vec{F}_{net}$ is pointing in the same direction as the displacement, therefore the angle between the force and displacement is $\theta = 0^\circ$.

$$\begin{aligned}W_{net} &= F_{net} \Delta x \cos \theta \\ &= (200) (20) \cos (0) \\ &= 4000 \text{ J}\end{aligned}$$

IMPORTANT!

The two different approaches give the same result but it is very important to treat the signs correctly. The forces are vectors but work is a scalar so they shouldn't be interpreted in the same way.

5.3 Work-energy theorem

ESCMD

Conservative and non-conservative forces

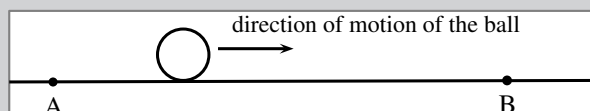
ESCMF

In Grade 10, you saw that mechanical energy was conserved in the **absence** of non-conservative forces. It is important to know whether a force is an conservative force or an non-conservative force in the system, because this is related to whether the force can change an object's total mechanical energy when it does work on an object.

When the only forces doing work are conservative forces (for example, gravitational and spring forces), energy changes forms - from kinetic to potential (or vice versa); yet the total amount of mechanical energy ($E_K + E_P$) is conserved. For example, as an object falls in a gravitational field from a high elevation to a lower elevation, some of the object's potential energy is changed into kinetic energy. However, the sum of the kinetic and potential energies remain constant.

Investigation: Non-conservative forces

We can investigate the effect of non-conservative forces on an object's total mechanical energy by rolling a ball along the floor from point A to point B.



Find a nice smooth surface (e.g. a highly polished floor), mark off two positions, A and B, and roll the ball between them.

The total mechanical energy of the ball, at each point, is the sum of its kinetic energy (E_k) and gravitational potential energy (E_p):

$$\begin{aligned} E_{\text{total},A} &= E_{k,A} + E_{p,A} \\ &= \frac{1}{2}mv_A^2 + mgh_A \\ &= \frac{1}{2}mv_A^2 + mg(0) \\ &= \frac{1}{2}mv_A^2 \end{aligned}$$

$$\begin{aligned} E_{\text{total},B} &= E_{k,B} + E_{p,B} \\ &= \frac{1}{2}mv_B^2 + mgh_B \\ &= \frac{1}{2}mv_B^2 + mg(0) \\ &= \frac{1}{2}mv_B^2 \end{aligned}$$

In the absence of friction and other non-conservative forces, the ball should slide along the floor and its speed should be *the same* at positions A and B. Since there are no non-conservative forces acting on the ball, its total mechanical energy at points A and B are equal.

$$\begin{aligned} v_A &= v_B \\ \frac{1}{2}mv_A^2 &= \frac{1}{2}mv_B^2 \\ E_{\text{total},A} &= E_{\text{total},B} \end{aligned}$$

Now, let's investigate what happens when there is friction (an *non-conservative force*) acting on the ball.

Roll the ball along a rough surface or a carpeted floor. What happens to the speed of the ball at point A compared to point B?

If the surface you are rolling the ball along is very rough and provides a large non-conservative frictional force, then the ball should be moving much slower at point B than at point A.

Let's compare the total mechanical energy of the ball at points A and B:

$$\begin{aligned} E_{\text{total},A} &= EK_A + PE_A \\ &= \frac{1}{2}mv_A^2 + mgh_A \\ &= \frac{1}{2}mv_A^2 + mg(0) \\ &= \frac{1}{2}mv_A^2 \end{aligned}$$

$$\begin{aligned} E_{\text{total},B} &= EK_B + PE_B \\ &= \frac{1}{2}mv_B^2 + mgh_B \\ &= \frac{1}{2}mv_B^2 + mg(0) \\ &= \frac{1}{2}mv_B^2 \end{aligned}$$

However, in this case, $v_A \neq v_B$ and therefore $E_{\text{total},A} \neq E_{\text{total},B}$. Since

$$\begin{aligned}v_A &> v_B \\E_{\text{total},A} &> E_{\text{total},B}\end{aligned}$$

Therefore, the ball has lost mechanical energy as it moves across the carpet. However, although the ball has lost mechanical energy, energy in the larger system has still been conserved. In this case, the missing energy is the work done by the carpet through applying a frictional force on the ball. In this case the carpet is doing negative work on the ball.

When a non-conservative force (for example friction, air resistance, applied force) does work on an object, the total mechanical energy ($E_k + E_p$) of that object changes. If positive work is done, then the object will gain energy. If negative work is done, then the object will lose energy.

When a net **force does work** on an object, then there is **always a change in the kinetic energy** of the object. This is because the object experiences an acceleration and therefore a change in velocity.

This leads us to the work-energy theorem.

DEFINITION: *Work-energy theorem*

The work-energy theorem states that the work done on an object by the net force is equal to the change in its kinetic energy:

$$W_{\text{net}} = \Delta E_k = E_{k,f} - E_{k,i}$$

The work-energy theorem is another example of the conservation of energy which you saw in Grade 10.

Worked example 6: Work-energy theorem

QUESTION

A 1 kg brick is dropped from a height of 10 m. Calculate the work that has been done on the brick between the moment it is released and the moment when it hits the ground. Assume that air resistance can be neglected.

SOLUTION

Step 1: Determine what is given and what is required

- Mass of the brick: $m = 1 \text{ kg}$.
- Initial height of the brick: $h_i = 10 \text{ m}$.
- Final height of the brick: $h_f = 0 \text{ m}$.
- We are required to determine the work done on the brick as it hits the ground.

Step 2: Determine how to approach the problem

The brick is falling freely, so energy is conserved. We know that the work done is equal to the difference in kinetic energy. The brick has no kinetic energy at the moment it is dropped, because it is stationary. When the brick hits the ground, all the brick's potential energy is converted to kinetic energy.

Step 3: Determine the brick's potential energy at h_i

$$\begin{aligned} E_p &= m \cdot g \cdot h \\ &= (1)(9,8)(10) \\ &= 98 \text{ J} \end{aligned}$$

Step 4: Determine the work done on the brick

The brick had 98 J of potential energy when it was released and 0 J of kinetic energy. When the brick hit the ground, it had 0 J of potential energy and 98 J of kinetic energy. Therefore $E_{k,i} = 0 \text{ J}$ and $E_{k,f} = 98 \text{ J}$.

From the work-energy theorem:

$$\begin{aligned} W_{net} &= \Delta E_k \\ &= E_{k,f} - E_{k,i} \\ &= 98 - 0 \\ &= 98 \text{ J} \end{aligned}$$

Hence, 98 J of work was done on the brick.

Worked example 7: Work-energy theorem 2**QUESTION**

The driver of a 1000 kg car travelling at a speed of $16,7 \text{ m}\cdot\text{s}^{-1}$ applies the car's brakes when he sees a red light. The car's brakes provide a frictional force of 8000 N. Determine the stopping distance of the car.

SOLUTION**Step 1: Determine what is given and what is required**

We are given:

- mass of the car: $m = 1000 \text{ kg}$
- speed of the car: $v = 16,7 \text{ m}\cdot\text{s}^{-1}$
- frictional force of brakes: $\vec{F} = 8000 \text{ N}$

We are required to determine the stopping distance of the car.

TIP

A force only does work on an object for the time that it is in contact with the object. For example, a person pushing a trolley does work on the trolley, but the road does no work on the tyres of a car if they turn without slipping (the force is not applied over any distance because a different piece of tyre touches the road every instant).

Step 2: Determine how to approach the problem

We apply the work-energy theorem. We know that all the car's kinetic energy is lost to friction. Therefore, the change in the car's kinetic energy is equal to the work done by the frictional force of the car's brakes.

Therefore, we first need to determine the car's kinetic energy at the moment of braking using: $E_k = \frac{1}{2}mv^2$

This energy is equal to the work done by the brakes. We have the force applied by the brakes, and we can use: $W = F\Delta x \cos \theta$

to determine the stopping distance.

Step 3: Determine the kinetic energy of the car

$$\begin{aligned} E_k &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}(1000)(16,7)^2 \\ &= 139\,445 \text{ J} \end{aligned}$$

Step 4: Determine the work done

Assume the stopping distance is Δx_0 . Since the direction of the applied force and the displacement are in opposite directions, $\theta = 180^\circ$. Then the work done is:

$$\begin{aligned} W &= F\Delta x \cos \theta \\ &= (8000)(\Delta x_0) \cos(180) \\ &= (8000)(\Delta x_0)(-1) \\ &= (-8000)(\Delta x_0) \end{aligned}$$

Step 5: Apply the work-energy theorem

The change in kinetic energy is equal to the work done.

$$\begin{aligned} \Delta E_k &= W_{net} \\ E_{k,f} - E_{k,i} &= (-8000)(\Delta x_0) \\ 0 - 139\,445 &= (-8000)(\Delta x_0) \\ \therefore \Delta x_0 &= \frac{139445}{8000} \\ &= 17,4 \text{ m} \end{aligned}$$

Step 6: Write the final answer

The car stops in 17,4 m.

Worked example 8: Block on an inclined plane [credit: OpenStax College]**QUESTION**

A block of 2 kg is pulled up along a smooth incline of length 10 m and height 5 m by applying a non-conservative force. At the end of incline, the block is released from rest to slide down to the bottom. Find the

1. work done by the non-conservative force,

2. the kinetic energy of the block at the end of round trip, and
3. the speed at the end of the round trip.

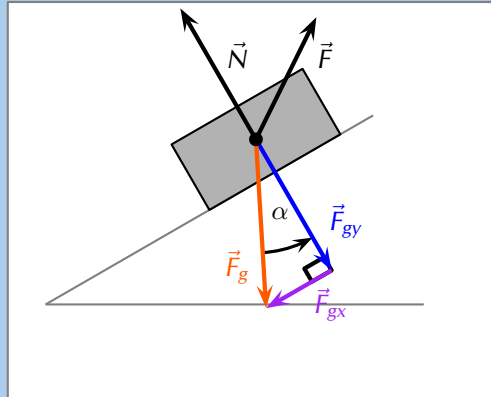
SOLUTION

Step 1: Analyse what is given and what is required

There are three forces on the block while going up:

1. weight of the block, $\vec{F}_g = m\vec{g}$,
2. normal force, $\vec{N}$, applied by the block and
3. non-conservative force, $\vec{F}$.

On the other hand, there are only two forces while going down. The non-conservative force is absent in downward journey. The force diagram of the forces is shown here for upward motion of the block.



Step 2: Work done by non-conservative force during round trip

The work done by the non-conservative force we will call W_F .

We have not been given a magnitude or direction for $\vec{F}$, all we do know is that it must result in the block moving up the slope.

We have represented the non-conservative force on the force diagram with an arbitrary vector. $\vec{F}$ acts only during upward journey. Note that the block is simply released at the end of the upward journey. We need to find the work done by the non-conservative force only during the upward journey.

$$W_F = W_{F(up)} + W_{F(down)} = W_{F(up)} + 0 = W_{F(up)}$$

The kinetic energies in the beginning and at the end of the motion up the slope are zero.

We can conclude that sum of the work done by all three forces is equal to zero during the upward motion. The change in kinetic energy is zero which implies that the net work done is zero.

$$\begin{aligned} W_{net} &= W_{F(up)} + W_{g(up)} + W_{N(up)} \\ 0 &= W_{F(up)} + W_{g(up)} + W_{N(up)} \end{aligned}$$

If we know the work done by the other two forces (normal force and gravity), then we can calculate the work done by the non-conservative force, F , as required.

Step 3: Work done by normal force during upward motion

The block moves up the slope, the normal force is perpendicular to the slope and, therefore, perpendicular to the direction of motion. Forces that are perpendicular to

the direction of motion do no work.

$$\begin{aligned} 0 &= W_{F(up)} + W_{g(up)} + W_{N(up)} \\ 0 &= W_{F(up)} + W_{g(up)} + (0) \\ W_{F(up)} &= -W_{g(up)} \end{aligned}$$

Thus, we need to compute work done by the gravity in order to compute work by the non-conservative force.

Step 4: Work done by gravity during upward motion

The component of the force due to gravity that is perpendicular to the slope, $\vec{F}_{gy}$, will do no work so we only need to consider the component parallel to the slope, $\vec{F}_{gx}$.

For the upward motion $\vec{F}_{gx}$ is in the opposite direction to the motion, therefore the angle between them is $\theta = 180^\circ$, which means $\cos \theta = -1$, so we expect to see negative work is done.

The magnitude of $\vec{F}_{gx}$ is $F_{gx} = mg \sin \alpha$. So the work done by gravity during the upward motion is :

$$\begin{aligned} W_{g(up)} &= F_{gx} \Delta x \cos \theta \\ &= mg \sin \alpha \cdot \Delta x \cos \theta \\ &= (2)(9.8)\left(\frac{5}{10}\right)(10) \cos(180) \\ &= -98 \text{ J} \end{aligned}$$

IMPORTANT!

Be careful not to be confused by which angle has been labelled α and which θ . α is not the angle between the force and the direction of motion but the incline of the plane in this particular problem. It is important to understand which symbol represents which physical quantity in the equations you have learnt.

Hence, the work done by the non-conservative force during the round trip is:

$$\begin{aligned} W_F &= W_{F(up)} = -W_{g(up)} \\ &= -(-98) \\ &= 98 \text{ J} \end{aligned}$$

Step 5: Kinetic energy at the end of round trip

The kinetic energy at the end of the upward motion was zero but it is not zero at the end of the entire downward motion.

We can use the work-energy theorem to analyse the whole motion:

$$\begin{aligned} W_{(round \text{ trip})} &= E_{k,f} - E_{k,i} \\ &= E_{k,f} - 0 \\ &= E_{k,f} \end{aligned}$$

To determine the net work done, $W_{(\text{roundtrip})}$, we need to sum the work done by each force acting during the period. We have calculated the work done by $\vec{F}$ already and we know that there is no work done by the normal force.

The total work done during round trip by gravity is the sum of the work done during the upward motion (where the force is in the opposite direction to the motion) and the downward motion (where the force is in the same direction as the motion).

The distance over which the force acts is the same during the upward and downward motion and the magnitude of the force is the same. The only difference between the calculation for the work done during the upward and downward motion is the sign because of the change of direction of the motion. Therefore:

$$\begin{aligned} W_{g(\text{round trip})} &= W_{g(\text{up})} + W_{g(\text{down})} \\ &= (mg \sin \alpha \cdot \Delta x \cos 180^\circ) + (mg \sin \alpha \cdot \Delta x \cos 0^\circ) \\ &= (-mg \sin \alpha \cdot \Delta x) + (mg \sin \alpha \cdot \Delta x) \\ &= 0 \text{ J} \end{aligned}$$

Hence, the total work done during round trip is:

$$\begin{aligned} W_{(\text{round trip})} &= W_F + W_g + W_N \\ &= W_F + W_{g(\text{up})} + W_{N(\text{up})} \\ &= (98) + (0) + (0) \\ &= 98 \text{ J} \end{aligned}$$

We can now use this in the equation for the work-energy theorem:

$$\begin{aligned} W_{(\text{round trip})} &= E_{k,f} \\ E_{k,f} &= 98 \text{ J} \end{aligned}$$

Step 6: Speed of the block

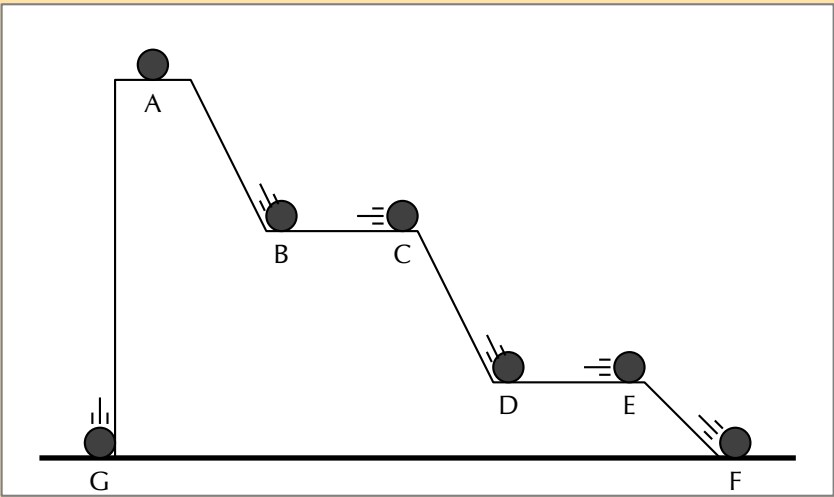
We know the kinetic energy of the block therefore we can determine its speed:

$$\begin{aligned} E_{k,f} &= \frac{1}{2}mv^2 \\ \frac{1}{2}mv^2 &= E_{k,f} \\ v &= \pm \sqrt{\frac{2}{m}E_{k,f}} \\ &= \pm \sqrt{\frac{2}{(2)}(98)} \\ &= \pm 9,89949493661 \\ &= 9,90 \text{ m}\cdot\text{s}^{-1} \end{aligned}$$

Note that the total work done during the upward motion is zero as the block is stationary at the beginning and at the end of the motion up the incline. The positive work done by the non-conservative force is cancelled by the fact that exactly the same amount of negative work is done by gravity. The net work comes from the work done during the downward motion by gravity. Net work done results in a change in kinetic energy as per the work-energy theorem.

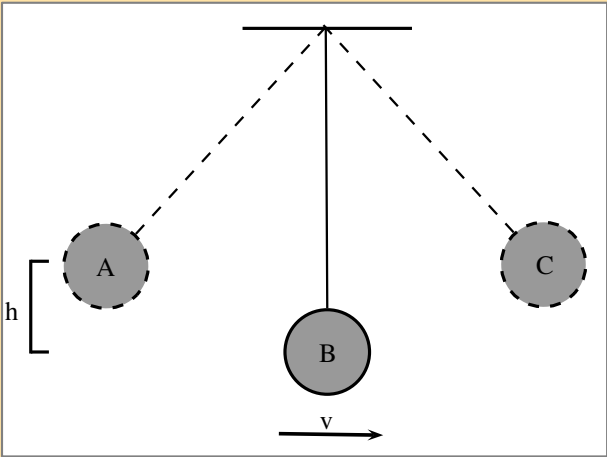
Exercise 5 – 2: Energy

1. Fill in the table with the missing information using the positions of the 1 kg ball in the diagram below combined with the work-energy theorem.



position	EK	PE	v
A		50 J	
B		30 J	
C			
D		10 J	
E			
F			
G			

2. A falling ball hits the ground at $10 \text{ m}\cdot\text{s}^{-1}$ in a vacuum. Would the speed of the ball be increased or decreased if air resistance were taken into account. Discuss using the work-energy theorem.
3. A pendulum with mass 300 g is attached to the ceiling. It is pulled up to point A which is a height $h = 30 \text{ cm}$ from the equilibrium position.



Calculate the speed of the pendulum when it reaches point B (the equilibrium point). Assume that there are no non-conservative forces acting on the pendulum.

4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27RV 2. 27RW 3. 27RX



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5.4 Conservation of energy

ESCMG

There are two categories of forces we will consider, conservative and non-conservative.

DEFINITION: *Conservative force*

A conservative force is one for which work done by or against it depends only on the starting and ending points of a motion and not on the path taken.

A **conservative force** results in stored or **potential energy** and we can define a potential energy (E_p) for any conservative force.

Gravity is a conservative force and we studied gravitational potential energy in Grade 10. We now have all the concepts we need to actually deduce this ourselves. Let us consider pushing a ball up a number of different slopes.

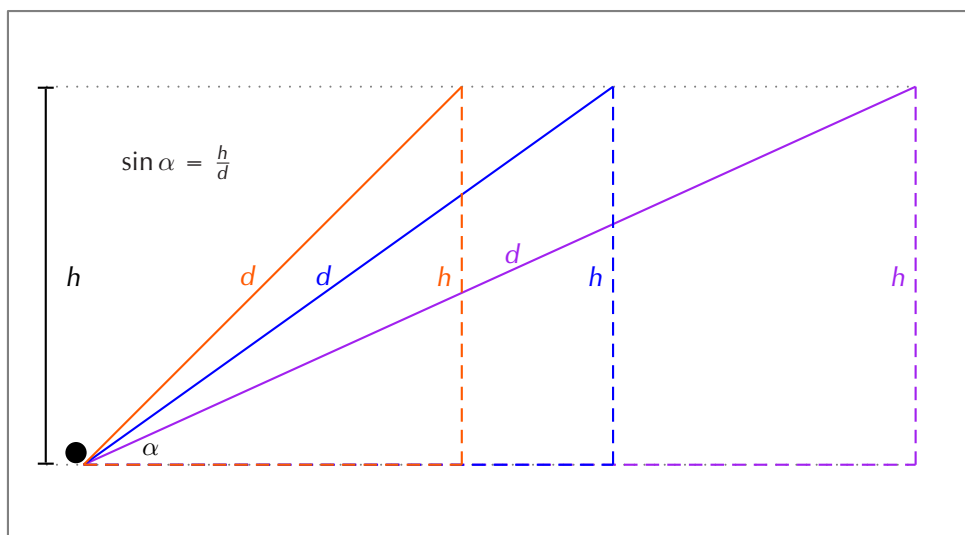
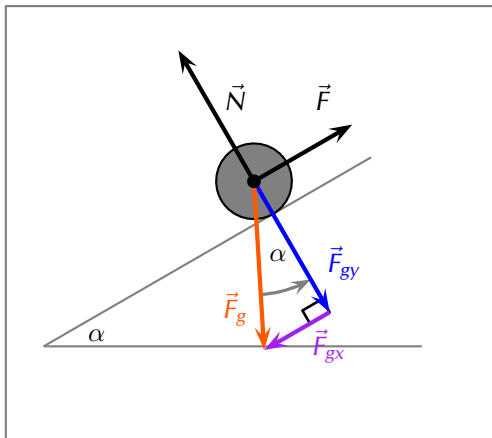


Figure 5.5: Three different slopes are shown, all rising to a height of h . The imaginary right-angled triangle is shown for each slope. d is the length of the slope. α is the angle the slope makes with the horizontal.

The slope, of length d is the hypotenuse of an imaginary right-angled triangle. The work done by gravity while pushing a ball of mass, m , up each of the slopes can be

calculated. We know that the component of the gravitational force parallel to the slope is $\vec{F}_{gx} = \vec{F}_g \sin \alpha$ down the slope.



The work done by gravity when the force $\vec{F}$ pushes the ball up the slope will be negative because the direction of the motion and $\vec{F}_g \sin \alpha$ are opposite.

$$\begin{aligned}
 W_g &= Fd \cos \theta \\
 &= (\vec{F}_g \sin \alpha)d(-1) \\
 &= -\vec{F}_g(\sin \alpha)d \\
 &= -\vec{F}_g \left(\frac{\text{opposite}}{\text{hypotenuse}} \right) d \\
 &= -\vec{F}_g \left(\frac{h}{d} \right) d \\
 &= -\vec{F}_g h
 \end{aligned}$$

This final result is independent of the angle of the slope. This is because $\sin \alpha = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{h}{d}$ and so the distance cancels out. If the ball moves down the slope the only change is the sign, the work done by gravity still only depends on the change in height. This is why mechanical energy includes gravitational potential energy and is conserved. If an object goes up a distance h gravity does negative work, if it moves back down h gravity does positive work, but the absolute amount of work is the same so you 'get it back', no matter what path you take!

This means that the work done by gravity will be same for the ball moving up any of the slopes because the end position is at the same height. The different slopes do not end in exactly the same position in the picture. If we break each slope into two sections as show in Figure 5.6 then we have 3 different paths to precisely the same end-point. In this case the total work done by gravity along each path is the sum of the work done on each piece which is just related to the height. The total work done is related to the total height.

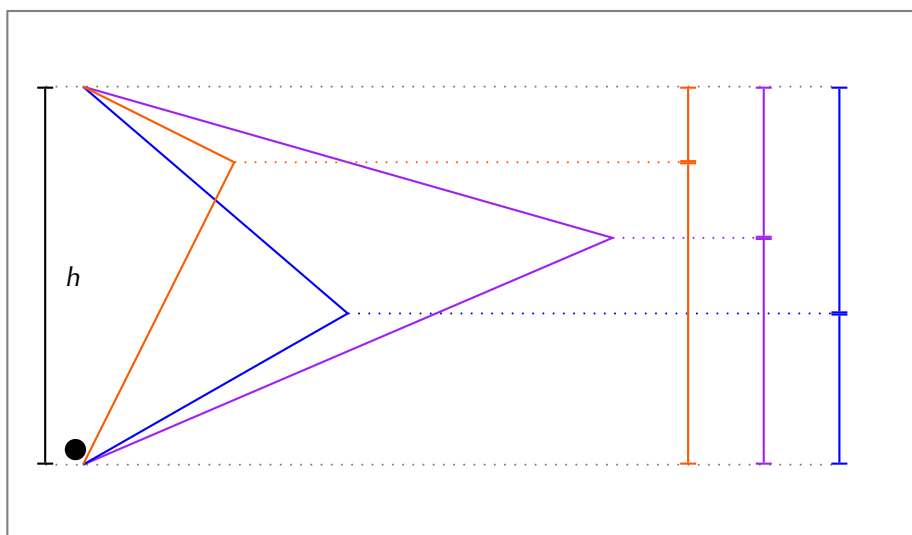


Figure 5.6: Three different paths that lead from the same start-point to the same-end point. Each path leads to the same overall change in height, h , and, therefore, the same work done by gravity.

There are other examples, when you wind up a toy, an egg timer, or an old-fashioned watch, you do work against its spring and store energy in it. (We treat these springs as ideal, in that we assume there is no friction and no production of thermal energy.) This stored energy is recoverable as work, and it is useful to think of it as potential energy contained in the spring.

The total work done by a conservative force results in a change in potential energy, ΔE_p . If the conservative force does positive work then the change in potential energy is negative. Therefore:

$$W_{\text{conservative}} = -\Delta E_p$$

DEFINITION: *Non-conservative force*

A non-conservative force is one for which work done on the object depends on the path taken by the object.

IMPORTANT!

Non-conservative forces do not imply that total energy is not conserved. Total energy is always conserved. Non-conservative forces mean that mechanical energy isn't conserved in a particular system which implies that the energy has been transferred in a process that isn't reversible.

Friction is a good example of a non-conservative force because it removes energy from the system so the amount of mechanical energy is not conserved. Non-conservative forces can also do positive work thereby increasing the total mechanical energy of the system.

The energy transferred to overcome friction depends on the distance covered and is converted to thermal energy which can't be recovered by the system.

Non-conservative forces and work-energy theorem

ESCMH

We know that the net work done will be the sum of the work done by all of the individual forces:

$$F_{\text{net}} = F_{\text{conservative}} + F_{\text{non-conservative}}$$

(the work done by these forces yields)

$$W_{\text{net}} = W_{\text{conservative}} + W_{\text{non-conservative}}$$

(using the work-energy theorem)

$$\Delta E_k = W_{\text{conservative}} + W_{\text{non-conservative}}$$

(using $W_{\text{conservative}} = -\Delta PE$)

$$\Delta E_k = -\Delta E_p + W_{\text{non-conservative}}$$

$$W_{\text{non-conservative}} = \Delta E_k + \Delta E_p$$

When the non-conservative forces oppose the motion, the work done by the non-conservative forces is negative, causing a decrease in the mechanical energy of the system. When the non-conservative forces do positive work, energy is added to the system. If the sum of the non-conservative forces is zero then mechanical energy is conserved.

QUESTION

Consider the situation shown where a football player slides to a stop on level ground. Using energy considerations, calculate the distance the 65,0 kg football player slides, given that his initial speed is $6,00 \text{ m}\cdot\text{s}^{-1}$ and the force of friction against him is a constant 450 N.



SOLUTION

Step 1: Analyse the problem and determine what is given

Friction stops the player by converting his kinetic energy into other forms, including thermal energy. In terms of the work-energy theorem, the work done by friction, which is negative, is added to the initial kinetic energy to reduce it to zero. The work done by friction is negative, because F_f is in the opposite direction of the motion (that is, $\theta = 180^\circ$, and so $\cos \theta = -1$). Thus $W_{\text{non-conservative}} = -F_f \Delta x$.

There is no change in potential energy.

Step 2: Next we calculate the distance using the conservation of energy

We begin with conservation of energy:

$$W_{\text{non-conservative}} = \Delta E_k + \Delta E_p$$

The equation expands to:

$$\begin{aligned} W_{\text{non-conservative}} &= \Delta E_k + \Delta E_p \\ &= E_{k,f} - E_{k,i} + (0) \\ -F_f \Delta x &= E_{k,f} - E_{k,i} \\ -F_f \Delta x &= (0) - E_{k,i} \\ F_f \Delta x &= \frac{1}{2} m v_i^2 \\ \Delta x &= \frac{m v_i^2}{2 F_f} \\ &= \frac{(65,0)(6,00)^2}{2(450)} \\ &= 2,60 \text{ m} \end{aligned}$$

Step 3: Quote the final answer

The footballer comes to a stop after sliding for 2,60 m.

Discussion

The most important point of this example is that the amount of non-conservative work

equals the change in mechanical energy. For example, you must work harder to stop a truck, with its large mechanical energy, than to stop a mosquito.

Worked example 10: Sliding up a slope [credit: OpenStax College Physics]

QUESTION

The same 65,0 kg footballer running at the same speed of $6,00 \text{ m}\cdot\text{s}^{-1}$ dives up the inclined embankment at the side of the field. The force of friction is still 450 N as it is the same surface, but the surface is inclined at 5° . How far does he slide now?

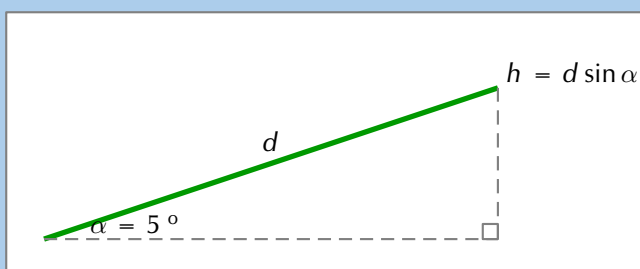
SOLUTION

Step 1: Analyse the question

Friction stops the player by converting his kinetic energy into other forms, including thermal energy, just in the previous worked example. The difference in this case is that the height of the player will change which means a non-zero change to gravitational potential energy.

The work done by friction is negative, because F_f is in the opposite direction of the motion (that is, $\theta = 180^\circ$).

We sketch the situation showing that the footballer slides a distance d up the slope.



In this case, the work done by the non-conservative friction force on the player reduces the mechanical energy he has from his kinetic energy at zero height, to the final mechanical energy he has by moving through distance d to reach height h along the incline. This is expressed by the equation:

$$\begin{aligned} W_{\text{non-conservative}} &= \Delta E_k + \Delta E_p \\ W_{\text{non-conservative}} &= E_{k,f} - E_{k,i} + E_{p,f} - E_{p,i} \\ W_{\text{non-conservative}} + E_{k,i} + E_{p,i} &= E_{k,f} + E_{p,f} \end{aligned}$$

We know that:

- the work done by friction is $W_{\text{non-conservative}} = -F_f d$,
- the initial potential energy is $E_{p,i} = mg(0) = 0$,
- the initial kinetic energy is $E_{k,i} = \frac{1}{2}mv_i^2$,
- the final kinetic energy is $E_{k,f} = 0$, and

- the final potential energy is $E_{p,f} = mgh = mgd \sin \theta$.

Step 2: Solve for the distance

$$\begin{aligned}
 W_{\text{non-conservative}} + E_{k,i} + E_{p,i} &= E_{k,f} + E_{p,f} \\
 (-F_f d) + \frac{1}{2}mv_i^2 + 0 &= 0 + mgd \sin \theta \\
 (-F_f d) - mgd \sin \theta &= -\frac{1}{2}mv_i^2 \\
 d(F_f + mg \sin \theta) &= \frac{1}{2}mv_i^2 \\
 d &= \frac{\frac{1}{2}mv_i^2}{F_f + mg \sin \theta} \\
 &= \frac{\frac{1}{2}(65,0)(6,00)^2}{450 + (65,0)(9,8) \sin(5,00)} \\
 &= 2,31 \text{ m}
 \end{aligned}$$

Step 3: Quote the final answer

The player slides for 2,31 m before stopping.

As might have been expected, the footballer slides a shorter distance by sliding uphill. Note that the problem could also have been solved in terms of the forces directly and the work energy theorem, instead of using the potential energy. This method would have required combining the normal force and force of gravity vectors, which no longer cancel each other because they point in different directions, and friction, to find the net force. You could then use the net force and the net work to find the distance d that reduces the kinetic energy to zero. By applying conservation of energy and using the potential energy instead, we need only consider the gravitational potential energy, without combining and resolving force vectors. This simplifies the solution considerably.

Exercise 5 – 3: Energy conservation

1. A 60,0 kg skier with an initial speed of $12,0 \text{ m}\cdot\text{s}^{-1}$ coasts up a 2,50 m-high rise as shown in the figure. Find her final speed at the top, given that the coefficient of friction between her skis and the snow is 0,0800. (Hint: Find the distance traveled up the incline assuming a straight-line path as shown in the figure.)



2.
 - a) How high a hill can a car coast up (engine disengaged) if work done by friction is negligible and its initial speed is $110 \text{ km}\cdot\text{h}^{-1}$?
 - b) If, in actuality, a 750 kg car with an initial speed of $110 \text{ km}\cdot\text{h}^{-1}$ is observed to coast up a hill to a height 22,0 m above its starting point, how much thermal energy was generated by friction?

- c) What is the average force of friction if the hill has a slope $2,5^\circ$ above the horizontal?
3. A bullet traveling at 100 m/s just pierces a wooden plank of 5 m. What should be the speed (in m/s) of the bullet to pierce a wooden plank of same material, but having a thickness of 10m?
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27RY 2. 27RZ 3. 27S2



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FACT

The unit watt is named after Scottish inventor and engineer James Watt (19 January 1736 - 19 August 1819) whose improvements to the steam engine were fundamental to the Industrial Revolution. A key feature of it was that it brought the engine out of the remote coal fields into factories.

FACT

Historically, the *horsepower* (symbol hp) was the unit used to describe the power delivered by a machine. One horsepower is equivalent to approximately 750 W. The horsepower was derived by James Watt to give an indication of the power of his steam engine in terms of the power of a horse, which was what most people used to for example, turn a mill wheel.

5.5 Power

ESCMJ

Now that we understand the relationship between work and energy, we are ready to look at a quantity related the rate of energy transfer. For example, a mother pushing a trolley full of groceries can take 30 s or 60 s to push the trolley down an aisle. She does the same amount of work, but takes a different length of time. We use the idea of *power* to describe the rate at which work is done.

DEFINITION: Power

Power is defined as the rate at which work is done or the rate at which energy is transferred to or from a system. The mathematical definition for power is:

$$P = \frac{W}{t}$$

Power is easily derived from the definition of work. We know that: $W = F\Delta x \cos \theta$

Power is defined as the rate at which work is done. Therefore,

$$\begin{aligned} P &= \frac{W}{t} \\ &= \frac{F\Delta x \cos \theta}{t} \end{aligned}$$

in the case where F and Δx are in the same direction

$$\begin{aligned} &= \frac{F\Delta x}{t} \\ &= F \frac{\Delta x}{t} \\ &= Fv \end{aligned}$$

IMPORTANT!

In the case where the force and the velocity are in opposite directions the power will be negative.

The unit of power is watt (symbol W).

Worked example 11: Power calculation 1

QUESTION

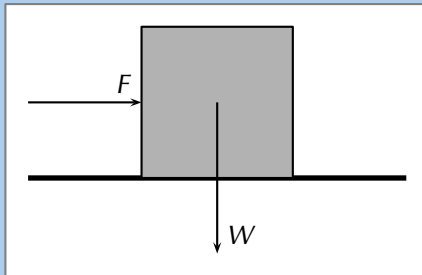
Calculate the power required for a force of 10 N applied to move a 10 kg box at a speed of $1 \text{ m} \cdot \text{s}^{-1}$ over a frictionless surface.

SOLUTION

Step 1: Determine what is given and what is required.

- We are given the force, $F = 10 \text{ N}$.
- We are given the speed, $v = 1 \text{ m} \cdot \text{s}^{-1}$.
- We are required to calculate the power required.

Step 2: Draw a force diagram



Step 3: Determine how to approach the problem

From the force diagram, we see that the weight of the box is acting at right angles to the direction of motion. The weight does not contribute to the work done and does not contribute to the power calculation. We can therefore calculate power from: $P = F \cdot v$.

Step 4: Calculate the power required

$$\begin{aligned} P &= F \cdot v \\ &= (10 \text{ N}) (1 \text{ m} \cdot \text{s}^{-1}) \\ &= 10 \text{ W} \end{aligned}$$

Step 5: Write the final answer

10 W of power are required for a force of 10 N to move a 10 kg box at a speed of $1 \text{ m} \cdot \text{s}^{-1}$ over a frictionless surface.

Machines are designed and built to do work on objects. All machines usually have a power rating. The power rating indicates the rate at which that machine can do work upon other objects.

A car engine is an example of a machine which is given a power rating. The power rating relates to how rapidly the car can accelerate. Suppose that a 50 kW engine could accelerate the car from $0 \text{ km} \cdot \text{hr}^{-1}$ to $60 \text{ km} \cdot \text{hr}^{-1}$ in 16 s. Then a car with four times the power rating (i.e. 200 kW) could do the same amount of work in a quarter of the time. That is, a 200 kW engine could accelerate the same car from $0 \text{ km} \cdot \text{hr}^{-1}$ to $60 \text{ km} \cdot \text{hr}^{-1}$ in 4 s.

Worked example 12: Power calculation 2

QUESTION

A forklift lifts a crate of mass 100 kg at a constant velocity to a height of 8 m over a time of 4 s. The forklift then holds the crate in place for 20 s. Calculate how much power the forklift exerts in lifting the crate? How much power does the forklift exert in holding the crate in place?

SOLUTION

Step 1: Determine what is given and what is required

We are given:

- mass of crate: $m = 100 \text{ kg}$
- height that crate is raised: $h = 8 \text{ m}$
- time taken to raise crate: $t_r = 4 \text{ s}$
- time that crate is held in place: $t_s = 20 \text{ s}$

We are required to calculate the power exerted.

Step 2: Determine how to approach the problem

We can use: $P = Fv = F \frac{\Delta x}{\Delta t}$

to calculate power. The force required to raise the crate is equal to the weight of the crate.

Step 3: Calculate the power required to raise the crate

$$\begin{aligned} P &= F \frac{\Delta x}{\Delta t} \\ &= m \cdot g \frac{\Delta x}{\Delta t} \\ &= (100 \text{ kg}) (9,8 \text{ m} \cdot \text{s}^{-2}) \frac{8 \text{ m}}{4 \text{ s}} \\ &= 1960 \text{ W} \end{aligned}$$

Step 4: Calculate the power required to hold the crate in place

While the crate is being held in place, there is no displacement. This means there is no work done on the crate and therefore there is no power exerted.

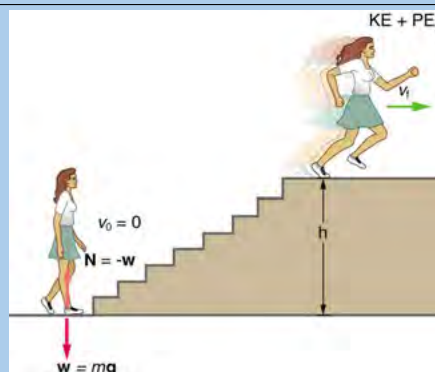
Step 5: Write the final answer

1960 W of power is exerted to raise the crate and no power is exerted to hold the crate in place.

Worked example 13: Stair climb

QUESTION

What is the power output for a 60,0 kg woman who runs up a 3,00 m high flight of stairs in 3,50 s, starting from rest but having a final speed of $2,00 \text{ m} \cdot \text{s}^{-1}$? Her power output depends on how fast she does this.



SOLUTION

Step 1: Analyse the question

The work going into mechanical energy is $W = E_k + E_p$. At the bottom of the stairs, we take both E_k and the potential energy due to gravity, $E_{p,g}$, as initially zero; thus, $W = E_{k,f} + E_{p,g} = \frac{1}{2}mv_f^2 + mgh$, where h is the vertical height of the stairs. Because all terms are given, we can calculate W and then divide it by time to get power.

Step 2: Calculate power

Substituting the expression for W into the definition of power given in the previous equation, $P = \frac{W}{t}$ yields:

$$\begin{aligned} P &= \frac{W}{t} \\ &= \frac{\frac{1}{2}mv_f^2 + mgh}{t} \\ &= \frac{\frac{1}{2}(60,0)(2,00)^2 + (60,0)(9,80)(3,00)}{3,50} \\ &= \frac{120 + 1764}{3,50} \\ &= 538,3 \text{ W} \end{aligned}$$

Step 3: Quote the final answer

The power generated is 538,0 W.

The woman does 1764 J of work to move up the stairs compared with only 120 J to increase her kinetic energy; thus, most of her power output is required for climbing rather than accelerating.

Worked example 14: A borehole

QUESTION

What is the power required to pump water from a borehole that has a depth $h = 15,0$ m at a rate of $20,0 \text{ l}\cdot\text{s}^{-1}$?

SOLUTION

Step 1: Analyse the question

We know that we will have to do work on the water to overcome gravity to raise it a certain height. If we ignore any inefficiencies we can calculate the work, and power, required to raise the mass of water the appropriate height.

We know how much water is required in a single second. We can first determine the mass of water: $20,0 \text{ l} \times \frac{1 \text{ kg}}{1 \text{ l}} = 20,0 \text{ kg}$.

The water will also have non-zero kinetic energy when it gets to the surface because it needs to be flowing. The pump needs to move 20,0 kg from the depth of the borehole

every second, we know the depth so we know the speed that the water needs to be moving is $v = \frac{h}{t} = \frac{15,0}{1} = 15,0 \text{ m}\cdot\text{s}^{-1}$.

Step 2: Work done to raise the water

We can use

$$\begin{aligned} W_{\text{non-conservative}} &= \Delta E_k + \Delta E_p \\ &= E_{k,f} - E_{k,i} + E_{p,f} - E_{p,i} \\ &= \frac{1}{2}mv^2 - (0) + mgh - 0 \\ &= \frac{1}{2}(20)(15)^2 + (20)(9,8)(15) \\ &= 2,25 \times 10^3 + 2,94 \times 10^3 \\ &= 5,19 \times 10^3 \text{ J} \end{aligned}$$

Step 3: Calculate power

$$\begin{aligned} P &= \frac{W}{t} \\ &= \frac{5,19 \times 10^3}{1} \\ &= 5,19 \times 10^3 \text{ W} \end{aligned}$$

Step 4: Quote the final answer

The minimum power required from the pump is $5,19 \times 10^3 \text{ W}$.

Experiment: Simple measurements of human power

You can perform various physical activities, for example lifting measured weights or climbing a flight of stairs to estimate your output power, using a stop watch. Note: the human body is not very efficient in these activities, so your actual power will be much greater than estimated here.

Exercise 5 – 4: Power

- [IEB 2005/11 HG] Which of the following is equivalent to the SI unit of power:
 - $\text{V}\cdot\text{A}$
 - $\text{V}\cdot\text{A}^{-1}$
 - $\text{kg}\cdot\text{m}\cdot\text{s}^{-1}$
 - $\text{kg}\cdot\text{m}\cdot\text{s}^{-2}$
- Two students, Bill and Bob, are in the weight lifting room of their local gym. Bill lifts the 50 kg barbell over his head 10 times in one minute while Bob lifts the 50 kg barbell over his head 10 times in 10 seconds. Who does the most work? Who delivers the most power? Explain your answers.
- Jack and Jill ran up the hill. Jack is twice as massive as Jill; yet Jill ascended the same distance in half the time. Who did the most work? Who delivered the most power? Explain your answers.
- When doing a chin-up, a physics student lifts her 40 kg body a distance of 0,25 m in 2 s. What is the power delivered by the student's biceps?
- The unit of power that is used on a monthly electricity account is *kilowatt-hours* (symbol kWh). This is a unit of energy delivered by the flow of 1 kW of electricity for 1 hour. Show how many joules of energy you get when you buy 1 kWh of electricity.

6. An escalator is used to move 20 passengers every minute from the first floor of a shopping mall to the second. The second floor is located 5-meters above the first floor. The average passenger's mass is 70 kg. Determine the power requirement of the escalator in order to move this number of passengers in this amount of time.
7. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27S3 2. 27S4 3. 27S5 4. 27S7 5. 27S8 6. 27S9



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5.6 Chapter summary

ESCMK

▶ See presentation: 27SC at www.everythingscience.co.za

- **Kinetic energy:** $E_k = \frac{1}{2}mv^2$
- **Conservation of Energy:** Energy is never created nor destroyed, but is only transformed from one form to another.
- **Conservation of Mechanical Energy:** in the absence of non-conservative forces mechanical energy is conserved.
- When a force acting on an object has a component along the line of motion, work is done.
- Work is the process of transferring energy from object or system to another.
- Energy is the ability to do work.

Units					
Quantity	Symbol	Unit	S.I. Units		Direction
Velocity	$\vec{v}$	—	$\text{m}\cdot\text{s}^{-1}$	or $\text{m}\cdot\text{s}^{-1}$	✓
Force	$\vec{F}$	—	$\text{kg}\cdot\text{m}\cdot\text{s}^{-2}$	or N	✓
Energy	E	J	$\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}$	or $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}$	—
Work	W	J	N·m	or $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}$	—
Kinetic Energy	E_k	J	N·m	or $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}$	—
Potential Energy	E_p	J	N·m	or $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}$	—
Mechanical Energy	U	J	N·m	or $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-2}$	—
Power	P	W	$\text{N}\cdot\text{m}\cdot\text{s}^{-1}$	or $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-3}$	—

Exercise 5 – 5:

1. How much work does a person do in pushing a shopping trolley with a force of 200 N over a distance of 80 m in the direction of the force?
2. How much work does the force of gravity do in pulling a 20 kg box down a 45° frictionless inclined plane of length 18 m?

3. [IEB 2001/11 HG1] Of which one of the following quantities is $\text{kg}\cdot\text{m}^2\cdot\text{s}^{-3}$ the base S.I. unit?

a	Energy	c	Power
b	Force	d	Momentum

4. [IEB 2003/11 HG1] A motor is used to raise a mass m through a vertical height h in time t . What is the power of the motor while doing this?

a	$mght$	c	$\frac{mgt}{h}$
b	$\frac{mgh}{t}$	d	$\frac{ht}{mg}$

5. [IEB 2002/11 HG1] An electric motor lifts a load of mass M vertically through a height h at a constant speed v . Which of the following expressions can be used to calculate the power transferred by the motor to the load while it is lifted at constant speed?

a	Mgh	c	Mgv
b	$Mgh + \frac{1}{2} Mv^2$	d	$Mgv + \frac{1}{2} \frac{Mv^3}{h}$

6. A set 193 kg containers need to be lifted onto higher floors during a building operation. The distance that they need to be raised is 7.5 m, at constant speed. The project manager has determined that in order to keep to budget and time this has to happen in as close to 5,0 s as possible. The power ratings for three motors are listed as 1,0 kW, 3,5 kW, and 5,5 kW. Which motor is best for the job?
7. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27SF 2. 27SG 3. 27SH 4. 27SJ 5. 27SK 6. 27SM



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Doppler effect

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6.1 Introduction

ESCMM

Have you noticed how the pitch of a police car or ambulance siren changes as it passes where you are standing, or how an approaching car or train sounds different to when it is travelling away from you? If you haven't, try to do an experiment by paying extra careful attention the next time it happens to see if you can notice a difference in pitch. This doesn't apply to just vehicles and trains but anything that emits waves, be those sound waves or any other electromagnetic (EM) waves.

The effect actually occurs if you move towards or away from the source of the sound as well. This effect is known as the **Doppler effect** and will be studied in this chapter.

Investigation: Creating the Doppler effect in class

You can create the Doppler effect in class. One way of doing this is to get:

- string, and
- a tuning fork.

Tie the string to the base of the tuning fork. Strike the tuning fork to create a note and then hold the other end of the string and swing the tuning fork in circles in the air in a horizontal plane.

WARNING!

The string needs to be very securely tied to the tuning fork to ensure that it does not come loose during the demonstration.

The class should be able to hear that the frequency heard when the tuning fork is moving is different to the frequency heard when it is stationary.

▶ See video: 27SP at www.everythingscience.co.za

▶ See video: 27SQ at www.everythingscience.co.za

▶ See video: 27SR at www.everythingscience.co.za

Key linked concepts

- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Sound waves — Physical Sciences, Grade 10, Sound
- Electromagnetic radiation — Physical Sciences, Grade 10, Electromagnetic radiation

You, the person hearing the sounds, are called the *observer* or *listener* and the thing emitting the sound is called the *source*. As mentioned in the introduction, there are two situations which lead to the Doppler effect:

1. When the source moves relative to a stationary observer.
2. When the observer moves relative to a stationary source.

In points 1 and 2 above there is relative motion between the source and the observer. Both the source and the observer can be moving at the same time but we won't deal with that case in this chapter.

DEFINITION: Doppler effect

The Doppler effect is the change in the observed frequency of a wave when the source or the detector moves relative to the transmitting medium.

The Doppler effect occurs when a source of waves and/or observer move relative to each other, resulting in the observer measuring a different frequency of the waves than the frequency that the source is emitting. The medium that the waves are travelling through, the transmitting medium, is also stationary in the cases we will study.

The question that probably comes to mind is: "*How does the Doppler effect come about?*". We can understand what is happening by thinking through the situation in detail.

Case 1: Moving source, stationary observer

Let us consider a source of sound waves with a constant frequency and amplitude. The sound waves can be represented as concentric circles where each circle represents a crest or peak as the wavefronts radiate away from the source. This is because the waves travel away from the source in all directions and the distance between consecutive crests or consecutive troughs in a wave is constant (the wavelength as we learnt in Grade 10). In this figure the crests are represented by the black lines and the troughs by the orange lines.

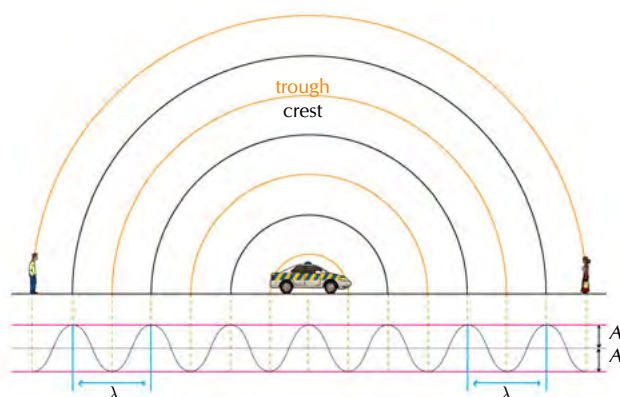


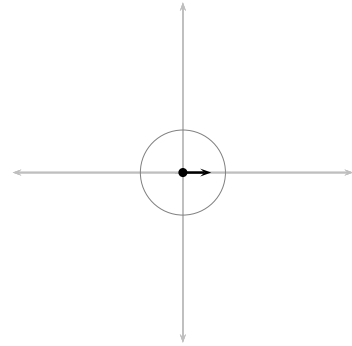
Figure 6.1: Stationary sound source as more wavefronts are emitted.

FACT

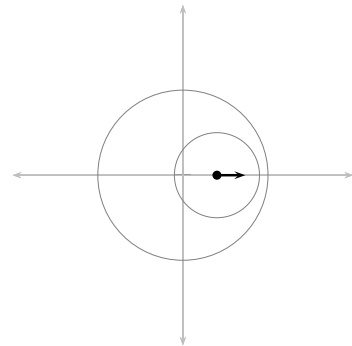
The Doppler effect is named after Johann Christian Andreas Doppler (29 November 1803 - 17 March 1853), an Austrian mathematician and physicist who first explained the phenomenon in 1842.

The sound source is the police car in the middle and is stationary. For the Doppler effect to take place (manifest), the source must be moving *relative* to the observer.

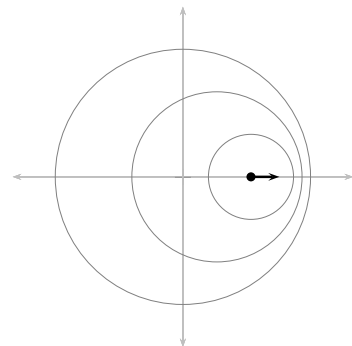
Let's consider the following situation: The source (represented by the black dot) emits one wave (the black circles represent the crests of the sound wave) that moves away from the source at the same rate in all directions. The distance between the crests represents the wavelength (λ) of the sound. The closer together the crests, the higher the frequency (or pitch) of the sound according to $f = \frac{v}{\lambda}$, where v (speed of sound) is constant.



As this crest moves away, the source also moves and then emits more crests. Now the two circles are not concentric any more, but on the one side they are closer together and on the other side they are further apart. This is shown in the next diagram.



If the source continues moving at the same speed in the same direction, then the distance between crests on the right of the source is constant. The distance between crests on the left is also constant. The distance between successive crests on the left is constant but larger than the distance between successive crests on the right.



When a car approaches you, the sound waves that reach you have a shorter wavelength and a higher frequency. You hear a sound with a higher pitch. When the car moves away from you, the sound waves that reach you have a longer wavelength and lower frequency. You hear a sound with a lower pitch.

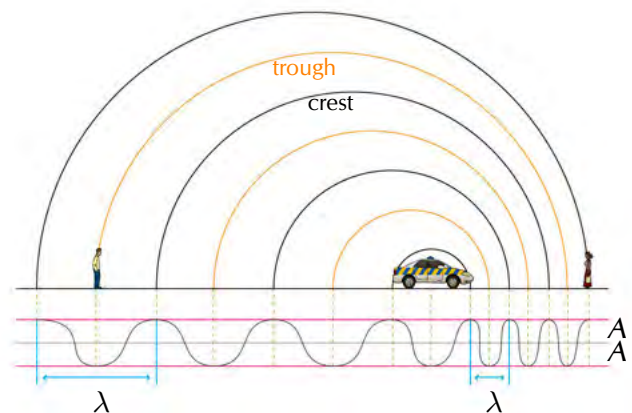


Figure 6.2: Moving sound source as more wavefronts are emitted.

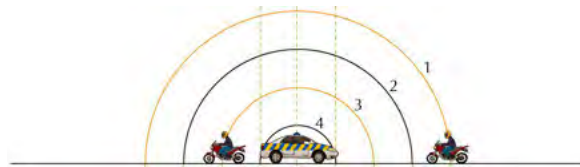
Just as we did before, let us consider a source (a police car) of sound waves with a constant frequency and amplitude. There are two observers, one on the left that will move away from the source and one on the right that will move towards the source. We have three diagrams:

1. shows the overall situation with the siren starting at time t_1 ;
2. shows the situation at time t_2 when the observers are moving; and
3. shows the situation at t_3 after the observers have been moving for a time interval, $\Delta t = t_3 - t_2$.

The crests and troughs are numbered so you can see how they move further away and so that we can track which ones an observer has measured.

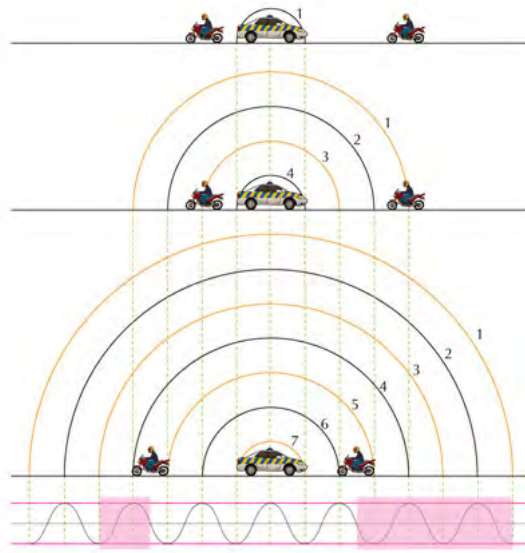


The observers can hear the sound waves emitted by the police car and they start to move (we ignore the time it takes them to accelerate).



The frequency of the wave that an observer measures is the number of complete waves cycles per unit time. By numbering the crests and troughs we can see which complete wave cycles have been measured by each of the observers in time, Δt . To find the frequency we divide the number of wave cycles by Δt .

In the time interval that passed, the observer moving towards the police car observed the crests and troughs numbered 1 through 5 (the portion of the wave is highlighted below). The observer moving away encountered a smaller portion of the wavefront, crest 3 and trough 4. The time interval for each of them is the same. To the observers this will mean that the frequency they measured is different.



The motion of the observer will alter the frequency of the measured sound from a stationary source:

- An observer moving **towards** the source measures a **higher** frequency.
- An observer moving **away from** the source measures a **lower** frequency.

It is important to note that we have only looked at the cases where the source and observer are moving directly towards or away from each other and these are the only cases we will consider.

NOTE:

We didn't actually need to analyse both cases. We could have used either explanation because of relative motion. The case of a stationary source with moving observer is the same as the case of the stationary observer and the moving source because the relative motion is the same. Do you agree? Discuss with your friends and try to convince yourselves that this is the case. Being able to explain work to each other will help you understand it better. If you don't understand it, you won't be able to explain it convincingly.

For a real conceptual test, discuss what you think will happen if the source and the observer are both moving, in the same direction and at the same speed.

The formula that provides the relationship between the frequency emitted by the source and the frequency measured by the listener is:

$$f_L = \left(\frac{v \pm v_L}{v \pm v_S} \right) f_S$$

- where f_L is the frequency perceived by the observer (listener),
- f_S is the frequency of the source,
- v is the speed of the waves,
- v_L the speed of the listener and
- v_S the speed of the source.

Note: The signs show whether or not the relative motion of the source and observer is towards each other or away from each other:

Source moves towards listener	v_S : negative
Source moves away from listener	v_S : positive
Listener moves towards source	v_L : positive
Listener moves away from source	v_L : negative

We only deal with one of the source or observer moving in this chapter. To understand the sign choice you can think about the pictures of the motion. For the listener/observer we are using the numerator in the equation. A fraction gets larger when the numerator gets larger so if we expect the frequency to increase we expect addition in the numerator ($f_L = (\frac{v+v_L}{v})f_S$). If the numerator gets smaller the fraction gets smaller so if we expect the frequency to decrease then it is subtraction in the numerator ($f_L = (\frac{v-v_L}{v})f_S$).

For the denominator the reverse is true because of the fact that we divide by the denominator. The larger the denominator the smaller the fraction and vice versa. So if we expect the motion of the source to increase the frequency we expect subtraction in the denominator ($f_L = (\frac{v}{v-v_S})f_S$) and if we expect the frequency to decrease we expect addition in the denominator ($f_L = (\frac{v}{v+v_S})f_S$).

► See video: 27SS at www.everythingscience.co.za

Worked example 1: Ambulance siren

QUESTION

The siren of an ambulance emits sound with a frequency of 700 Hz. You are standing on the pavement.

If the ambulance drives past you at a speed of $20 \text{ m}\cdot\text{s}^{-1}$, what frequency will you hear, when

1. a) the ambulance is approaching you
2. b) the ambulance is driving away from you

Take the speed of sound in air to be $340 \text{ m}\cdot\text{s}^{-1}$.



SOLUTION

Step 1: Analyse the question

The question explicitly asks what frequency you will hear when the source is moving at a certain speed. This tells you immediately that the question is related to the Doppler effect. The values given in the question are all in S.I. units so no conversions are required.

Step 2: Determine how to approach the problem based on what is given

We know that we are looking for the observed frequency with a moving source. The change in frequency can be calculated using:

$$f_L = \left(\frac{v \pm v_L}{v \pm v_S} \right) f_S$$

To correctly apply this we need to confirm that it is valid and determine what signs we need to use for the various speeds. You (the listener) are not moving but we have to consider two different cases, when the ambulance is moving towards you (a) and away from you (b). We have been told that if the source is moving towards the observer then we will use subtraction in the denominator and if it is moving away, addition. This means:

$$f_S = 700 \text{ Hz}$$

$$v = 340 \text{ m} \cdot \text{s}^{-1}$$

$$v_L = 0 \text{ because you, the observer, are not moving}$$

$$v_S = -20 \text{ m} \cdot \text{s}^{-1} \text{ for (a) and}$$

$$v_S = +20 \text{ m} \cdot \text{s}^{-1} \text{ for (b)}$$

Step 3: Determine f_L when ambulance is approaching

$$\begin{aligned} f_L &= \left(\frac{v \pm v_L}{v \pm v_S} \right) f_S \\ f_L &= \left(\frac{340 + 0}{340 - 20} \right) (700) \\ &= 743,75 \text{ Hz} \end{aligned}$$

Step 4: Determine f_L when ambulance has passed

$$\begin{aligned} f_L &= \left(\frac{v \pm v_L}{v \pm v_S} \right) f_S \\ f_L &= \left(\frac{340 + 0}{340 + 20} \right) (700) \\ &= 661,11 \text{ Hz} \end{aligned}$$

Step 5: Quote the final answer

When the ambulance is approaching you, you hear a frequency of 743,75 Hz and when it is going away you hear a frequency of 661,11 Hz

Worked example 2: Moving observer**QUESTION**

What is the frequency heard by a person driving at $15 \text{ m} \cdot \text{s}^{-1}$ toward a factory whistle that is blowing at a frequency of 800 Hz. Assume that the speed of sound in air is $340 \text{ m} \cdot \text{s}^{-1}$.

SOLUTION**Step 1: Analyse the question**

The question explicitly asks what frequency you will hear when the observer is moving at a certain speed. This tells you immediately that the question is related to the Doppler shift. The values given in the question are all in S.I. units so no conversions

are required.

Step 2: Determine how to approach the problem based on what is given

We can use:

$$f_L = \left(\frac{v \pm v_L}{v \pm v_S} \right) f_S$$

with:

$$v = 340 \text{ m} \cdot \text{s}^{-1}$$

$$v_L = +15 \text{ m} \cdot \text{s}^{-1}$$

$$v_S = 0 \text{ m} \cdot \text{s}^{-1}$$

$$f_S = 800 \text{ Hz}$$

$$f_L = ?$$

The listener is moving towards the source, so v_L is positive and the source is stationary so $v_S = 0$.

Step 3: Calculate the frequency

$$\begin{aligned} f_L &= \left(\frac{v + v_L}{v + v_S} \right) f_S \\ &= \left(\frac{340 + 15}{340 + 0} \right) (800) \\ &= 835,29 \text{ Hz} \end{aligned}$$

Step 4: Write the final answer

The driver hears a frequency of 835,29 Hz.

Worked example 3: Doppler effect [NSC 2011 Paper 1]

QUESTION

A train approaches a station at a constant speed of $20 \text{ m} \cdot \text{s}^{-1}$ with its whistle blowing at a frequency of 458 Hz. An observer, standing on the platform, hears a change in pitch as the train approaches him, passes him and moves away from him.

1. Name the phenomenon that explains the change in pitch heard by the observer. (1 mark)
2. Calculate the frequency of the sound that the observer hears while the train is approaching him. Use the speed of sound in air as $340 \text{ m} \cdot \text{s}^{-1}$. (4 marks)
3. How will the observed frequency change as the train passes and moves away from the observer? Write down only INCREASES, DECREASES or REMAINS THE SAME. (1 mark)
4. How will the frequency observed by the train driver compare to that of the sound waves emitted by the whistle? Write down only GREATER THAN, EQUAL TO or LESS THAN. Give a reason for the answer. (2 marks)

[TOTAL: 8 marks]

SOLUTION

Question 1:

Doppler effect (1 mark)

Question 2:

$$\begin{aligned} f_L &= \frac{v \pm v_L}{v \pm v_S} f_S \\ \therefore f_L &= \frac{340 \pm 0}{340 - 20} (458) \\ &= 486,63 \text{ Hz} \end{aligned}$$

(4 marks)

Question 3:

Decreases (1 mark)

Question 4

Equal to, because ...

- the velocity of train driver relative to the whistle is zero.
- the train driver has same velocity as the whistle.
- there is no relative motion between source and observer.

OR

OR

(2 marks)

[TOTAL: 11 marks]

Ultrasound and the Doppler effect

ESCMR

Ultrasonic waves (ultrasound) are sound waves with a frequency greater than 20 000 Hz (the upper limit of human hearing). These waves can be used in medicine to determine the direction of blood flow. The device, called a Doppler flow meter, sends out sound waves. The sound waves can travel through skin and tissue and will be reflected by moving objects in the body (like blood). The reflected waves return to the flow meter where its frequency (received frequency) is compared to the transmitted frequency. Because of the Doppler effect, blood that is moving towards the flow meter will change the sound to a higher frequency and blood that is moving away from the flow meter will cause a lower frequency.

Ultrasound can be used to determine whether blood is flowing in the right direction in the circulation system of unborn babies, or identify areas in the body where blood flow is restricted due to narrow veins. The use of ultrasound equipment in medicine is called sonography or ultrasonography.

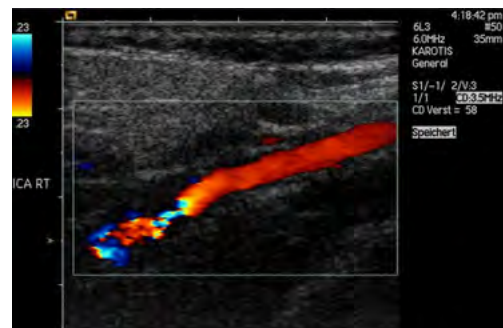


Figure 6.3: Colour Doppler imaging of cervico-cephalic fibromuscular dysplasia

Exercise 6 – 1: The Doppler effect with sound

1. Suppose a train is approaching you as you stand on the platform at the station. As the train approaches the station, it slows down. All the while, the engineer is sounding the hooter at a constant frequency of 400 Hz. Describe the pitch of the hooter and the changes in pitch of the hooter that you hear as the train approaches you. Assume the speed of sound in air is $340 \text{ m}\cdot\text{s}^{-1}$.
2. Passengers on a train hear its whistle at a frequency of 740 Hz. Anja is standing next to the train tracks. What frequency does Anja hear as the train moves

directly toward her at a speed of $25 \text{ m}\cdot\text{s}^{-1}$? Assume the speed of sound in air is $340 \text{ m}\cdot\text{s}^{-1}$.

3. A small plane is taxiing directly away from you down a runway. The noise of the engine, as the pilot hears it, has a frequency 1,15 times the frequency that you hear. What is the speed of the plane? Assume the speed of sound in air is $340 \text{ m}\cdot\text{s}^{-1}$.
4. In places like Canada during winter temperatures can get as low as -35°C . This affects the speed of sound in air and you can use the Doppler effect to determine what the speed of sound is. On a winter's day in Canada with a temperature of -35°C , a source emits sound at a frequency of 1050 Hz and moves away from an observer at $25 \text{ m}\cdot\text{s}^{-1}$. The frequency that the observer measures is $971,41 \text{ Hz}$, what is the speed of sound?
5. Cecil approaches a source emitting sound with a frequency of $437,1 \text{ Hz}$.
 - a) How fast does Cecil need to move to observe a frequency that is 20 percent higher?
 - b) If he passes the source at this speed, what frequency will he measure when he is moving away?
 - c) What is a practical means of achieving this speed?

Assume the speed of sound in air is $340 \text{ m}\cdot\text{s}^{-1}$.

6. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27ST 2. 27SV 3. 27SW 4. 27SX 5. 27SY



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6.3 The Doppler effect with light

ESCMS

Light is a wave and earlier you learnt how you can study the properties of one wave and apply the same ideas to another wave. The same applies to sound and light. We know the Doppler effect is relevant in the context of sound waves when the source is moving. Therefore, in the context of light (EM waves), the frequency of observed light should be different to the emitted frequency when the source of the light is moving relative to the observer.

A frequency shift of light in the visible spectrum could result in a change of colour which could be observable with the naked eye. There will still be a frequency shift for frequencies of EM radiation we cannot see.

We can apply all the ideas that we learnt about the Doppler effect to light. When talking about light we use slightly different terminology to describe what happens. If you look at the colour spectrum (more details in Chapter 12) then you will see that blue light has a shorter wavelength than red light. Since for light, $c = f\lambda$, shorter wavelength equals higher frequency. Relative to the middle of the visible spectrum (approximately green light) longer wavelengths (or lower frequencies) are redder and shorter wavelengths (or higher frequencies) are bluer. So we call shifts towards longer wavelengths "redshifts" and shifts towards shorter wavelengths "blueshifts".

TIP

1. If the light source is moving away from the observer (positive velocity) then the observed frequency is lower and the observed wavelength is greater (redshifted).
2. If the source is moving towards the observer (negative velocity), the observed frequency is higher and the wavelength is shorter (blueshifted).

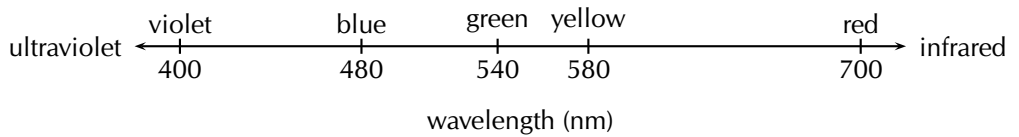


Figure 6.4: Blue light has a shorter wavelength than red light.

A shift in wavelength implies that there is also a shift in frequency. Longer wavelengths of light have lower frequencies and shorter wavelengths have higher frequencies. From the Doppler effect we know that when the source moves towards the observer any waves they emit that you measure are shifted to shorter wavelengths (blueshifted). If the source moves away from the observer, the shift is to longer wavelengths (redshifted).

The expanding universe

ESCMT

Stars emit light, which is why we can see them at night. Galaxies are huge collections of stars. An example is our own Galaxy, the Milky Way, of which our sun is only one of the billions of stars!

Using large telescopes like the Southern African Large Telescope (SALT) in the Karoo, astronomers can measure the light from distant galaxies. The spectrum of light can tell us what elements are in the stars in the galaxies because each element has unique energy levels and therefore emits or absorbs light at particular wavelengths. These characteristic wavelengths are called spectral lines because the lines show up as discrete frequencies in the spectrum of light from the star.

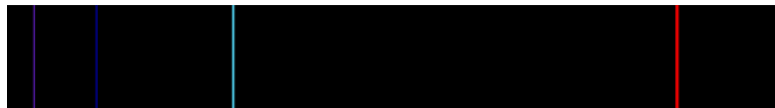


Figure 6.5: A subset of the spectral lines of hydrogen

If these lines are observed to be shifted from their usual wavelengths to shorter wavelengths, then the light from the galaxy is said to be *blueshifted*. If the spectral lines are shifted to longer wavelengths, then the light from the galaxy is said to be *redshifted*. If we think of the blueshift and redshift in Doppler effect terms, then a blueshifted galaxy would appear to be moving *towards* us (the observers) and a redshifted galaxy would appear to be moving *away* from us.

Edwin Hubble (20 November 1889 - 28 September 1953) measured the Doppler shift of a large sample of galaxies. He found that the light from distant galaxies is *redshifted* and he discovered that there is a proportionality relationship between the *redshift* and the *distance* to the galaxy. Galaxies that are further away always appear more redshifted than nearby galaxies. Remember that a redshift in Doppler terms means a velocity of the light source directed away from the observer. So why do all distant galaxies appear to be moving away from our Galaxy? None of them seem to be moving towards us.

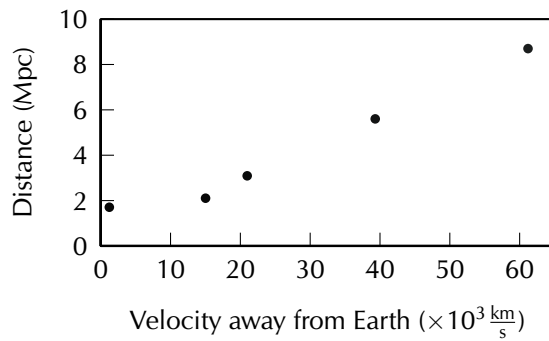


Figure 6.6: The distance to galaxies plotted against their speed away from us. The distance Mpc is a megaparsec which is very big, $1 \text{ Mpc} = 3.3 \text{ million light years}$. A light year is the distance that light can travel in one year, $365 \text{ days} \times 24 \text{ hours} \times 60 \text{ minutes} \times 60 \text{ s} \times 3 \times 10^8 \text{ m} \cdot \text{s}^{-1} = 9.5 \times 10^{15} \text{ m}$.

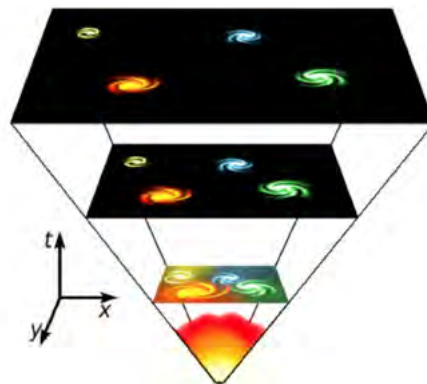
The reason is that the **universe is expanding!** Some of the galaxies will be moving in our direction but more slowly than the space between us and them is expanding. The expansion is so large that it is the primary effect that we observe. The primary reason the light is redshifted isn't actually because all of the Doppler effect, it is redshifted because the space is expanding, the waves are being stretched out. If the Doppler effect were a larger effect then some of the galaxies would still be blueshifted (just less than if space were not expanding).

IMPORTANT!

You might think that this means we are at the centre of the universe. This isn't correct, the situation will look the same from every galaxy because space is expanding in all directions.

There are two things you can do to help you visualise this a little better. One thing to try is to get a balloon and draw some dots on it with a marker. As you blow the balloon up all the dots get further away from all the other dots. The dots represent galaxies in a two-dimensional, expanding universe (the balloon surface). Another thing to imagine is baking raisin bread. As the bread rises, the distance between all the raisins gets larger. Every raisin thinks that all the other raisins are moving away from it.

In this picture the bottom vertex represents the beginning of time, the flat surface represents space. As you move up through the panels you are moving later in time and the expansion of the the flat surface shows the expansion of the universe. The galaxies shown on the surface get further away from each other just because of the expansion of space.



📺 See video: [27SZ](https://www.youtube.com/watch?v=27SZ) at www.everythingscience.co.za

FACT

Hubble Law is:

$$v = H_0 \times d$$

where latest value of H_0 is $67,15 \text{ km} \cdot \text{s}^{-1} \cdot \text{Mpc}^{-1}$ (rate of expansion of the Universe). Latest value from Planck mission, 2013.

TIP

Cool exercise that can be done with Sloan Digital Sky Survey data:

http://skyserver.sdss3.org/dr8/en/proj/basic/universe_original/simple.asp

📺 See presentation: 27T2 at www.everythingscience.co.za

- The Doppler effect is a change in observed frequency due to the relative motion of a source and an observer.
- The following equation can be used to calculate the frequency of the wave according to the observer or listener:

$$f_L = \left(\frac{v + v_L}{v + v_S} \right) f_S$$

- If the direction of the wave from the listener to the source is chosen as positive, the velocities have the following signs:

Source moves towards listener	v_S : negative
Source moves away from listener	v_S : positive
Listener moves towards source	v_L : positive
Listener moves away from source	v_L : negative

- The Doppler effect can be observed in all types of waves, including ultrasound, light and radiowaves.
- Sonography makes use of ultrasound and the Doppler effect to determine the direction of blood flow.
- Light is emitted by stars. Due to the Doppler effect, the frequency of this light decreases and the stars appear redder than if they were stationary. This is called a red shift and means that the stars are moving away from the Earth. This is one of the reasons we can conclude that the Universe is expanding.

Physical Quantities		
Quantity	Unit name	Unit symbol
Frequency (f)	hertz	Hz
Speed (v)	metres per second	$\text{m}\cdot\text{s}^{-1}$

Table 6.1: Units used in **Doppler effect**.

Exercise 6 – 2:

- Write a definition for each of the following terms.
 - Doppler effect
 - Redshift
 - Ultrasound
- The hooter of an approaching taxi has a frequency of 500 Hz. If the taxi is travelling at $30 \text{ m}\cdot\text{s}^{-1}$ and the speed of sound is $340 \text{ m}\cdot\text{s}^{-1}$, calculate the frequency of sound that you hear when
 - the taxi is approaching you.

b) the taxi passed you and is driving away.

Draw a sketch of the observed frequency as a function of time.

3. A truck approaches you at an unknown speed. The sound of the truck's engine has a frequency of 210 Hz, however you hear a frequency of 220 Hz. The speed of sound is $340 \text{ m}\cdot\text{s}^{-1}$.
 - a) Calculate the speed of the truck.
 - b) How will the sound change as the truck passes you? Explain this phenomenon in terms of the wavelength and frequency of the sound.
4. **[Extension question]** A police car is driving towards a fleeing suspect at $\frac{v}{35} \text{ m}\cdot\text{s}^{-1}$, where v is the speed of sound. The frequency of the police car's siren is 400 Hz. The suspect is running away at $\frac{v}{68}$. What frequency does the suspect hear?
5. Explain how the Doppler effect is used to determine the direction of flow of blood in veins.
6. An person in a car travelling at $22,2 \text{ m}\cdot\text{s}^{-1}$ approaches a source emitting sound waves with a frequency of 410 Hz. What is the frequency of the sound waves observed by the person? Assume the speed of sound in air is $340 \text{ m}\cdot\text{s}^{-1}$.
7. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27T3 2. 27T4 3. 27T5 4. 27T6 5. 27T7 6. 27T8



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Rate and Extent of Reaction

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7.1 Introduction

ESCMW

A car slowly rusting away, a match burning, a tablet dissolving in water. All of these are chemical reactions and all of these take place at different rates. If you were to hold a race between these three reactions you would find that the match wins (it burns quickly), while the car rusting would lose (it may take years to rust).

In this chapter we will look at why reactions proceed at different rates (speeds) and how we can change the rate of the reaction.

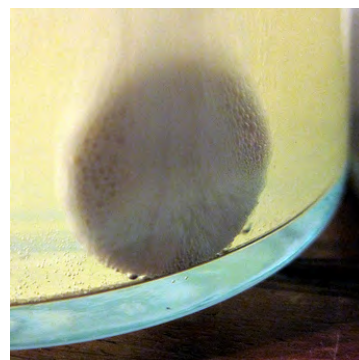


Figure 7.1: A rusting car, a match burning and a tablet dissolving in water.

7.2 Rates of reaction and factors affecting rate

ESCMX

Firstly, let's think about some different types of reactions and how quickly or slowly they occur.

Exercise 7 – 1: Thinking about reaction rates

- Think about each of the following reactions (look at your Grade 11 textbook for a reminder on these processes):
 - corrosion (e.g. the rusting of iron)
 - weathering of rocks
 - photosynthesis
 - combustion
 - For each of the reactions above, write a balanced chemical equation for the reaction that takes place.
 - Rank these reactions in order from the fastest to the slowest.
 - How did you decide which reaction was the fastest and which was the slowest?
 - Think of some other examples of chemical reactions. How fast or slow is each of these reactions, compared with those listed earlier?
- More questions. Sign in at Everything Science online and click 'Practise Science'. Check answers online with the exercise code below or click on 'show me the answer'.

1. 27T9



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This video is a simple demonstration of how a change in **surface area** can affect the average rate of a reaction.

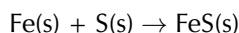
▶ See video: 27TB at www.everythingscience.co.za

You can see how quickly the fuel burns when spread over the table. Think about how much more fuel would be needed to cook a meal if you had it spread over a large surface area rather than kept in a container with a small surface area.

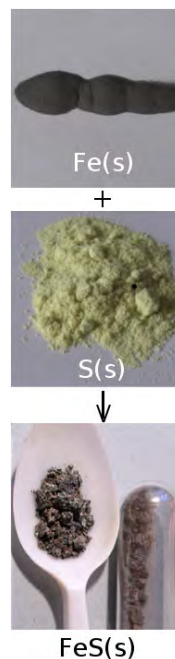
What is a reaction rate?

ESCMY

In a chemical reaction, the substances that are undergoing the reaction are called the **reactants**, while the substances that form as a result of the reaction are called the **products**. The **reaction rate** describes how quickly or slowly the reaction takes place. So how do we know whether a reaction is slow or fast? One way of knowing is to look either at how quickly the *reactants are used* during the reaction or at how quickly the *products form*. For example, iron and sulfur react according to the following equation:



In this reaction, we can observe the speed of the reaction by measuring how long it takes before there is no iron or sulfur left in the reaction vessel. In other words, the reactants have been used. Alternatively, one could see how quickly the iron sulfide (the product) forms. Since iron sulfide looks very different from either of its reactants, this is easy to do.



In another example: $2\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{MgO(s)}$

In this case, the reaction rate depends on the speed at which the reactants (oxygen gas and solid magnesium) are used, or the speed at which the product (magnesium oxide) is formed.

DEFINITION: Reaction rate

The average rate of a reaction describes how quickly *reactants are used* or how quickly *products are formed* during a chemical reaction.

The average rate of a reaction is expressed as the number of moles of reactant used, divided by the total reaction time, or as the number of moles of product formed, divided by the total reaction time.

Average reaction rate for:

- the use of a reactant = $\frac{\text{moles reactant used}}{\text{reaction time (s)}}$
- the formation of a product = $\frac{\text{moles product formed}}{\text{reaction time (s)}}$

Using the magnesium reaction shown earlier:

- Average reaction rate of Mg being used = $\frac{\text{moles Mg used}}{\text{reaction time (s)}}$
- Average reaction rate of O_2 being used = $\frac{\text{moles O}_2 \text{ used}}{\text{reaction time (s)}}$

FACT

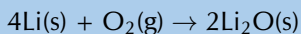
In physics, speed is the scalar measurement that tells us how fast or slow an object is moving (position vs. time). For reactions the speed (or rate) of a reaction means how fast or slow the reaction is and indicates the amount of product formed and the amount of reactants used per unit time.

- Average reaction rate of MgO being formed = $\frac{\text{moles MgO formed}}{\text{reaction time (s)}}$

Worked example 1: Reaction rates

QUESTION

The following reaction takes place:



After two minutes, 4 g of lithium has been used. Calculate the rate of the reaction.

SOLUTION

Step 1: Calculate the number of moles of lithium that are used in the reaction

$$n = \frac{m}{M} = \frac{4 \text{ g}}{6,94 \text{ g.mol}^{-1}} = 0,58 \text{ mol}$$

Step 2: Calculate the time (in seconds) for the reaction

$$t = 2 \text{ minutes} = 2 \times 60 \text{ s} = 120 \text{ seconds}$$

Step 3: Calculate the rate of the reaction

$$\text{Rate of reaction of Li used} = \frac{\text{moles of lithium used}}{\text{time}} = \frac{0,58 \text{ mol}}{120 \text{ s}} = 0,005 \text{ mol.s}^{-1}$$

The rate of the reaction is $0,005 \text{ mol.s}^{-1}$

Exercise 7 – 2: Reaction rates

1. A number of different reactions take place. The table below shows the number of moles of reactant that are used in a particular time for each reaction.

Reaction	Reactants used (mol)	Time (s)	Reaction rate (mol.s^{-1})
1	2	30	
2	5	120	
3	1	90	
4	3,2	90	
5	5,9	30	

- a) Complete the table by calculating the average rate of each reaction.
 - b) Which is the *fastest* reaction?
 - c) Which is the *slowest* reaction?
2. Iron reacts with oxygen as shown in the balanced reaction:

$$2\text{Fe(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{FeO(s)}$$

2 g of Fe and 0,57 g of O_2 are used during the reaction. 2,6 g of FeO is produced. The reaction takes 30 minutes to go to completion.

Calculate the average rate of reaction for:

 - a) the use of Fe.
 - b) the use of O_2 .
 - c) the formation of FeO.
 3. Two reactions occur simultaneously in separate reaction vessels. The reactions are as follows:

$$\text{Mg(s)} + \text{Cl}_2\text{(g)} \rightarrow \text{MgCl}_2\text{(s)}$$

$$2\text{Na(s)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{NaCl(s)}$$

After 1 minute, 2 g of MgCl_2 has been produced in the first reaction.

- a) How many moles of MgCl_2 are produced after 1 minute?
- b) Calculate the average rate of the reaction, using the amount of product that is produced.
- c) Assuming that the second reaction also proceeds at the same rate, calculate:
- the number of moles of NaCl produced after 1 minute.
 - the minimum mass (in g) of sodium that is needed for this reaction to take place for 1 min.
4. More questions. Sign in at Everything Science online and click 'Practise Science'. Check answers online with the exercise code below or click on 'show me the answer'.
1. 27TC 2. 27TD 3. 27TF



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Reaction rates and collision theory

ESCMZ

It should be clear now that the average rate of a reaction varies depending on a number of factors. But how can we explain why reactions take place at different speeds under different conditions? **Collision theory** is used to explain the rate of a reaction.

For a reaction to occur, the particles that are reacting must collide with one another. Only a fraction of all the collisions that take place actually cause a chemical change. These are called *successful* or *effective* collisions.

DEFINITION: Collision theory

Reactant particles must collide with the correct energy and orientation for the reactants to change into products.

Collision theory explains how chemical reactions occur and why reaction rates differ for different reactions. It states that for a reaction to occur the reactant particles must:

- collide
- have enough energy
- have the right orientation at the moment of impact

These *successful collisions* are necessary to break the existing bonds (in the reactants) and form new bonds (in the products).

Activity: Collision Theory

Aim:

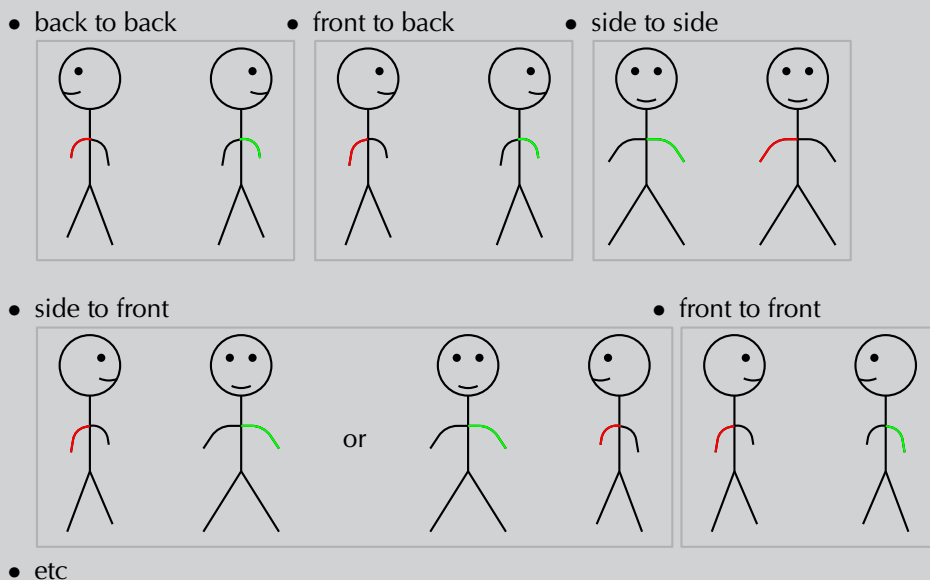
To determine the best way to approach your friend, in order to link your **right** arm with their **left** arm.

Method:

Try different ways of approaching your friend:

FACT

Oxalic acid is abundant in many plants. The leaves of the tea plant (*Camellia sinensis*) contain very high concentrations of oxalic acid relative to other plants. Oxalic acid also occurs in small amounts in foods such as parsley, chocolate, nuts and berries. Oxalic acid irritates the lining of the gut when it is eaten, and can be fatal in very large doses.



• etc

Results:

Determine how hard it is to link arms in each of these positions.

Discussion:

If you approach your friend from behind (facing their back) it is hard to link arms. Approaching from their left (sideways so that your right side is on their left), it is easy to link up.

Conclusion:

You should have found that each method had a different level of difficulty for linking arms. This is similar to how molecules (compounds) approach in a reaction. The different ways you approached your friend represent the different orientations of the molecules. The correct orientation makes successful collisions possible.

Factors affecting reaction rates

ESCN2

Several factors affect the average rate of a reaction. It is important to know these factors so that reaction rates can be controlled. This is particularly important when it comes to industrial reactions, where greater productivity leads to greater profits for companies. The following are some of the factors that affect the average rate of a reaction.

Nature of reactants

Substances have different chemical properties and therefore react differently, and at different rates (e.g. the rusting of iron vs. the tarnishing of silver).

▶ See video: 27TG at www.everythingscience.co.za

Experiment: The nature of reactants

Aim:

To determine the effect of the nature of reactants on the average rate of a reaction.

Apparatus:

You will need the following items for this experiment:

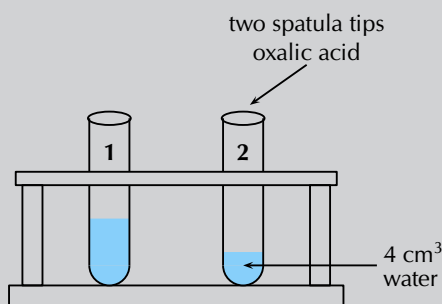
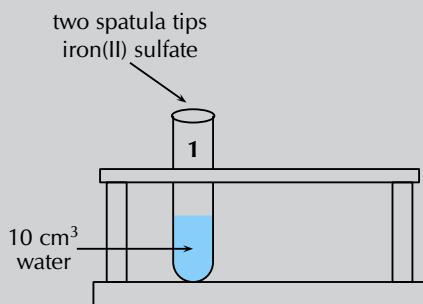
- Oxalic acid ($(\text{COOH})_2$), iron(II) sulfate (FeSO_4), potassium permanganate (KMnO_4) and concentrated sulfuric acid (H_2SO_4)
- a spatula, two test tubes, a medicine dropper, a glass beaker and a glass rod.

Method:

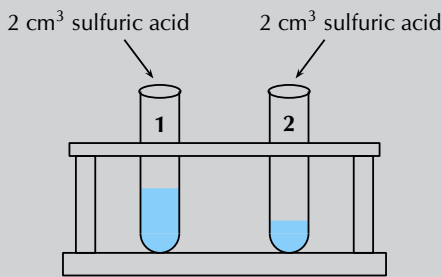
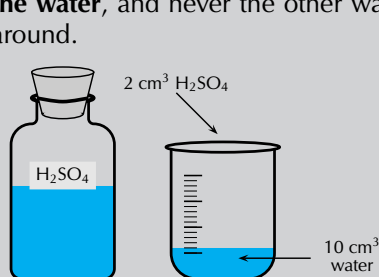
WARNING!

Concentrated H_2SO_4 can cause serious burns. We suggest using gloves and safety glasses whenever you work with an acid. Remember to add the acid to the water and to avoid sniffing the acid. Handle all chemicals with care.

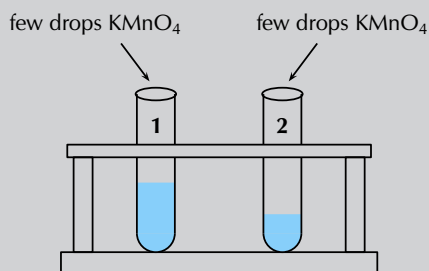
1. Label one test tube **1**. Prepare an iron(II) sulfate solution in test tube 1 by dissolving two spatula tips of iron(II) sulfate in 10 cm^3 of water.
2. Label the other test tube **2**. Prepare a solution of oxalic acid in test tube 2 in the same way.



3. Prepare a separate solution of sulfuric acid by adding 2 cm^3 of the concentrated acid to 10 cm^3 of water. Remember always to add the **acid to the water**, and never the other way around.
4. Add 2 cm^3 of the sulfuric acid solution to the iron(II) sulfate and oxalic acid solutions respectively.



5. Using the medicine dropper, add a few drops of potassium permanganate to the two test tubes. Observe how quickly the potassium permanganate solution discolours in each solution.



Results:

- You should have seen that the the potassium permanganate discolours in the oxalic acid solution much more slowly than in the iron(II) sulfate solution.

FACT

The KMnO_4 with oxalic acid and iron(II) sulfate pictures are screenshots from videos by katalofuromai and Aaron Huggard on Youtube.

KMnO_4 + oxalic acid



reaction takes a few minutes

KMnO_4 + iron(II) sulfate



reaction takes a few seconds

These reactions can be seen in the following videos:

▶ See video: 27TH at www.everythingscience.co.za

▶ See video: 27TJ at www.everythingscience.co.za

- It is the oxalate ions ($\text{C}_2\text{O}_4^{2-}$) and the Fe^{2+} ions that cause the discolouration. It is clear that the Fe^{2+} ions react much more quickly with the permanganate than the ($\text{C}_2\text{O}_4^{2-}$) ions. The reason for this is that there are no covalent bonds to be broken in the iron ions before the reaction can take place. In the case of the oxalate ions, covalent bonds between carbon and oxygen atoms must be broken first.

Conclusions:

Despite the fact that both these reactants (oxalic acid and iron(II) sulfate) are in aqueous solutions, with similar concentrations and at the same temperature, the reaction rates are very different. This is because the nature of the reactants can affect the average rate of a reaction.

The nature of the iron(II) sulfate in solution (iron ions, ready to react) is very different to the nature of oxalic acid in solution (oxalate ions with covalent bonds that must be broken). This results in significantly different reaction rates.

Surface area (of solid reactants)

Experiment: Surface area and reaction rate

Marble (CaCO_3) reacts with hydrochloric acid (HCl) to form calcium chloride, water and carbon dioxide gas according to the following equation:



Aim:

To determine the effect of the surface area of reactants on the average rate of the reaction.

Apparatus:

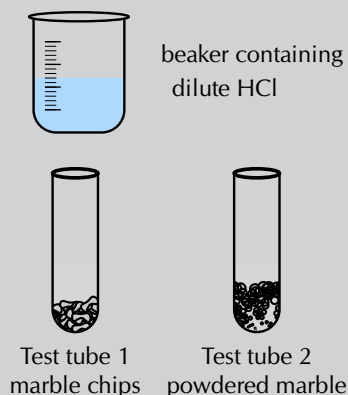
- 2 g marble chips, 2 g powdered marble, concentrated hydrochloric acid (HCl)
- one beaker, two test tubes.

Method:

WARNING!

Concentrated HCl can cause serious burns. We suggest using gloves and safety glasses whenever you work with an acid. Remember to add the acid to the water and handle with care.

1. Prepare a solution of hydrochloric acid in the beaker by adding 2 cm^3 of the concentrated acid to 20 cm^3 of water.
2. Place the marble chips into one test tube and the powdered marble into a separate test tube.
3. Add 10 cm^3 of the dilute hydrochloric acid to each of the test tubes and observe the rate at which carbon dioxide gas (CO_2) is produced (you should see bubbles of CO_2).



Results:

Note (write down) what you observe.

Questions and discussion:

- Which reaction proceeds faster?
- Can you explain this?

Conclusion:

The reaction with powdered marble is faster. The smaller the pieces of marble are (in this case the powdered form is smallest), the greater the surface area for the reaction to take place.

Only the molecules at the surface of the solid can react with the hydrochloric acid. The next layer of molecules can only react once the surface molecules have reacted. That is, the next layer of molecules becomes the surface.

The chips of marble are relatively large, so only a small percentage of the molecules are at the surface and can react initially. The powdered marble has much smaller solid pieces, so there are many more surface molecules exposed to the hydrochloric acid. The more molecules exposed on the surface (the greater the surface area) the faster the reaction will be.

For the same amount of mass, smaller pieces of solid react faster (shown in Figure 7.2).

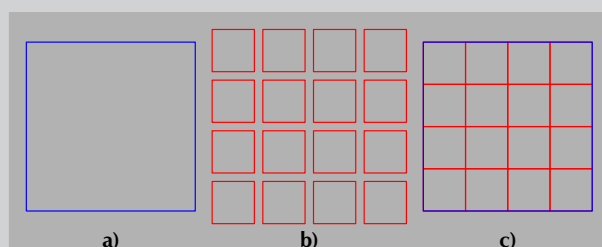


Figure 7.2: **a)** A large particle, **b)** small particles with the same volume as the large particle. **c)** The surface area of large particles (shown in blue) is much smaller than that of small particles (shown in red).

► See video: 27TK at www.everythingscience.co.za

Calcium carbonate reacts with hydrochloric acid according to the following reaction:



Consider the solid calcium carbonate.

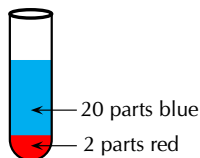
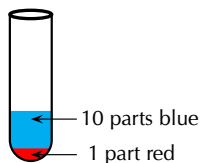
If we react 1 g of CaCO_3 we find that the reaction is faster if the CaCO_3 is powdered when compared with the CaCO_3 being large lumps.

FACT

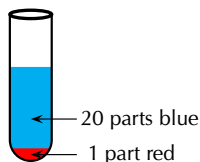
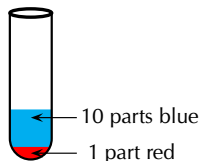
Surface area, concentration and pressure all have the same effect on reaction rate (an increase leads to a faster reaction rate). This is because in each case an increase in the property leads to an increase in the number of collisions in that phase of matter.

FACT

both solutions have the same concentration



these solutions have different concentrations



Explanation:

The large lump of CaCO_3 has a small surface area relative to the same mass of powdered CaCO_3 . This means that more particles of CaCO_3 will be in contact with HCl in the powdered CaCO_3 than in the lumps. As a result, there can be more *successful* collisions per unit time and the reaction of powdered CaCO_3 is faster.

Increasing the surface area of the reactants increases the rate of the reaction.

The following video shows the effect of surface area on the time an effervescent tablet takes to fully dissolve. The tablet is fully dissolved once the bubbles (CO_2 gas) stop forming.

▶ See video: 27TM at www.everythingscience.co.za

Concentration (of solutions)

As the concentration of the reactants increases, so does the reaction rate.

Experiment: Concentration and reaction rate

Aim:

To determine the effect of reactant concentration on reaction rate.

Apparatus:

- Concentrated hydrochloric acid (HCl), magnesium ribbon
- Two beakers, two test tubes and a measuring cylinder.

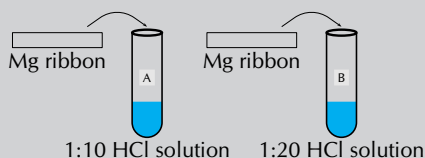
Method:

WARNING!

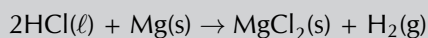
Do not get hydrochloric acid (HCl) on your hands. We suggest you use gloves and safety glasses whenever handling acids and handle with care.

1. When diluting a solution remember that if you want a 1:10 solution (1 part original solution in 10 parts water) measure 10 cm^3 of water in a measuring cylinder and pour it into a beaker, then add 1 cm^3 of the original solution to the beaker as well. 2 parts concentrated acid to 20 parts water will also be a 1:10 solution. Remember to always add the *acid* to the *water*, and not the other way around.
 - Prepare a solution of 1 part acid to 10 parts water (1:10). Label a test tube **A** and pour 10 cm^3 of this solution into the test tube.
 - Prepare a solution of 1 part acid to 20 parts water (1:20). Label a test tube **B** and pour 10 cm^3 of this solution into the test tube.
2. Take two pieces of magnesium ribbon of the **same length**. At the same time, put one piece of magnesium ribbon into test tube A and the other into test tube B, and pay close attention to what happens.

Make sure that the magnesium ribbon is long enough so that your hand is not close to the HCl .



The equation for the reaction is:



Results:

Write down what happened (what did you observe?) in each test tube.

Questions and discussion:

- Which of the two solutions is more concentrated, the 1:10 or 1:20 hydrochloric acid solution?
- In which of the test tubes is the reaction faster? Suggest a reason for this.
- How can you measure the average rate of this reaction?
- Name the gas that is produced?
- Why is it important that the same length of magnesium ribbon is used for each reaction?

Conclusions:

The 1:10 solution is more concentrated and therefore this reaction proceeds faster. The greater the concentration of the reactants, the faster the average rate of the reaction. The average rate of the reaction can be measured by the rate at which the magnesium ribbon disappears.

▶ See video: 27TN at www.everythingscience.co.za

Explanation:

The greater concentration of the reactant means that there are more particles of reactant (HCl) per unit volume of solution. Therefore the chance that HCl particles will collide with the Mg particles will be higher for the solution with the greater concentration. The number of successful collisions per unit time will be higher and so the rate of the reaction will be faster.

▶ See video: 27TP at www.everythingscience.co.za

Project: Concentration and rate

Design an experiment to determine the effect of concentration on rate using vinegar and baking soda.

Hint: mix water and vinegar to change concentration but keep the total volume constant.

Pressure (of gaseous reactants)

As the pressure of the reactants increase, so does the reaction rate.

The higher the pressure, the more particles of gas per unit volume. Therefore there are more collisions per unit time. The number of successful collisions per unit time will be higher and so the rate of the reaction will be faster.

▶ See video: 27TQ at www.everythingscience.co.za

Temperature

If the temperature of the reaction increases, so does the average rate of the reaction.

Experiment: Temperature and reaction rate

Aim:

To determine the effect of temperature on reaction rate.

Apparatus:

- Two effervescent tablets (e.g. Cal-C-Vita)
- An ice-bath, two test tubes
- Two balloons, two rubber bands

Method:

1. Half fill two large test tubes with water. Label them A and B.
2. Break two effervescent tablets in two or three pieces and place them in the two balloons. Fit one of these balloons tightly to test tube A and one to test tube B, being careful not to drop the contents into the water. You can stand the test tube in a beaker to help you do this.
3. Place *only* test tube A into an ice-bath and leave to equilibrate (come to the same temperature). Approximately 10 minutes should be enough.
4. At the same time lift the balloons on test tubes A and B so that the tablets go into the water.
Do not shake either test tube.



$\text{CO}_2(\text{g})$ is released during this reaction.

Observe how quickly the balloons increase in size and write down your observations (which increases in size faster).

Results:

Note (write down) your observations.

Questions and discussion:

- Which balloon expanded faster?
- Suggest a reason for the difference in rates.

Conclusions:

The balloon on test tube B will expand faster. This is because the higher temperature (room temperature rather than an ice bath) leads to an increase in the average rate of CO_2 gas production.

The video below shows how much pressure can build up when $\text{CO}_2(\text{g})$ is released during the reaction of an effervescent tablet with water.

▶ See video: 27TR at www.everythingscience.co.za

The higher the temperature, the greater the average kinetic energy of the particles, which means that the particles are moving faster. Therefore:

- particles moving faster means more collisions per unit time (collision theory)
- particles with higher kinetic energy are also more likely to react on colliding

as they have enough energy for the reaction to occur (see Section 7.4 on the mechanism of reaction).

▶ See video: 27TS at www.everythingscience.co.za

Catalyst

Adding a catalyst increases the reaction rate by lowering the energy required for a successful reaction to take place. A catalyst speeds up a reaction and is released at the end of the reaction, completely unchanged.

Experiment: Catalysts and reaction rate

Aim:

Hydrogen peroxide decomposes slowly over time into water and oxygen. The aim of this experiment is to determine the effect a catalyst has on the reaction rate.

Apparatus:

- 3% hydrogen peroxide (H_2O_2), manganese dioxide (MnO_2) powder, yeast powder
- two beakers or large measuring cylinders

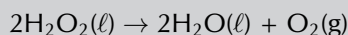
Method:

WARNING!

Be careful when handling H_2O_2 as it can burn you. We recommend wearing gloves and safety glasses.

1. Pour 30 cm³ H_2O_2 into two separate containers.
2. Add a spatula tip of yeast to one container.
3. Time how long it takes for the bubbles to stop.
4. Repeat with MnO_2 in the second container.
5. Compare the effect of the two catalysts.

The balanced equation for this reaction is:



This can also be written:

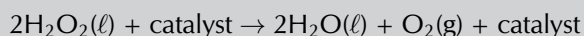


Figure 7.3: H_2O_2 before the addition of MnO_2 (left) and after the addition of MnO_2 (right).

Results:

- Which chemical compounds are acting as catalysts in these reactions?
- What causes the bubbles that form in the reaction?

Conclusions:

The bubbles that form are oxygen gas formed through the decomposition of hydrogen peroxide. This would happen over time without the presence of the catalyst. The manganese dioxide speeds up the reaction significantly. The yeast speeds up the reaction, but not as much as the manganese dioxide.

▶ See video: 27TT at www.everythingscience.co.za

Experiment: Catalysts and reaction rate

Aim:

To determine the effect of a catalyst on the average rate of a reaction

Apparatus:

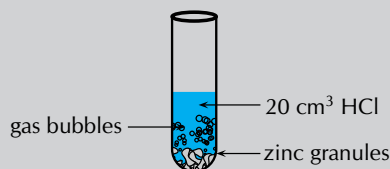
- Zinc granules, 0,1 mol.dm⁻³ hydrochloric acid, copper pieces
- One test tube, a glass beaker, tongs

Method:

WARNING!

Do not get hydrochloric acid (HCl) on your hands. We suggest you use gloves and safety glasses whenever handling acids. Be especially careful when removing the copper pieces from the test tube.

1. Place a few of the zinc granules in the test tube, using tongs.
2. Measure the mass of a few pieces of copper and, using tongs, keep them separate from the rest of the copper.
3. Add 20 cm³ of HCl to the test tube. You will see that a gas is released. Take note of how quickly or slowly this gas is released (use a stopwatch or your cellphone to time this). Write a balanced chemical equation for the chemical reaction that takes place.
4. Now add the copper pieces to the same test tube. What happens to the rate at which the gas is produced?
5. Carefully remove the copper pieces from the test tube (use tongs), rinse them in water and alcohol and then weigh them again. Has the mass of the copper changed since the start of the experiment?



Results:

During the reaction, the gas that is released is hydrogen. The rate at which the hydrogen is produced *increases* when the copper pieces (the catalyst) are added. The mass of the copper does not change during the reaction.

Conclusions:

The copper acts as a *catalyst* during the reaction. It speeds up the average rate of the reaction, but is not changed itself in any way.

We will return to catalysts in more detail once we have explored the mechanism of reactions later in this chapter.

Experiment: Temperature, concentration and reaction rate

Aim:

To determine the effect of temperature and concentration on the average reaction rate of the iodine clock experiment. This experiment is best done in groups.

Apparatus:

- Potassium iodide (KI), soluble starch, sodium thiosulfate solution (Na₂S₂O₃),

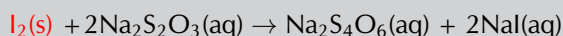
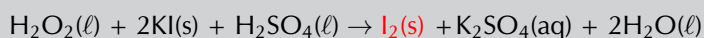
dilute (around $0,2 \text{ mol.dm}^{-3}$) sulfuric acid (H_2SO_4), 3% hydrogen peroxide (H_2O_2) solution

- Five beakers, a measuring cylinder, a hotplate, an ice bath, a glass stirring rod, a stop-watch

Method:

- Preheat the hotplate to 40°C
- Label a beaker solution 1. Measure 75 ml H_2SO_4 into the beaker. Add 25 ml 3% H_2O_2 . Remember to use dilute ($0,2 \text{ mol.dm}^{-3}$) sulfuric acid.

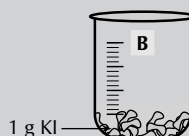
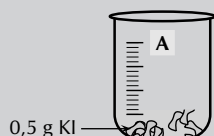
The equations for what is occurring in this reaction are given below:



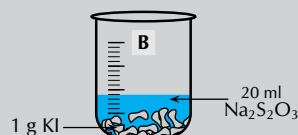
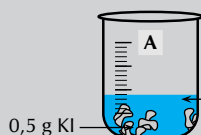
It is good scientific practice to vary only one factor at a time during an experiment. Therefore, this experiment has two parts. First we will vary the concentration of KI, then we will vary the temperature:

• Varying the concentration

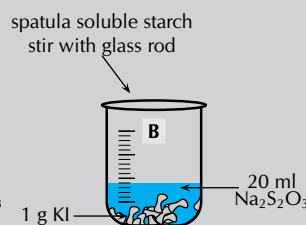
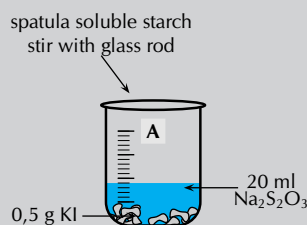
1. Weigh out 0,5 g of KI into a beaker and label it A.
2. Weigh out 1 g of KI into a different beaker and label it B.



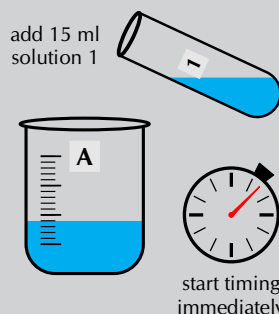
3. Add 20 ml $\text{Na}_2\text{S}_2\text{O}_3$ to both beaker A and beaker B.



4. Add a spatula of soluble starch to both beaker A and beaker B and stir with a glass rod.



5. Measure 15 ml of solution 1 with the measuring cylinder. Get your stopwatch ready. Pour the 15 ml of solution 1 into beaker A and start timing.



Stop timing when the solution starts to change colour. Write down your time in the table below.

6. Repeat step 5 with beaker B.

Beaker	Concentration (M)	Temperature ($^\circ\text{C}$)	Time (s)
A	approx. 0.15	room temperature	
B	approx. 0.3	room temperature	

- **Varying the temperature**

1. Weigh out 0,5 g of KI into a new beaker and label it C.
2. Add 20 ml $\text{Na}_2\text{S}_2\text{O}_3$ to beaker C.
3. Add a spatula of soluble starch to beaker C and stir with a glass rod.
4. Measure 15 ml of solution 1 with the measuring cylinder.
5. Place beaker C in the ice bath.
6. Get your stopwatch ready. Pour the 15 ml of solution 1 into beaker C and start timing. Stop timing when the solution starts to change colour. Write down your time in the table below.
7. Repeat steps 1 - 4 (label the beaker D).
Place beaker D on the hotplate. Then repeat step 6

Beaker	Concentration (M)	Temperature ($^{\circ}\text{C}$)	Time (s)
A	approx. 0.15	room temperature	
C	approx. 0.15	0	
D	approx. 0.15	40	

Beaker A has been included here because it has the same concentration as beakers C and D, but is at a different temperature.

Results:

Make a table with the information for all the beakers. Include columns for concentration, temperature, time, and reaction rate.

Questions and discussion:

- Did beaker A or B have the faster reaction rate?
- Why did it have a faster reaction rate?
- Did beaker A, C or D have the fastest reaction rate? Why?
- Did beaker A, C or D have the slowest reaction rate? Why?

Conclusions:

You will notice that the faster reaction rate occurs in the beaker with the higher concentration of KI. You should also see that the higher the temperature, the faster the reaction rate.

▶ See video: 27TV at www.everythingscience.co.za

This video shows how this experiment can be used as a clock with the concentration chosen so that the experiment changes colour at a specific time (or with a particular part of a song). This is why this experiment is known as the *iodine clock reaction*.

▶ See video: 27TW at www.everythingscience.co.za

Worked example 2: Reaction rates

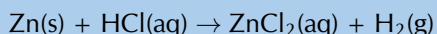
QUESTION

Write a balanced equation for the exothermic reaction between Zn(s) and HCl(l) . Also name three ways to increase the rate of this reaction.

SOLUTION

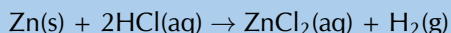
Step 1: Write the equation for zinc and hydrochloric acid

The products must be a salt and hydrogen gas. Zinc ions have a charge of 2+ while chloride ions have a charge of 1-. Therefore the salt must be ZnCl_2 .



Step 2: Balance the equation if necessary

There are more chloride ions and hydrogen atoms on the right side of the equation. Therefore there must be 2 HCl on the left side of the equation.



Step 3: Think about the methods mentioned in this section that would increase reaction rate

- A catalyst could be added
- The zinc solid could be ground into a fine powder to increase its surface area
- The HCl concentration could be increased

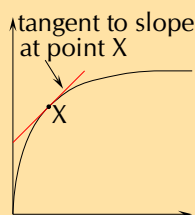
Exercise 7 – 3: Rates of reaction

1. Hydrochloric acid and calcium carbonate react according to the following equation: $\text{CaCO}_3\text{(s)} + 2\text{HCl}(\ell) \rightarrow \text{CaCl}_2\text{(s)} + \text{H}_2\text{O}(\ell) + \text{CO}_2\text{(g)}$

The volume of carbon dioxide that is produced during the reaction is measured at different times. The results are shown in the table below.

Time (mins)	Total Volume of CO_2 produced (cm^3)
1	14
2	26
3	36
4	44
5	50
6	58
7	65
8	70
9	74
10	77

*Note: On a graph of production against time, it is the gradient of the tangent to the graph that shows the rate of the reaction at that time.
e.g.*



- Use the data in the table to draw a graph showing the volume of gas that is produced in the reaction, over a period of 10 minutes.
(Remember to label the axes and plot the graph on graphing paper)
- At which of the following times is the reaction *fastest*: 1 minute; 6 minutes or 8 minutes. Explain.
- Suggest a reason why the reaction slows down over time.

- d) Use the graph to estimate the volume of gas that will have been produced after 11 minutes.
- e) How long do you think the reaction will take to stop (give a time in minutes)?
- f) If the experiment was repeated using a more concentrated hydrochloric acid solution:
 - i. would the average rate of the reaction increase or decrease from the one shown in the graph?
 - ii. draw a line on the same set of axes to show how you would expect the reaction to proceed with a more concentrated HCl solution.

2. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [27TX](#)



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7.3 Measuring rates of reaction

ESCN3

How the average rate of a reaction is measured will depend on what the reaction is, what the reactants are, and what product forms. Look back at the reactions that have been discussed so far. In each case, how was the average rate of the reaction measured? The following examples will give you some ideas about other ways to measure the average rate of a reaction:

Measuring the volume of gas produced per unit time

ESCN4

The volume of gas produced in a reaction may be measured by collecting the gas in a gas syringe (Figure 7.4).

As more gas is produced, the plunger is pushed out and the volume of the gas in the syringe can be recorded.

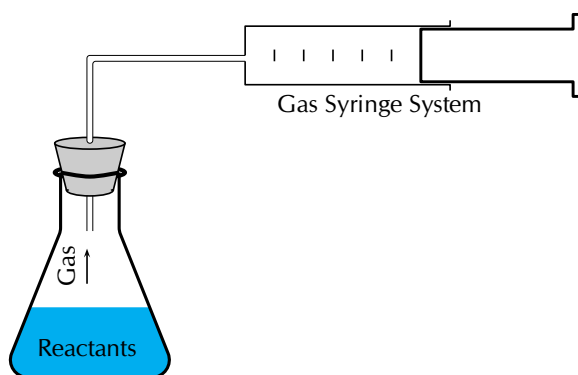


Figure 7.4: The gas syringe method.

▶ See video: [27TY](#) at www.everythingscience.co.za

By measuring the volume at set time intervals, we can graph the data (Figure 7.5) and hence determine the rate of the reaction.

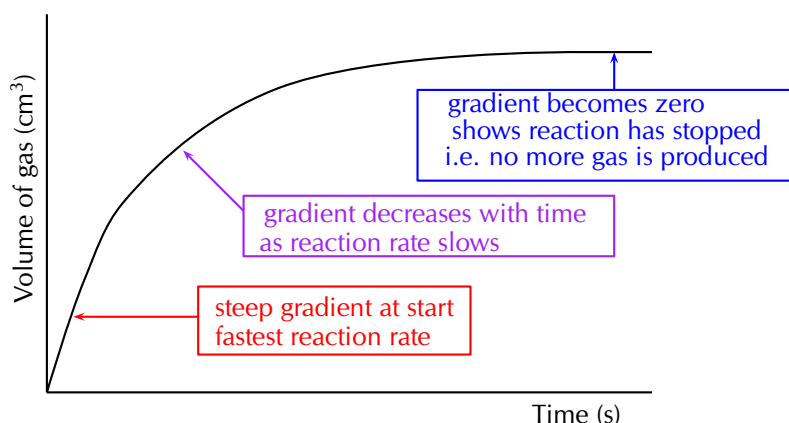


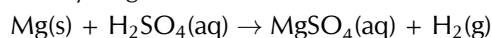
Figure 7.5: A plot of the volume of gas collected at set time intervals.

Examples of reactions that produce gas are listed below:

- *Reactions that produce hydrogen gas:*

When a *metal reacts with an acid*, hydrogen gas is produced. The hydrogen can be collected in a test tube. A **lit splint** can be used to test for hydrogen. The '**pop**' sound shows that hydrogen is present.

For example, magnesium reacts with sulfuric acid to produce magnesium sulfate and hydrogen.



- *Reactions that produce carbon dioxide:*

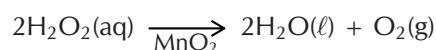
When a *carbonate reacts with an acid*, carbon dioxide gas is produced. When carbon dioxide is passed through **limewater**, it turns the limewater **milky**. A burning splint will also **stop burning** (be extinguished) in the presence of CO_2 gas. These are a simple tests for the presence of carbon dioxide.

For example, calcium carbonate reacts with hydrochloric acid to produce calcium chloride, water and carbon dioxide.



- *Reactions that produce oxygen:*

Hydrogen peroxide decomposes in the presence of a manganese(IV) oxide catalyst to produce oxygen and water.



Experiment: Measuring reaction rates

Aim:

To measure the effect of concentration on the average rate of a reaction.

Apparatus:

- Solid zinc granules, 1 mol.dm^{-3} hydrochloric acid (HCl)
- Two conical flasks, two beakers, two balloons, bunsen burner, splint of wood

Method:

WARNING!

Do not get hydrochloric acid (HCl) on your hands. We suggest you use gloves and safety glasses whenever handling acids and handle with care.

FACT

When the reaction between two liquids forms an insoluble compound, solid material will form in the liquid. This solid is called a precipitate.

1. Label a conical flask **A**. Weigh 5 g zinc granules into it.
Repeat with the second conical flask but label it **B**.

2. Label a beaker **1**. Pour 10 cm³ HCl into it.

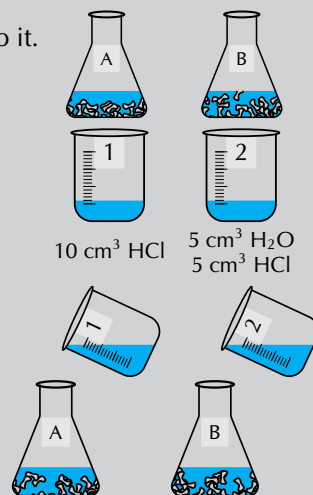
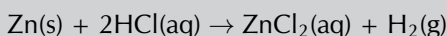
Label the other beaker **2**. Pour 5 cm³ H₂O into it.
Add 5 cm³ HCl to this second beaker.

3. **Quickly:** Pour the liquid in beaker **1** into conical flask **A** and pour the liquid in beaker **2** into conical flask **B**. Attach one balloon firmly to each conical flask.

4. Note which balloon filled more quickly.

5. Fill a test tube with the gas formed. Light **only** the gas in the test tube. Keep open flames away from the balloons.

The equation for this reaction is:



Results:

- Which beaker contained the more concentrated solution of HCl?
- Which balloon filled more quickly?
- What happened when you lit the gas in the test tube?

Conclusions:

The more concentrated solution led to a faster reaction rate (i.e. the balloon filling with H₂ gas more quickly). The test for hydrogen gas would have resulted in a loud *pop* when the lit splint was placed near the mouth of a test tube.

▶ See video: 27TZ at www.everythingscience.co.za

Precipitate reactions

ESCN5

In reactions where a *precipitate* is formed, the amount of precipitate formed in a period of time can be used as a measure of the reaction rate. For example, when sodium thiosulfate reacts with an acid, a yellow precipitate of sulfur is formed. The reaction is as follows:



One way to estimate the average rate of this reaction is to carry out the investigation in a conical flask and to place a piece of paper with a black cross underneath the bottom of the flask. At the beginning of the reaction, the cross will be clearly visible when you look into the flask (Figure 7.6). However, as the reaction progresses and more precipitate is formed, the cross will gradually become less clear and will eventually disappear altogether. Measuring the time that it takes for this to happen will give an idea of the reaction rate. Note that it is not possible to collect the SO₂ gas that is

produced in the reaction, because it is very soluble in water.

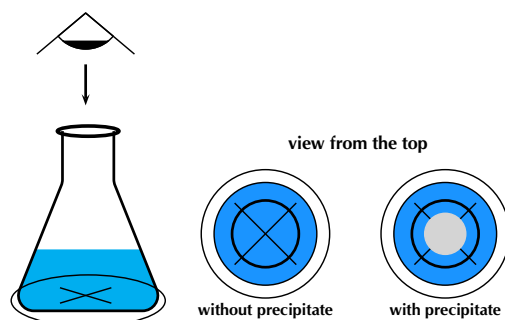


Figure 7.6: At the beginning of the reaction between sodium thiosulfate and hydrochloric acid, when no precipitate has been formed, the cross at the bottom of the conical flask can be clearly seen. As the precipitate forms less of it can be seen.

Informal experiment: Measuring reaction rates

Aim:

To measure the effect of concentration on the average rate of a reaction.

Apparatus:

- 300 cm³ of sodium thiosulfate Na₂S₂O₃ solution. (Prepare a solution of sodium thiosulfate by adding 12 g of Na₂S₂O₃ to 300 cm³ of water). This is solution 'A'.
- 300 cm³ of water
- 100 cm³ of 1:10 dilute hydrochloric acid. This is solution 'B'.
- Six 100 cm³ glass beakers, measuring cylinders, paper and marking pen, stop-watch or timer

Method:

WARNING!

Do not get hydrochloric acid (HCl) on your hands. We suggest you use gloves and safety glasses whenever handling acids and that you handle with care.

One way to measure the average rate of this reaction is to place a piece of paper with a cross underneath the reaction beaker to see how long it takes until the cross cannot be seen due to the formation of the sulfur precipitate.

1. Set up six beakers on a flat surface and label them 1 to 6.
2. Pour 60 cm³ solution A into the first beaker and add 20 cm³ of water
3. Place the beaker on a piece of paper with a large black cross on it.
4. Use the measuring cylinder to measure 10 cm³ HCl. Now add this HCl to the solution that is already in the first beaker (NB: Make sure that you always clean the measuring cylinder you have used before using it for another chemical).
5. Using a stopwatch with seconds, write down the time it takes for the precipitate that forms to block out the cross.
6. Now measure 50 cm³ of solution A into the second beaker and add 30 cm³ of water. Place the beaker over the black cross on the paper. To this second beaker, add 10 cm³ HCl, time the reaction and write down the results as you did before.
7. Continue the experiment by diluting solution A as shown below.

Beaker	Solution A (cm ³)	Water (cm ³)	Solution B (cm ³)	Time (s)
1	60	20	10	
2	50	30	10	
3	40	40	10	
4	30	50	10	
5	20	60	10	
6	10	70	10	

The equation for the reaction between sodium thiosulfate and hydrochloric acid is:



Results:

- Calculate the reaction rate in each beaker. Remember that:

$$\text{rate of the formation of product} = \frac{\text{moles product formed}}{\text{reaction time (s)}}$$

In this experiment you are stopping each experiment when the same approximate amount of precipitate is formed (the cross is blocked out by precipitate). So a relative reaction rate can be determined using the following equation:

$$\text{reaction rate} = \frac{1}{\text{time (s)}}$$

- Represent your results on a graph. **Concentration** will be on the x-axis and **reaction rate** on the y-axis. Note that the original volume of Na₂S₂O₃ can be used as a measure of concentration.
- Why was it important to keep the volume of HCl constant?
- Describe the relationship between concentration and reaction rate.

Conclusions:

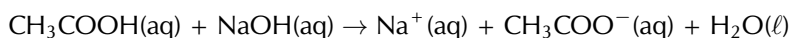
The higher the concentration of the reactants, the faster the average reaction rate.

Changes in colour

ESCN6

In some reactions there is a change in colour which tells us that the reaction is occurring. The faster the colour change the faster the reaction rate.

For example, when ethanoic acid (acetic acid) is titrated with sodium hydroxide, an indicator such as phenolphthalein is added. The solution is clear in an acidic solution and changes to pink when the reaction is complete. If the concentration of the base were increased, the colour change would happen faster (after a smaller volume of base was added), showing that a higher concentration of base increased the reaction rate.



Changes in mass

ESCN7

For a reaction that produces gas, the mass of the reaction vessel can be measured over time. The mass loss indicates the amount of gas that has been produced and escaped from the reaction vessel (Figure 7.7).

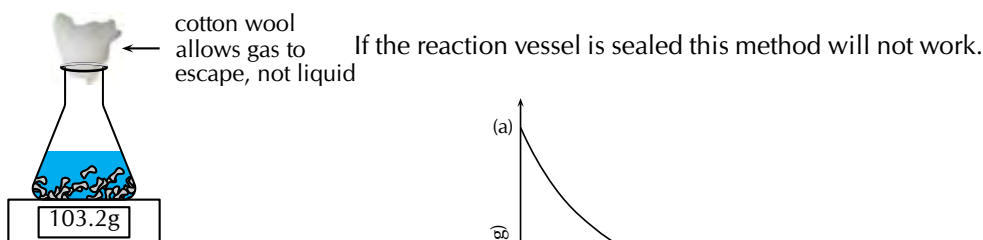


Figure 7.7: A change in mass indicating the loss of a gas

(photo by tjmwatson on Flickr).

Notice the difference between the two graphs in Figure 7.8. The mass of the sample will decrease as the reaction proceeds. The mass lost from the sample will increase as the reaction proceeds. The material lost (that accounts for the mass loss) can be collected and measured (for example with the gas syringe method as shown in Figure 7.4).

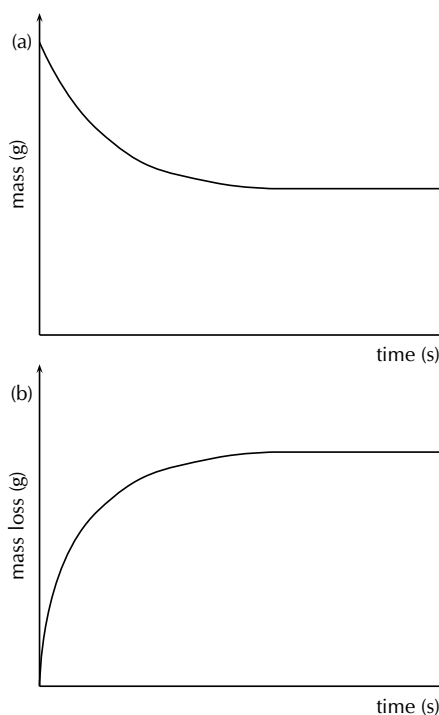


Figure 7.8: **a)** A graph of the sample mass versus time and **b)** a graph of the mass loss versus time.

7.4 Mechanism of reaction and catalysis

ESCN8

Earlier it was mentioned that it is the *collision* of particles that causes reactions to occur and that only some of these collisions are successful. This is because the reactant particles have a wide range of kinetic energies, and only a small fraction of the particles will have enough energy (and the correct orientation) to actually break bonds so that a chemical reaction can take place. The minimum energy that is needed for a reaction to take place is called the **activation energy**. For more information on the energy of reactions, refer to Grade 11 ([Chapter 12](#)).

Activation Energy

ESCN9

DEFINITION: *Activation energy*

The minimum energy required for a chemical reaction to proceed.

Even at a fixed temperature, the energy of the particles varies, meaning that only some of them will have enough energy to be part of the chemical reaction, depending on the activation energy for that reaction (Figure 7.9). Increasing the reaction temperature has the effect of increasing the number of particles with enough energy to take part in the reaction, and increasing the reaction rate.

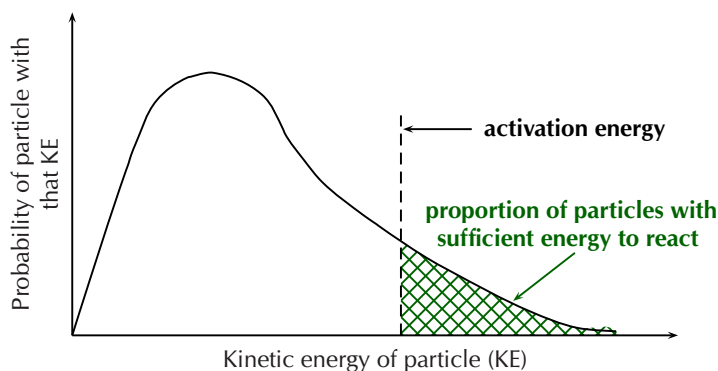


Figure 7.9: The distribution of particle kinetic energies at a fixed temperature.

Remember that a molecule must have energy greater than the activation energy, as well as the correct orientation, for the reaction to take place.

Increasing the temperature of a reaction mixture raises the average kinetic energy of the particles. As can be seen on the graph (Figure 7.10), a higher proportion of the particles can now react making the reaction faster. With the increased movement of the molecules the chances of a molecule having the correct orientation is also increased.

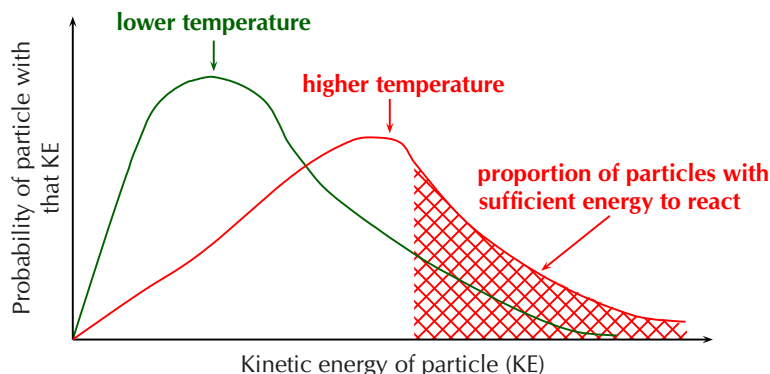


Figure 7.10: The distribution of particle kinetic energies with an increase in temperature. There is an increased number of particles with sufficient energy due to the higher temperature.

An *endothermic* reaction can be represented by:



This can be shown in an activation energy diagram (Figure 7.11). These graphs are also sometimes called a reaction profile or a potential energy graph.

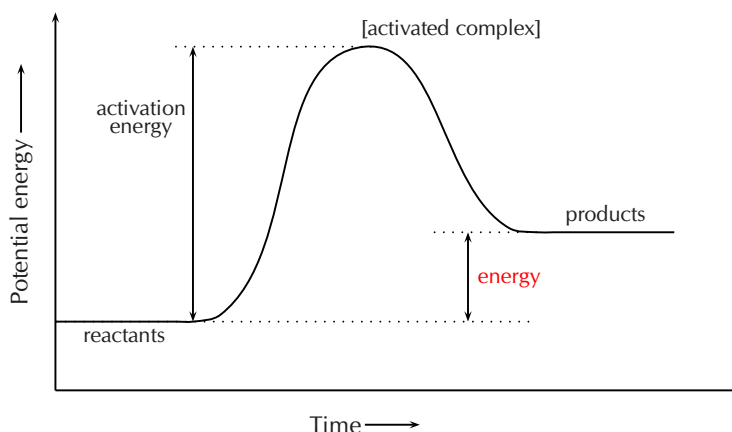


Figure 7.11: An activation energy diagram with reactant energy lower than product energy, i.e. *endothermic*

An *exothermic* reaction can be represented by:

Reactants $\rightarrow$ Products + **Energy** i.e. a reaction that releases energy

This can be shown in an activation energy diagram (Figure 7.12):

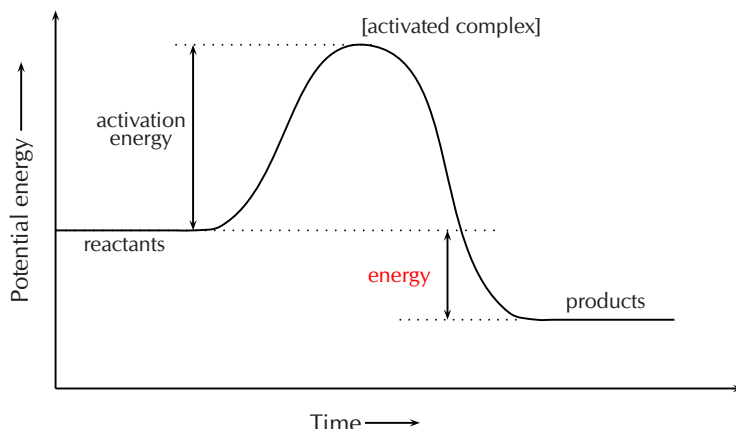


Figure 7.12: An activation energy diagram with reactant energy greater than product energy, i.e. *exothermic*

Revise this section from Grade 11 ([Activation energy, Section 12.3](#)).

How do catalysts work?

ESCNB

A **catalyst** increases reaction rates in a slightly different way from other methods of increasing reaction rate. The function of a catalyst is to lower the activation energy so that a greater proportion of the particles have enough energy to react. A catalyst can lower the activation energy for a reaction by:

- orienting the reacting particles in such a way that successful collisions are more likely
- reacting with the reactants to form an intermediate that requires lower energy to form the product

Some *metals* e.g. platinum, copper and iron can act as catalysts in certain reactions. In our own bodies, we have *enzymes* that are catalysts, which help to speed up biological reactions. Catalysts generally react with one or more of the reactants to form a chemical intermediate, which then reacts to form the final product. The chemical intermediate is sometimes called the **activated complex**.

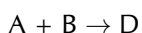
The following is an example of how a reaction involving a catalyst might proceed. A and B are reactants, C is the catalyst, and D is the product of the reaction of A and B.

Step 1: $A + C \rightarrow AC$ **Step 2:** $B + AC \rightarrow ACB$ **Step 3:** $ACB \rightarrow C + D$

ACB represents the intermediate chemical. Although the catalyst (C) is consumed by reaction 1, it is later released again by reaction 3, so that the overall reaction with a catalyst is as follows:



You can see from this that the catalyst is released at the end of the reaction, completely unchanged. Without a catalyst the overall reaction would be:



TIP

An activated complex occurs in reactions without catalysts as well as those with catalysts.

The catalyst has provided an alternative set of reaction steps, which we refer to as an alternative pathway. The pathway involving the catalyst requires less activation energy and is therefore faster. This can be seen in the following diagram (Figure 7.13).

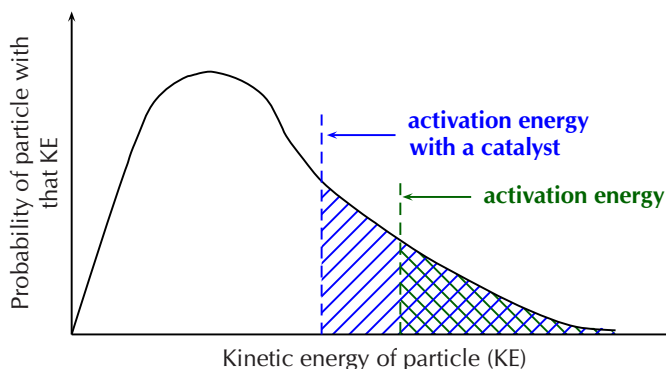


Figure 7.13: The proportion of particles that have enough energy to react is increased in the presence of a catalyst.

DEFINITION: *Catalyst*

A catalyst speeds up a chemical reaction, without being consumed by the reaction. It increases the reaction rate by lowering the activation energy for a reaction.

Energy diagrams are useful to illustrate the effect of a **catalyst** on reaction rates. Catalysts decrease the activation energy required for a reaction to proceed (shown by the smaller magnitude of the activation energy on the energy diagram in Figure 7.14), and therefore increase the reaction rate. Remember that with a catalyst, the average kinetic energy of the molecules remains the same but the required energy decreases (Figure 7.13).

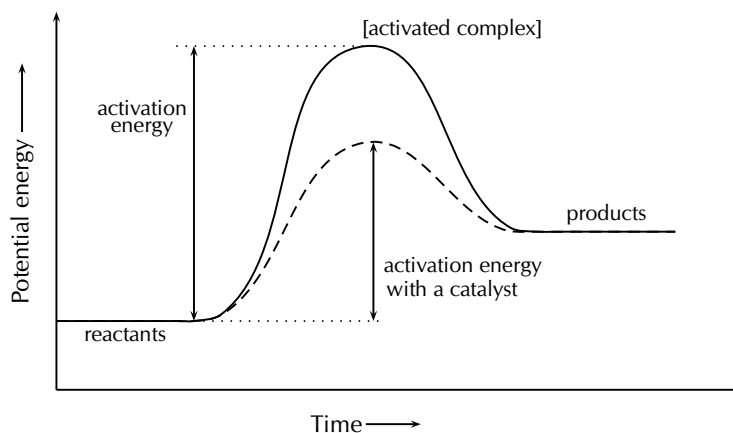


Figure 7.14: The effect of a catalyst on the activation energy of an endothermic reaction. The catalyst would act in the same way for an exothermic reaction.

Exercise 7 – 4: Reaction rates

1. For each of the following, say whether the statement is **true** or **false**. If it is false, re-write the statement correctly.
 - a) A catalyst increases the energy of reactant molecules so that a chemical reaction can take place.

- b) Increasing the temperature of a reaction has the effect of increasing the number of reactant particles that have more energy than the activation energy.
- c) A catalyst does not become part of the final product in a chemical reaction.
2. 5 g of zinc pieces are added to 400 cm³ of 0,5 mol.dm⁻³ hydrochloric acid. To investigate the average rate of the reaction, the change in the mass of the flask containing the zinc and the acid is measured by placing the flask on a direct reading balance. The reading on the balance shows that there is a decrease in mass during the reaction. The reaction which takes place is given by the following equation:
- $$\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$$
- a) Why is there a decrease in mass during the reaction?
- b) The experiment is repeated, this time using 5 g of powdered zinc instead of pieces of zinc. How will this influence the average rate of the reaction?
- c) The experiment is repeated once more, this time using 5 g of zinc pieces and 600 cm³ of 0,5 mol.dm⁻³ hydrochloric acid. How does the average rate of this reaction compare with the **original** reaction rate?
- d) What effect would a catalyst have on the average rate of this reaction?
- (IEB Paper 2 2003)
3. 5 g of calcium carbonate powder reacts with 20 cm³ of a 0,1 mol.dm⁻³ solution of hydrochloric acid. The gas that is produced at a temperature of 25°C is collected in a gas syringe.
- a) Write a balanced chemical equation for this reaction.
- b) The average rate of the reaction is determined by measuring the volume of gas that is produced in the first minute of the reaction. How would the average rate of the reaction be affected if:
- a lump of calcium carbonate of the same mass is used
 - 20 cm³ of 0,2 mol.dm⁻³ hydrochloric acid is used
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. [27V2](#) 2. [27V3](#) 3. [27V4](#)



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7.5 Chapter summary

ESCNC

► See presentation: [27V5](#) at www.everythingscience.co.za

- The **average rate of a reaction** describes how quickly reactants are used, or how quickly products form. The units used are usually moles per second.
- A number of factors can affect the average rate of a reaction. These include the **nature of the reactants**, the **concentration** of solutions, **surface area** of solids or **pressure** of gases, the **temperature** of the reaction and the presence or absence of a **catalyst**.
- **Collision theory** provides one way of explaining why each of the factors mentioned in the previous bullet can affect the average rate of a reaction. For example, higher temperatures mean increased reaction rates because the reactant

particles have more energy and are more likely to collide successfully with each other.

- Different methods can be used to **measure the average rate of a reaction**. The method used will depend on the nature of the product or products. Reactions that produce gases can be measured by collecting the gas in a syringe or by measuring a change in mass of the reaction vessel. In reactions that produce a precipitate the mass of the dried precipitate can be measured.
- A **change in colour** can show that the **product has formed** or that the **reaction** has reached **completion**. (This is known as colorimetry).
- For any reaction to occur, a minimum amount of energy is needed so that bonds in the reactants can break, and new bonds can form in the products. The minimum energy that is required is called the **activation energy** of a reaction.
- In reactions where the particles do not have enough energy to overcome this activation energy, one of two methods can be used to facilitate a reaction to take place: increase the temperature of the reaction or add a catalyst.
- **Increasing the temperature of a reaction** means that the average energy of the reactant particles increases and they are more likely to have enough energy to overcome the activation energy and the chances of two molecules having the correct orientation are increased.
- A **catalyst** is used to lower the activation energy so that the reaction is more likely to take place. A catalyst does this by providing an alternative, lower energy pathway for the reaction.
- A catalyst therefore **speeds up a reaction** but remains unchanged after the reaction is complete.

Exercise 7 – 5:

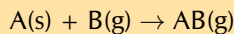
1. Explain the following:
 - a) Reaction rate
 - b) Collision theory
 - c) Activation energy
 - d) Catalyst
2. $2\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{MgO(s)}$
 1 g of Mg is oxidised. The reaction takes 1 day to reach completion. Which is the correct reaction rate for the oxidation of Mg?
 - a) $2,87 \times 10^{-7} \text{ mol.s}^{-1}$
 - b) $1,74 \times 10^{-3} \text{ mol.s}^{-1}$
 - c) $4,76 \times 10^{-7} \text{ mol.s}^{-1}$
 - d) $3,62 \times 10^{-7} \text{ mol.s}^{-1}$
3. 10 g of magnesium ribbon reacts with a $0,15 \text{ mol.dm}^{-3}$ solution of hydrochloric acid at a temperature of 25°C .
 - a) Write a balanced chemical equation for the reaction.
 - b) State **two** ways of increasing the average rate of production of H_2 (g).
 - c) A table of the results is given:

Time elapsed (min)	Vol of $\text{H}_2\text{(g)}$ (cm^3)
0	0
0,5	17
1,0	25
1,5	30
2,0	33
2,5	35
3,0	35

- i. Plot a graph of volume versus time for these results.
- ii. Explain the shape of the graph during the following two time intervals:
 $t = 0$ to $t = 120$ s and then $t = 150$ and $t = 180$ s by referring to the volume of H_2 (g) produced.

(IEB Paper 2, 2001)

4. Consider the following reaction, which takes place in a closed container:

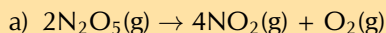


If you wanted to increase the average rate of the reaction, which of the following would you do?

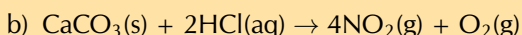
- a) decrease the concentration of B
- b) grind A into a fine powder
- c) decrease the pressure

(IEB Paper 2, 2002)

5. Consider the following reaction and calculate the reaction rates for the situations that follow:



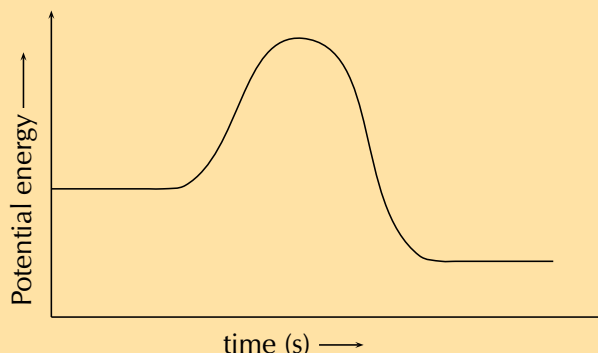
- i. 1,35 mol N_2O_5 is used and the reaction goes to completion in 30 minutes.
- ii. 2,7 mol N_2O_5 is used instead and the reaction goes to completion in 60 minutes.



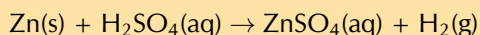
- i. 2 g of CaCO_3 is placed in excess HCl. The reaction goes to completion over 4 hours.
- ii. The concentration of the HCl is increased and so the reaction takes 2,5 hours instead.
- iii. 3,3 g of CaCO_3 is used with a dilute HCl. 0,6 g of solid CaCO_3 remains after 4,7 hours.

6. Given:

A catalyst is added to this reaction. On the same set of axes draw the graph to represent this.

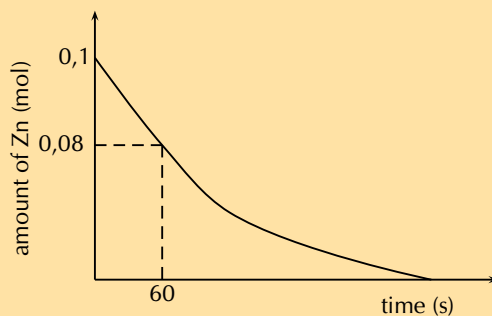


7. A group of learners use the reaction between zinc and sulfuric acid to investigate one of the factors that affects reaction rate. The equation below represents the reaction that takes place.



They add 6,5 g of zinc granules to excess *dilute* sulfuric acid and measure the mass of zinc used per unit time.

The learners then repeat the experiment using excess *concentrated* sulfuric acid. The results obtained for the reaction using *dilute* sulfuric acid are represented in the graph below:



- Define the term *reaction rate*
- Give a reason why the acid must be in excess
- Write down a hypothesis for this investigation.
- Give a reason why the learners must use the same amount of *zinc granules* in both experiments.
- Using the graph, calculate the mass of zinc used from $t = 0 \text{ s}$ to $t = 60 \text{ s}$.
- Calculate the average rate of the reaction (in grams per second) during the first 60 s.
- Copy the above graph and *on the same set of axes* use a dotted line to show the curve that will be obtained when concentrated sulfuric acid is used. No numerical values are required.

(NSC, Paper 2, 2011)

8. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 27V6
- 27V7
- 27V8
- 27V9
- 27VB
- 27VC
- 27VD



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Chemical equilibrium

8.1	<i>What is chemical equilibrium?</i>	300
8.2	<i>The equilibrium constant</i>	304
8.3	<i>Le Chatelier's principle</i>	313
8.4	<i>Chapter Summary</i>	331

Systems in equilibrium are balanced. For example, if you are running on a treadmill you are in constant motion. However, you are not moving forward or backwards. As fast as you run forward the treadmill is moving you backwards. You are in equilibrium with the treadmill.



In the same way chemical reactions can be in equilibrium (the products and reactants are produced at the same rate).

We will begin by exploring chemical equilibrium in more detail. Ways of measuring equilibrium and the factors that affect equilibrium will be covered.

8.1 What is chemical equilibrium?

ESCND

To define chemical equilibrium we need to ask some important questions about reactions:

1. Does a reaction always run its full course so that all the reactants are used up?
 - When all the reactants in a reaction are used up the reaction is said to have **gone to completion**. However, in some reactions not all the reactants are used. **Not all reactions go to completion**.
2. Does a reaction always proceed in the same direction or can it be reversed? In other words, does a reaction always proceed *reactants* $\rightarrow$ *products*, or is it possible that a reaction will reverse and go *products* $\rightarrow$ *reactants*?
 - Reactions that go to completion are *irreversible*. However, in some reactions the reactants form products (in a **forward reaction**), and the products can change back into reactants (in a **reverse reaction**).
3. Can a reaction reach a point where reactants are still present, but there does not seem to be any further change taking place in the reaction?
 - In all reactions, as the amount of reactant in a reaction decreases the product is formed more slowly. In a reversible reaction, as the amount of product increases the reactant is formed more quickly. Eventually the rate of the forward reaction (reactants $\rightarrow$ products) equals the rate of the reverse reaction (products $\rightarrow$ reactants).

At this point there **are still reactants present** but there does not *appear* to be any further change taking place. The reaction is said to be in **chemical equilibrium**.

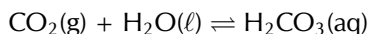
Chemical equilibrium is the state of a reversible reaction where the rate of the forward reaction equals the rate of the reverse reaction. While a reaction is in equilibrium the concentration of the reactants and products are constant.

DEFINITION: Chemical equilibrium

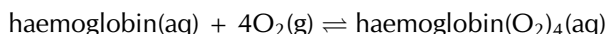
A reaction is in chemical equilibrium when the rate of the forward reaction equals the rate of the reverse reaction.

There are many examples of chemical equilibrium all around you. One example is a bottle of fizzy cooldrink. In the bottle there is carbon dioxide (CO_2) dissolved in the liquid. There is also CO_2 gas in the space between the liquid and the cap. There is a

constant movement of CO_2 from the liquid to the gas phase, and from the gas phase into the liquid. However, if you look at the bottle there does not appear to be any change. The system is in equilibrium.



Without chemical equilibrium life as we know it would not be possible. Another example of equilibrium in our everyday lives goes on within our very bodies. Haemoglobin is a macromolecule that transports oxygen around our bodies. Without it we would not survive. The haemoglobin has to be able to take up oxygen, but also to release it and this is done through changes in the chemical equilibrium of this reaction in different places in our bodies.

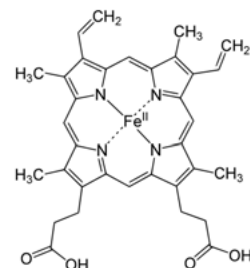


This video is a good demonstration of how a reaction reaches equilibrium with time.

▶ See video: 27VF at www.everythingscience.co.za

FACT

Haemoglobin, inside red blood cells in the lungs, bonds with oxygen. This *oxyhaemoglobin* moves with the red blood cells through the blood stream to cells throughout the body.



Open and closed systems

ESCNF

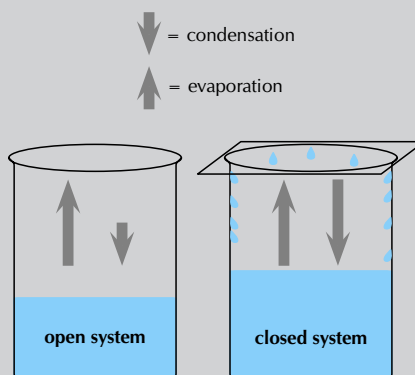
Experiment: Liquid-gas phase equilibrium

Apparatus:

- 2 beakers, glass cover
- water

Method:

1. Half-fill two beakers with water and mark the level of the water in each case.
2. Cover one of the beakers with a glass cover.
3. Leave the beakers and, over the course of a day or two, observe how the water levels in the two beakers change. What do you notice? Note: You could speed up this demonstration by placing the two beakers over a Bunsen burner, or in direct sunlight, to heat the water.



Observations:

You should notice that in the beaker that is uncovered, the water level drops more quickly than in the covered beaker. This is because of evaporation. In the beaker that is covered, there is an initial drop in the water level, but after a while evaporation appears to stop and the water level in this beaker is higher than that in the one that is open.

Discussion:

In the first beaker, liquid water becomes water vapour as a result of evaporation and the water level drops. A small amount of gas molecules will condense again, but because the gas molecules can escape from the system there is much less condensation than evaporation.

FACT

Evaporation is when a substance goes from the liquid phase to the gas phase (it evaporates).

Condensation is when a substance goes from the gas phase to the liquid phase (it condenses).

The bottle contains water vapour (gas) and water droplets are condensing on the side of the bottle.



FACT

Generally, some of the molecules at the surface of liquids and solids are moving into the gas phase. This means that matter is leaving the system. However, this is such a small fraction of the total volume of the liquid or solid that a reaction involving only solids or liquids can be considered a closed system.

FACT

It is useful to simplify situations in science by dividing the world into the *system* we are studying, and the surrounding environment that might influence the reaction, but is not part of it.

In the second beaker, evaporation also takes place. However, in this case, the vapour comes into contact with the surface of the glass cover and it cools and condenses to form liquid water again. This water is returned to the beaker. Once condensation has begun, the rate at which the water level drops will start to decrease. At some point, the rate of evaporation will be equal to the rate of condensation, and there will be no change in the water level in the beaker. This can be represented as follows:

liquid $\rightleftharpoons$ vapour

In this example, the reaction (in this case, a change in the phase of water) can proceed in either direction. In the forward direction there is a change in phase from liquid to gas, present here as water vapour. A reverse change can also take place, when vapour condenses to form liquid again.

An **open system** is one in which matter or energy can flow into or out of the system. In the liquid-gas demonstration we used, the first beaker was an example of an open system because the beaker could be heated or cooled (a change in *energy*), and water vapour (the *matter*) could evaporate from the beaker.

DEFINITION: *Open system*

An *open system* is one whose borders allow the movement of **energy and matter** into and out of the system.

A **closed system** is one in which energy can enter or leave, but matter cannot. The second beaker with the glass cover is an example of a closed system. The beaker can still be heated or cooled, but water vapour cannot leave the system because the glass cover is a barrier. Condensation changes the vapour to liquid and returns it to the beaker. In other words, there is no loss of matter from the system.

DEFINITION: *Closed system*

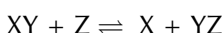
A *closed system* is one in which **only energy** can move into and out of the system. Matter cannot be gained by the system nor lost from the system.

In a **closed system** it is possible for reactions to be reversible, such as in the demonstration above. In a closed system, it is also possible for a chemical reaction to reach **equilibrium**.

Reversible reactions

ESCN

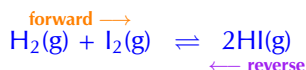
Some reactions can take place in two directions. In one direction the reactants combine to form the products. This is called the **forward reaction**. In the other direction, the products react to form the reactants again. This is called the **reverse reaction**. A special double-headed arrow ($\rightleftharpoons$) is used to show this type of **reversible reaction**:



DEFINITION: *A reversible reaction*

A reversible reaction is a chemical reaction that can proceed in both the forward and reverse directions. In other words, the reactants and products of the reaction may reverse roles.

So, in the following reversible reaction:



$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$ is the *forward* reaction.

The forward reaction is always taken from the given equation and written: **left to right**.

$2\text{HI}(\text{g}) \rightarrow \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ is the *reverse* reaction.

The reverse reaction is always taken from the given equation and written: **right to left**.

FACT

Remember that $\rightarrow$ implies that A goes to B and cannot become A again. $\rightleftharpoons$ implies that the reaction is reversible. A can go to B and B can go to A.

Exercise 8 – 1: Chemical equilibrium

1. Which of the following situations describes a closed system?

- A pot of water (without a lid) heated to 80°C
 - A pot of water (with a lid) heated to 80°C
- $\text{NaCl}(\text{aq}) + \text{AgNO}_3(\text{aq}) \rightarrow \text{NaNO}_3(\text{aq}) + \text{AgCl}(\text{s})$ in an open container.
 - $2\text{CO}(\text{g}) + \text{MoO}_2(\text{s}) \rightarrow 2\text{CO}_2(\text{g}) + \text{Mo}(\text{s})$ in an open container.

2. For the reaction: $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$

- Write down the forward reaction
- Write down the reverse reaction
- This reaction is said to be (fill in the missing word)

3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27VG 2. 27VH



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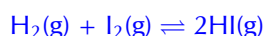


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Dynamic equilibrium

ESCNH

We will use the same reversible reaction that we used in an earlier example:



The *forward* reaction is: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$

The *reverse* reaction is: $2\text{HI}(\text{g}) \rightarrow \text{H}_2(\text{g}) + \text{I}_2(\text{g})$

When the rate of the forward reaction and the rate of the reverse reaction are equal, the system is said to be in **equilibrium**. Figure 8.1 shows this. Initially (time = 0), the rate of the *forward reaction* is high (fast). With time, the rate of the *forward reaction decreases*. As the reaction gets closer to equilibrium the rate of decrease levels out until the forward reaction has a constant rate.

Initially the rate of the *reverse reaction* is low (slow). As the reaction proceeds with time, the rate of the *reverse reaction increases*. As the reaction progresses the rate of increase levels out until the reverse reaction has a constant rate.

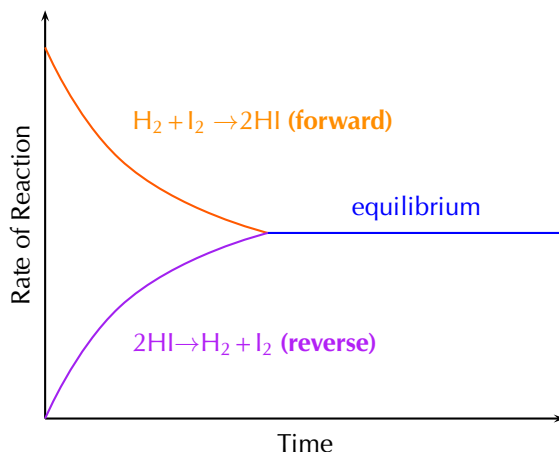
At this point the forward and reverse reaction rates are equal and this is called **equilibrium**.

FACT

In the equilibrium constant (K_c), the subscript c refers to the concentration of all the substances at equilibrium.

FACT

The equation for the equilibrium constant uses the concentrations of the reactants and products. Square brackets mean the concentration of the compound written within the square brackets: $[x]$ means the concentration of x.



Although it is not always possible to observe any macroscopic changes, this does not mean that the reaction has stopped. The forward and reverse reactions continue to take place and so microscopic changes still occur in the system. This state is called **dynamic equilibrium**.

Figure 8.1: The change in rate of forward and reverse reactions in a closed system.

DEFINITION: *Dynamic equilibrium*

There is a dynamic equilibrium in a reversible reaction when the rate of the forward reaction equals the rate of the reverse reaction. The amounts of reactants and products remain constant.

In the liquid-gas phase equilibrium demonstration, dynamic equilibrium was reached when there was no observable change in the level of the water in the second beaker even though evaporation and condensation continued to take place.

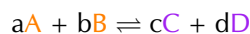
For more information on dynamic equilibria watch this video:

▶ See video: 27VJ at www.everythingscience.co.za

8.2 The equilibrium constant

ESCNJ

Consider the reversible chemical reaction:



A and B are the reactants, C and D are the products and a, b, c, and d are the coefficients from the balanced reaction.

When the rate of the forward reaction equals the rate of the reverse reaction, the system is in chemical equilibrium. It is useful to know how much of each substance is in the container - in particular the amount of products compared to the amount of reactants. A simple ratio from the balanced chemical equation gives us a number called the equilibrium constant (K_c).

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

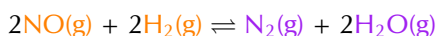
$[A]$, $[B]$, $[C]$ and $[D]$ are the molar concentrations of each substance present at equilibrium.

DEFINITION: *The equilibrium constant*

The equilibrium constant (K_c) is the ratio between the concentration of products and reactants in a chemical reaction.

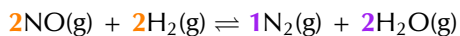
When the concentration of the reactants is much larger than the concentration of the products K_c will be small (normally less than one). When the concentration of the reactants is much less than that of the products K_c will be large (normally greater than one).

For example:



The **reactants** are NO and H₂. The **products** are N₂ and H₂O.

It is important to look at the coefficients of the equation as well:

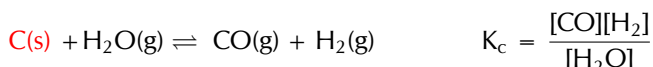


K_c for this equation will therefore be written as follows:

$$K_c = \frac{[\text{N}_2]^1 [\text{H}_2\text{O}]^2}{[\text{NO}]^2 [\text{H}_2]^2}$$

In the expression for K_c the concentration of a product or reactant is taken to the power of its coefficient in the balanced reaction. So if the coefficient of a product (C) is 3 in the balanced equation, then the concentration of C ([C]) will be written [C]³ in the expression for K_c.

We leave out reactants or products that are either pure **liquids** or in the **solid** phase when calculating K_c. For example:



Calculating the equilibrium constant

ESCNK

When calculations involving the equilibrium constant are done, the following tips may help:

1. Always *read the question carefully* to be sure that you understand what you have been asked to calculate.
2. If the equilibrium constant is involved, make sure that the concentrations you use are the concentrations **at equilibrium**, and not the concentrations or quantities that are present at some other time in the reaction.
3. When you are doing more complicated calculations, it helps to draw up a RICE table (Table 8.1). This is gone into in more detail later in this section.

Remember that the general form of the expression for K_c of a reaction at chemical equilibrium (aA + bB ⇌ cC + dD) is:

$$K_c = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$$

Worked example 1: Writing expressions for K_c

QUESTION

For the reaction: $9\text{X}(\text{g}) + \text{Y}_3(\text{g}) \rightleftharpoons 3\text{X}_3\text{Y}(\text{g})$

Write an expression for the equilibrium constant K_c.

SOLUTION

Step 1: Write the general expression for K_c

$$K_c = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$$

Step 2: Determine the reactants and the products of the reaction

X(g) and Y₃(g) are both reactants. They are gases and will be included in the expression. X₃Y(g) is a product. It is a gas and will be included in the expression.

Step 3: Write the general expression for K_c for this reaction

$$K_c = \frac{[\text{X}_3\text{Y}]^z}{[\text{X}]^x [\text{Y}_3]^y}$$

Step 4: What are the coefficients in the balanced equation?

The compound X has a coefficient of **9**. The compound Y₃ has a coefficient of **1**. The compound X₃Y has a coefficient of **3**.

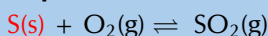
Step 5: Write the expression for K_c for this reaction

$$K_c = \frac{[X_3Y]^3}{[X]^9[Y_3]^1}$$

Worked example 2: Calculating reagent concentration**QUESTION**

For the reaction: $S(s) + O_2(g) \rightleftharpoons SO_2(g)$

1. Write an expression for the equilibrium constant.
2. Calculate the equilibrium concentration of O₂ if:
K_c = 6 and [SO₂] = 3 mol.dm⁻³ at equilibrium.

SOLUTION**Step 1: Determine which compounds will be part of the K_c expression**

O₂ and SO₂ are gases and so will be part of the expression for K_c. S is solid and so will not be part of the expression for K_c.

Step 2: Write the expression for K_c

$$K_c = \frac{[SO_2]}{[O_2]}$$

Step 3: Re-arrange the expression so that oxygen is on its own on one side of the expression

$$[O_2] = \frac{[SO_2]}{K_c}$$

Step 4: Fill in the values you know and calculate [O₂]

$$[O_2] = \frac{3 \text{ mol.dm}^{-3}}{6} = 0,5 \text{ mol.dm}^{-3}$$

Worked example 3: Calculating K_c**QUESTION**

For the reaction: $SO_2(g) + NO_2(g) \rightleftharpoons NO(g) + SO_3(g)$

The concentrations of the compounds used are:

$$[SO_2] = 0,2 \text{ mol.dm}^{-3}, [NO_2] = 0,1 \text{ mol.dm}^{-3}$$

$$[NO] = 0,4 \text{ mol.dm}^{-3}, [SO_3] = 0,2 \text{ mol.dm}^{-3}$$

Calculate the value of K_c.

SOLUTION**Step 1: Write the general expression for K_c**

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Step 2: Determine the reactants and the products of the reaction

SO₂(g) and NO₂(g) are both reactants. They are gases and will be included in the expression.

NO(g) and SO₃(g) are both products. They are gases and will be included in the expression.

Step 3: Write the expression for K_c for this reaction

All four compounds have a coefficient of 1.

$$K_c = \frac{[\text{NO}]^1[\text{SO}_3]^1}{[\text{SO}_2]^1[\text{NO}_2]^1}$$

Step 4: Fill in the values you know for this expression and calculate K_c

$$K_c = \frac{(0,4)(0,2)}{(0,2)(0,1)} = 4$$

TIP

You do not need to fill in the coefficients in a K_c calculation when they are 1. They are shown here so that you do not forget where the coefficients are reflected in the equation.

When you are doing more complicated calculations, it helps to draw up a RICE table (Table 8.1). A RICE table is an easy way of organising information in equilibrium calculations.

R	Reaction	the balanced chemical equation
I	Initial quantity	the moles of reactants and products at the beginning of the reaction
C	Change	how much the moles of the reactants and products changed between the beginning of the reaction and equilibrium
E	Equilibrium quantity	the moles of the reactants and products at equilibrium
E	Equilibrium concentration	to calculate K _c you need the concentration of the reactants and products at equilibrium

Table 8.1: The use of each row in a RICE.

Here are some guidelines on how to use a RICE table:

1. Fill in the balanced chemical equation: $aA + bB \rightleftharpoons cC + dD$

Reaction	aA	bB	cC	dD
Initial quantity (mol)				
Change (mol)				
Equilibrium quantity (mol)				
Equilibrium concentration (mol.dm ⁻³)				

2. In the **initial** row fill in the number of **moles** of each substance present at the beginning of the reaction. For example, if there are 0,3 moles of A, 0,7 moles of B and no moles of product initially:

Reaction	aA	bB	cC	dD
Initial quantity (mol)	0,3	0,7	0	0
Change (mol)				
Equilibrium quantity (mol)				
Equilibrium concentration (mol.dm ⁻³)				

3. In the **change** row fill in the number of moles of reactant used, or the number of moles of product formed, in terms of x. Use the balanced equation for this. For example, if dx moles of D is produced, bx moles of B will be used:

Reaction	aA	bB	cC	dD
Initial quantity (mol)	0,3	0,7	0	0
Change (mol)	-ax	-bx	+cx	+dx
Equilibrium quantity (mol)				
Equilibrium concentration (mol.dm ⁻³)				

4. In the **equilibrium** row fill in the number of **moles** of each substance present at equilibrium in terms of x. For example, there will be 0,3 - ax moles of A at equilibrium.

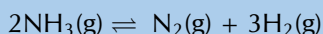
Reaction	aA	bB	cC	dD
Initial quantity (mol)	0,3	0,7	0	0
Change (mol)	-ax	-bx	+cx	+dx
Equilibrium quantity (mol)	0,3 - ax	0,7 - bx	+cx	+dx
Equilibrium concentration (mol.dm ⁻³)				

5. Use any extra information you have been given to calculate x, and then finish filling in the table.
For example, if the number of moles of B at equilibrium is 0,2:
 $0,7 - bx = 0,2$ and you can solve for x.
6. In the last row fill in the concentration of each substance at equilibrium. You will have to calculate this value.

Worked example 4: Equilibrium calculations

QUESTION

1,4 moles of NH₃(g) is introduced into a sealed 2,0 dm³ reaction vessel. The ammonia decomposes when the temperature is increased to 600 K and reaches equilibrium as follows:



When the equilibrium mixture is analysed, the concentration of NH₃(g) is 0,3 mol.dm⁻³.

- Calculate the concentration of N₂(g) and H₂(g) in the equilibrium mixture.
- Calculate the equilibrium constant for the reaction at 600 K.

SOLUTION

Step 1: Draw a RICE table

Reaction			
Initial quantity (mol)			
Change (mol)			
Equilibrium quantity (mol)			
Equilibrium concentration (mol.dm ⁻³)			

Step 2: Fill in the balanced chemical equation on the table:

Reaction	2NH ₃	1N ₂	3H ₂
Initial quantity (mol)			
Change (mol)			
Equilibrium quantity (mol)			
Equilibrium concentration (mol.dm ⁻³)			

Make sure you know which compounds are the **reactants**, and which are the **products**.

Step 3: Fill in the number of moles of NH₃, N₂ and H₂ present at the beginning of the reaction

Reaction	2NH ₃	1N ₂	3H ₂
Initial quantity (mol)	1,4	0	0
Change (mol)			
Equilibrium quantity (mol)			
Equilibrium concentration (mol.dm ⁻³)			

There are 1,4 mol of NH₃, and no moles of N₂ or H₂ at the beginning of the reaction.

Step 4: Fill in the change in the number of moles of NH₃, N₂ and H₂ in terms of x

The mole ratio of NH₃(g) : N₂ : H₂ is 2 : 1 : 3.

Therefore, for every 2 moles of NH₃ used, 1 mole of N₂ and 3 moles of H₂ will be formed. If the number of moles of NH₃ decreases by 2x (−2x), then the number of moles of N₂ increases by 1x (+1x) and the number of moles of H₂ increases by 3x (+3x).

Reaction	2NH ₃	1N ₂	3H ₂
Initial quantity (mol)	1,4	0	0
Change (mol)	−2x	+x	+3x
Equilibrium quantity (mol)			
Equilibrium concentration (mol.dm ⁻³)			

Step 5: Fill in the number of moles of NH₃, N₂ and H₂ at equilibrium in terms of x

Reaction	2NH ₃	1N ₂	3H ₂
Initial quantity (mol)	1,4	0	0
Change (mol)	−2x	+x	+3x
Equilibrium quantity (mol)	1,4 − 2x	+x	+3x
Equilibrium concentration (mol.dm ⁻³)			

To get the number of moles of a substance at equilibrium you take the number of moles initially, and add the change in number of moles in terms of x.

Step 6: Calculate the value of x

You know that the concentration of NH₃ is 0,3 mol.dm⁻³ at equilibrium and the reaction vessel has a volume of 2 dm³.

$$C = \frac{n}{V}, \text{ therefore } n = C \times V$$

$$n(\text{NH}_3 \text{ at equilibrium}) = 0,3 \text{ mol.dm}^{-3} \times 2,0 \text{ dm}^3 = 0,6 \text{ mol}$$

From the RICE table we can see that there are 1,4 − 2x moles of NH₃ at equilibrium.

$$0,6 \text{ mol} = 1,4 \text{ mol} - 2x$$

$$2x = 1,4 \text{ mol} - 0,6 \text{ mol} = 0,8 \text{ mol}$$

$$\text{Therefore, } x = 0,4 \text{ mol}$$

Step 7: Calculate the concentration of N₂ formed (at equilibrium).

From the RICE table we can see that the number of moles of N₂ at equilibrium = x

$$\text{therefore } n(\text{N}_2) = 0,4 \text{ mol} \quad C(\text{N}_2) = \frac{n}{V} = \frac{0,4 \text{ mol}}{2,0 \text{ dm}^3} = 0,2 \text{ mol.dm}^{-3}$$

Step 8: Calculate the concentration of H₂ formed (at equilibrium).

From the RICE table we can see that the number of moles of H₂ at equilibrium = 3x

$$n(\text{H}_2) = 3 \times 0,4 \text{ mol} = 1,2 \text{ mol} \quad C(\text{H}_2) = \frac{n}{V} = \frac{1,2 \text{ mol}}{2,0 \text{ dm}^3} = 0,6 \text{ mol.dm}^{-3}$$

Step 9: Complete the RICE table

Reaction	2NH ₃	1N ₂	3H ₂
Initial quantity (mol)	1,4	0	0
Change (mol)	-2x	+x	+3x
Equilibrium quantity (mol)	1,4 - 2x	+x	+3x
Equilibrium concentration (mol.dm ⁻³)	0,3	0,2	0,6

Step 10: Write the expression for K_c for this reaction

NH₃ is the reactant, N₂ and H₂ are the products. They are all in the gas phase, and so are included in the expression for K_c.

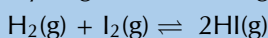
$$K_c = \frac{[N_2][H_2]^3}{[NH_3]^2}$$

Step 11: Calculate K_c

$$K_c = \frac{(0,2)(0,6)^3}{(0,3)^2} = 0,48$$

Worked example 5: Calculating K_c**QUESTION**

Hydrogen and iodine gas react according to the following expression:



When 0,496 mol H₂ and 0,181 mol I₂ are heated at 450°C in a 1 dm³ container, the equilibrium mixture is found to contain 0,00749 mol I₂. Calculate the equilibrium constant for the reaction at 450°C.

SOLUTION**Step 1: Draw a RICE table**

Reaction			
Initial quantity (mol)			
Change (mol)			
Equilibrium quantity (mol)			
Equilibrium concentration (mol.dm ⁻³)			

Step 2: Fill in the balanced chemical equation on the table:

Reaction	1H ₂	1I ₂	2HI
Initial quantity (mol)			
Change (mol)			
Equilibrium quantity (mol)			
Equilibrium concentration (mol.dm ⁻³)			

Step 3: Fill in the number of moles of H₂, I₂ and HI present at the beginning of the reaction

Reaction	1H ₂	1I ₂	2HI
Initial quantity (mol)	0,496	0,181	0
Change (mol)			
Equilibrium quantity (mol)			
Equilibrium concentration (mol.dm ⁻³)			

There are 0,496 moles of H₂, 0,181 moles of I₂ and no moles HI at the beginning of the reaction.

Step 4: Fill in the change in the number of moles of H₂, I₂ and HI in terms of x

The mole ratio of H₂ : I₂ : HI is 1 : 1 : 2.

Therefore, for every 1 mole of H₂ used, 1 mole of I₂ is used and 2 moles of HI will be formed. If the number of moles of H₂ decreases by x (−1x), then the number of moles of I₂ also decreases by x (−1x) and the number of moles of HI increases by 2x (+2x).

Reaction	1H ₂	1I ₂	2HI
Initial quantity (mol)	0,496	0,181	0
Change (mol)	-x	-x	+ 2x
Equilibrium quantity (mol)			
Equilibrium concentration (mol.dm ⁻³)			

Step 5: Fill in the number of moles of H₂, I₂ and HI at equilibrium in terms of x

To get the number of moles of a substance at equilibrium you take the number of moles initially, and add the change in number of moles in terms of x.

Reaction	1H ₂	1I ₂	2HI
Initial quantity (mol)	0,496	0,181	0
Change (mol)	-x	-x	+ 2x
Equilibrium quantity (mol)	0,496 - x	0,181 - x	+ 2x
Equilibrium concentration (mol.dm ⁻³)			

Step 6: Calculate the value of x

You know that number of moles of I₂ at equilibrium is 0,00749 mol. From the RICE table we can see that there are 0,181 - x moles of I₂ at equilibrium. Therefore:

$$0,00749 \text{ mol} = 0,181 \text{ mol} - x$$

$$x = 0,181 \text{ mol} - 0,00749 \text{ mol} = 0,1735 \text{ mol}$$

Step 7: Calculate the concentration of H₂ at equilibrium.

The volume of the container is 1 dm³. From the RICE table we can see that the number of moles of H₂ at equilibrium = 0,496 mol - x. Therefore:

$$n(\text{H}_2) = 0,496 \text{ mol} - 0,1735 \text{ mol} = 0,3225 \text{ mol}$$

$$C(\text{H}_2) = \frac{n}{V} = \frac{0,3225 \text{ mol}}{1 \text{ dm}^3} = 0,3225 \text{ mol.dm}^{-3}$$

Step 8: Calculate the concentration of I₂ at equilibrium.

$$C(\text{I}_2) = \frac{n}{V} = \frac{0,00749 \text{ mol}}{1 \text{ dm}^3} = 0,00749 \text{ mol.dm}^{-3}$$

Step 9: Calculate the concentration of HI at equilibrium.

From the RICE table we can see that the number of moles of HI at equilibrium = 2x. Therefore:

$$n(\text{HI}) = 2 \times 0,1735 \text{ mol} = 0,347 \text{ mol}$$

Step 10: Complete the RICE table

Reaction	1H ₂	1I ₂	2HI
Initial quantity (mol)	0,496	0,181	0
Change (mol)	-x	-x	+ 2x
Equilibrium quantity (mol)	0,496 - x	0,181 - x	+ 2x
Equilibrium concentration (mol.dm ⁻³)	0,3225	0,00749	0,347

Step 11: Write the expression for K_c for this reaction $K_c = \frac{[HI]^2}{[H_2][I_2]}$

Step 12: Calculate K_c $K_c = \frac{(0,347)^2}{(0,3225)(0,00749)} = 49,85$

The meaning of K_c values

ESCNM

The formula for K_c has:

- the concentration of the products in the numerator and the concentration of reactants in the denominator
- a K_c value **greater than 1** means that the equilibrium lies **to the right**
there are more products than reactants
- a K_c value **between 0 and 1** means that the equilibrium lies **to the left**
more reactants than products

This has been summarised in Table 8.2. Note that K_c cannot be 0 or less.

Yield describes the quantity of product in the container relative to the maximum product possible.

K_c value	Equilibrium	Yield
$K_c > 1$	to the right	more products than reactants
$0 < K_c < 1$	to the left	more reactants than products

Table 8.2: The meaning of K_c values.

A **high K_c** value means that:

- the concentration of products is high
- the equilibrium lies far to the right
- the reaction has a **high yield**

A **low K_c value** (close to 0) means that:

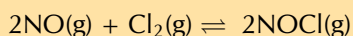
- the concentration of reactants is high
- the equilibrium lies far to the left
- the reaction has a **low yield**

Exercise 8 – 2: The equilibrium constant

1. Write the equilibrium constant expression, K_c , or balanced chemical equation for the following reactions:

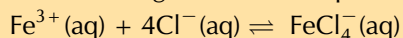
a) NO and Cl_2 are both reactants. NOCl is the product. They are all in the gas state and will be included in the equilibrium constant expression.

NO has a coefficient of **2**, Cl_2 has a coefficient of **1** and NOCl has a coefficient of **2**.



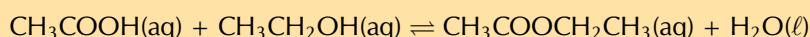
$$b) K_c = \frac{[H_2O(g)]^2}{[H_2(g)]^2[O_2(g)]}$$

2. The following reaction takes place:



K_c for the reaction is $7,5 \times 10^{-2}$. At equilibrium, the concentration of $FeCl_4^{-}(aq)$ is $0,95 \times 10^{-4} \text{ mol.dm}^{-3}$ and the concentration of $Fe^{3+}(aq)$ is $0,2 \text{ mol.dm}^{-3}$. Calculate the concentration of chloride ions at equilibrium.

3. Ethanoic acid (CH_3COOH) reacts with ethanol (CH_3CH_2OH) to produce ethyl ethanoate and water. The reaction is:



FACT

Henry Louis Le Chatelier.

At the beginning of the reaction, there is 0,5 mol of ethanoic acid and 0,5 mol of ethanol. At equilibrium, 0,3 mol of ethanoic acid was left unreacted. The volume of the reaction container is 2 dm³. Calculate the value of K_c .

4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27VK 2. 27VM 3. 27VN



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8.3 Le Chatelier's principle

ESCNN

Any factor that can affect the rate of either the forward or reverse reaction relative to the other can potentially affect the equilibrium position. The following factors can change the chemical equilibrium position of a reaction:

- **concentration**
- **temperature**
- **pressure** (for gaseous reactants)

It is important to understand what effect a change in one of these factors will have on a system that is in chemical equilibrium. However, performing an experiment every time to find out would waste a lot of time. Towards the end of the 1800s the French chemist Henry Louis Le Chatelier came up with principle to predict those effects.

Le Chatelier's Principle helps to predict what effect a change in temperature, concentration or pressure will have on the position of the equilibrium in a chemical reaction. This is very important, particularly in industrial applications, where yields must be accurately predicted and maximised. **IMPORTANT!**

Le Chatelier's Principle

When an external stress (change in pressure, temperature or concentration) is applied to a system in chemical equilibrium, the equilibrium will change in such a way as to reduce the effect of the stress.

So if the concentration of one (or more) of the reactants or products is increased the equilibrium will shift to decrease the concentration. Or if the temperature is decreased the equilibrium will shift to increase the temperature by favouring the exothermic reaction. Le Chatelier's principle is that:

- If you change the **concentration** of a reactant then the position of the equilibrium will shift to counteract that change.
- If you change the **temperature** of the reaction the equilibrium will shift to counteract that change.
- If you change the **pressure** of the system the position of the equilibrium will shift to counteract that change.

Each of these concepts is discussed in detail in the following pages.

The following video gives an example of Le Chatelier's principle in action.

▶ See video: 27VP at www.everythingscience.co.za

FACT

When the concentration of reactants is increased, the equilibrium shifts to the right and there will be more product than before. There will also be more reactants than before (more reactants were added). Once equilibrium has been reestablished (the rate of the forward and reverse reactions are equal again), K_c will be the same as it was before the change to the system. This concept is explained in more detail later in this chapter.

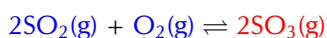
The effect of concentration on equilibrium

ESCNP

If the concentration of a substance is changed, the equilibrium will shift to minimise the effect of that change.

- If the concentration of a **reactant** is **increased** the equilibrium will shift in the direction of the reaction that **uses** the reactants, so that the reactant concentration decreases. **The forward reaction is favoured.**
- The forward reaction is also favoured if the concentration of the **product is decreased**, so that more product is formed.
- If the concentration of a **reactant is decreased** the equilibrium will shift in the direction of the reaction that **produces** the reactants, so that the reactant concentration increases. **The reverse reaction is favoured.**
- The reverse reaction is also favoured if the concentration of the **product is increased**, so that product is used.

For example, in the reaction between sulfur dioxide and oxygen to produce sulfur trioxide:



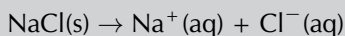
- If the SO_2 or O_2 concentration was **increased**:
 - Le Chatelier's principle predicts that equilibrium will shift to decrease the concentration of reactants.
 - Increasing the rate of the **forward reaction** will mean a decrease in reactants.
 - So some of the sulfur dioxide or oxygen is used to produce sulfur trioxide.
 - Equilibrium shifts to the **right**. That is, when a new equilibrium is reached (when the rate of forward and reverse reactions are equal again), there will be more product than before.
- If the SO_2 or O_2 concentration was **decreased**:
 - Le Chatelier's principle predicts that the equilibrium will shift to increase the concentration of reactants.
 - Increasing the rate of the **reverse reaction** will mean an increase in reactants.
 - So some sulfur trioxide would change back to sulfur dioxide and oxygen to restore equilibrium.
 - Equilibrium shifts to the **left**. That is, when a new equilibrium is reached there will be less product than before.
- If $[\text{SO}_3]$ **decreases**:
 - Le Chatelier's principle predicts that the equilibrium will shift to increase the concentration of products.
 - Increasing the rate of the **forward reaction** will mean an increase in products.
 - So some sulfur dioxide or oxygen is used to produce sulfur trioxide.
 - Equilibrium shifts to the **right**. That is, when a new equilibrium is reached there will be more product than before.
- If $[\text{SO}_3]$ **increases**:
 - Le Chatelier's principle predicts that the equilibrium will shift to decrease the concentration of products.
 - Increasing the rate of the **reverse reaction** will mean a decrease in products.
 - So some of the sulfur trioxide would change back to sulfur dioxide and oxygen to restore equilibrium.
 - Equilibrium shifts to the **left**. That is, when a new equilibrium is reached there will be less product than before.

Extension

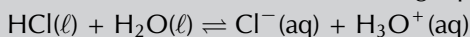
Common-ion effect

In solutions the change in equilibrium position can come about due to the **common-ion effect**. The common-ion effect is where one substance releases ions (upon dissociating or dissolving) which are already present in the equilibrium reaction.

If solid sodium chloride is added to an aqueous solution and dissolves, the following dissociation occurs:



If that solution contains the following equilibrium:



The added Cl^- ion (common-ion) interferes with the equilibrium by raising the concentration of the Cl^- ion. According to Le Chatelier's principle the reverse reaction speeds up as it tries to reduce the effect of the added Cl^- . As a result the equilibrium position shifts to the left.

FACT

Remember that heat is released during an exothermic reaction. It is a *product* of the reaction.

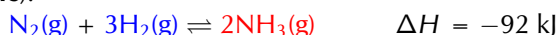
The effect of temperature on equilibrium

ESCNQ

If the temperature of a reaction mixture is changed, the equilibrium will shift to minimise that change.

- If the temperature is **increased** the equilibrium will shift to favour the reaction which will **reduce the temperature**. **The endothermic reaction is favoured**.
- If the temperature is **decreased** the equilibrium will shift to favour the reaction which will **increase the temperature**. **The exothermic reaction is favoured**.

For example the forward reaction shown below is exothermic (shown by the negative value for ΔH). This means that the forward reaction, where nitrogen and hydrogen react to form ammonia, gives off heat, increasing the temperature (**the forward reaction is exothermic**). In the reverse reaction, where ammonia decomposes into hydrogen and nitrogen gas, heat is taken in by the reaction, cooling the vessel (**the reverse reaction is endothermic**).



- An **increase** in temperature:
 - Favours the **endothermic** reaction because it **takes in** energy (cools the container).
 - The reverse reaction is endothermic, so the **reverse reaction** is favoured.
 - The **yield** of ammonia (NH_3) will **decrease**.
- A **decrease** in temperature:
 - Favours the **exothermic** reaction because it **releases** energy (warms the container).
 - The forward reaction is exothermic, so the **forward reaction** is favoured.
 - The **yield** of NH_3 will **increase**.

Informal experiment: Le Chatelier's principle

Aim:

To determine the effect of a change in concentration and temperature on chemical equilibrium

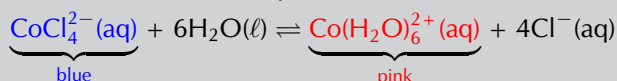
Apparatus:

- $0,2 \text{ mol.dm}^{-3}$ purple CoCl_2 in ethanol solution, concentrated HCl , water
- test tube, tongs
- ice-bath, water-bath, hot-plate or bunsen burner

Method:

1. Place the water bath on the hot-plate and heat.
2. During each step observe and record the colour change that takes place.
3. Put 4 – 5 drops of $0,2 \text{ mol.dm}^{-3}$ CoCl_2 solution into the test tube.
4. Add 10 – 12 drops of water.
5. Add 20 – 25 drops of concentrated HCl .
6. Place the test tube in the water-bath on the hot-plate (use tongs). Leave for 1 – 2 minutes. Record your observations.
7. Place the test tube in the ice-bath. Leave for 1 – 2 minutes. Record your observations.

The equation for the reaction that takes place is:



Results:

Complete your observations in the table below, noting the colour changes that take place, and also indicating whether the concentration of each of the ions in solution increases or decreases.

	Original colour	Final colour	$[\text{CoCl}_4^{2-}]$	$[\text{Co}(\text{H}_2\text{O})_6^{2+}]$	$[\text{Cl}^{-}]$
Add HCl					
Add H_2O					
Increase temp					
Decrease temp					

Conclusions:

- Use Le Chatelier's principle to explain the changes that you recorded in the table above.
- Draw a conclusion about the effect of a *change in concentration* of either the reactants or products on the equilibrium position.
- Also draw a conclusion about the effect of a *change in temperature* on the equilibrium position.

► See video: 27VQ at www.everythingscience.co.za

The effect of pressure on equilibrium

ESCNr

If the pressure of a gaseous reaction mixture is changed the equilibrium will shift to minimise that change.

- If the pressure is **increased** the equilibrium will shift to favour a **decrease** in pressure.
- If the pressure is **decreased** the equilibrium will shift to favour an **increase** in pressure.

When the volume of a system is decreased (and the temperature is constant), the pressure will increase. There are more collisions with the walls of the container. If there are fewer gas molecules there will be fewer collisions, and therefore lower pressure. The equilibrium will shift in a direction that reduces the number of gas molecules so that the pressure is also reduced. So, to predict in which direction the equilibrium will shift to change pressure you need to **look at the number of gas molecules in the balanced reactions**.

For example, the equation for the reaction between nitrogen and hydrogen is shown below:



The ratio in the balanced equation is 1 : 3 : 2. That is, for every 1 molecule of N_2 gas there are 3 molecules of H_2 gas and 2 molecules of NH_3 gas (from the balanced equation). Therefore the ratio is **4 molecules of reactant gas** to **2 molecules of product gas**.

- An **increase** in pressure will:
 - Favour the reaction that **decreases** the number of gas molecules.
 - There are **fewer molecules of product gas** than reactant gas, so the **forward reaction** is favoured.
 - The equilibrium will **shift to the right** and the **yield of NH_3** will **increase**.
- A **decrease** in pressure will:
 - Favour the reaction that **increases** the number of gas molecules.
 - There are **more molecules of reactant gas**, so the **reverse reaction** is favoured.
 - The equilibrium will **shift to the left** and the **yield of NH_3** will **decrease**.

Consider illustration in Figure 8.2.

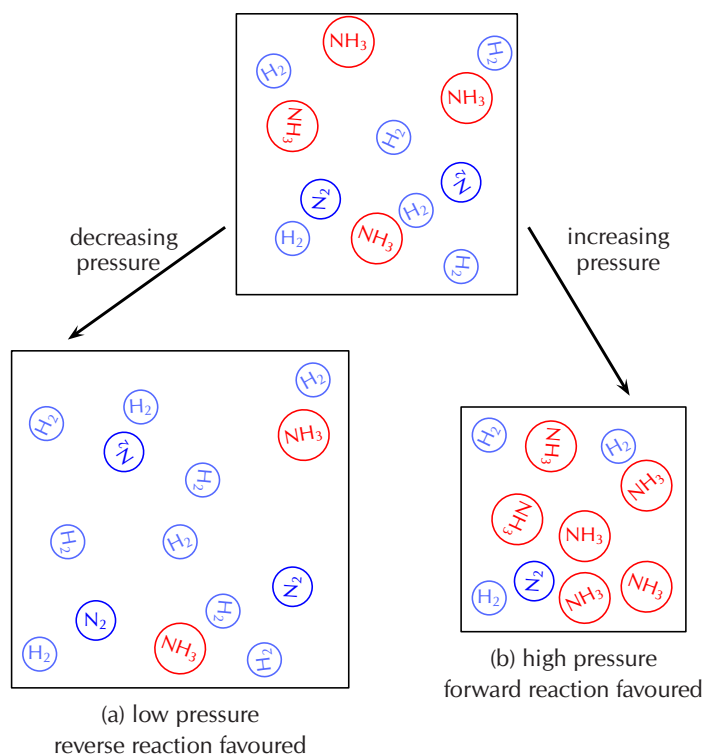


Figure 8.2: **(a)** A decrease in the pressure of this reaction favours the reverse reaction (more gas molecules), the equilibrium shifts to the left. **(b)** An increase in the pressure of this reaction favours the forward reaction (fewer gas molecules), the equilibrium shifts to the right.

TIP

Remember from [Grade 11](#) that:

$$p \propto \frac{T}{V}$$

That is, if the temperature remains constant, and the volume is increased, the pressure will decrease.

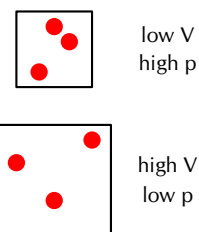


Figure 8.2 shows a decrease in pressure by an increase in the volume, and an increase in pressure by a decrease in the volume.

Figure 8.2 shows how changing the pressure of a system results in a shift in the equilibrium to counter that change. In the original system there are **12** molecules in total:



If you **decrease** the pressure (shown by an increase in volume), the equilibrium will shift to **increase** the number of gas molecules. That shift is to the left and the number of H_2 and N_2 molecules will increase while the number of NH_3 molecules will decrease:

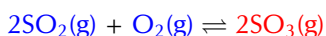


If you **increase** the pressure (shown by a decrease in volume), the equilibrium will shift to **decrease** the number of gas molecules. That shift is to the right and the number of H_2 and N_2 molecules will decrease while the number of NH_3 molecules will increase:



Note that the total number of nitrogen and hydrogen atoms remains the same in all three situations. Equations (a) and (b) are not balanced equations.

Another example is the reaction between sulfur dioxide and oxygen:



In this reaction, **two molecules of product gas** are formed for every **three molecules of reactant gas**.

- An **increase** in pressure will:
 - Favour the reaction that **decreases** the number of gas molecules.
 - There are **fewer molecules of product gas** than reactant gas, so the **forward** reaction is favoured.
 - The equilibrium will **shift to the right** and the **yield of SO_3** will **increase**.
- A **decrease** in pressure will:
 - Favour the reaction that **increases** the number of gas molecules.
 - There are **more molecules of reactant gas** than product gas, so the **reverse** reaction is favoured.
 - The equilibrium will **shift to the left** and the **yield of SO_3** will **decrease**.

The effect of a catalyst on equilibrium

ESCNS

If a catalyst is added to a reaction, both the **forward** and **reverse** reaction rates will be increased. If both rates are increased then the concentrations of the reactants and products will **remain the same**. This means that a catalyst has no effect on the equilibrium position.

However, a catalyst will affect how *quickly equilibrium is reached*. This is very important in industry where the longer a process takes, the more money it costs. So if a catalyst reduces the amount of time it takes to form specific products, it also reduces the cost of production.

Factors that influence the value of K_c

ESCNT

Concentration, pressure, and temperature all affect the equilibrium position of a reaction, and a catalyst affects reaction rates. However, only **temperature** affects the value of K_c .

- Changing **concentration**:
 - Changing the concentration of a reactant or product results in one of the reactions (forward or reverse) being favoured.

- This change in reaction rate minimises the effect of the change and restores the concentration ratio between reactants and products. There will just be more reactants and products.
- K_c will remain the same.
- Changing **pressure**:
 - Changing the pressure of the system will change the ratio between the reactant and product concentrations.
 - The equilibrium then shifts to minimise the effect of the change and restores the ratio between reactant and product concentrations.
 - K_c will remain the same.
- Adding a **catalyst** to the system:
 - Both the forward and reverse reactions rates are increased.
 - Therefore the ratio between reactant and product concentrations will remain the same.
 - K_c will remain the same.
- Changing **temperature**:
 - Changing the temperature will favour either the endothermic or exothermic reaction.
 - The ratio between the concentration of the reactants and products will change.
 - K_c will change.

IMPORTANT!

So make sure that when comparing K_c values for different reactions, the different reactions took place at the same temperature.

Using Le Chatelier's Principle

ESCNV

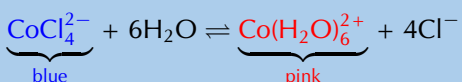
When a system is in chemical equilibrium, and there has been a change in conditions (e.g. concentration, pressure, temperature) the following steps are suggested:

1. Identify the disturbance or stress on the system.
For example, there is an **increase** in the concentration of reactant.
2. Use Le Chatelier's principle to decide how the system will respond.
Le Chatelier predicts a shift to **decrease** the concentration of reactant.
3. Look at the given equation and decide whether the rate of the forward reaction or the rate of the reverse reaction is increased. State the shift in equilibrium.
The **forward reaction will be favoured**. The equilibrium will shift **to the right**.
4. Where appropriate, link equilibrium shift to any observed change in the system.
This might result in a **colour change**.

Worked example 6: Using Le Chatelier's principle

QUESTION

Table salt is added to the (purple) solution in equilibrium:



1. Use Le Chatelier's principle to predict the change in equilibrium position.
2. What would be observed?

SOLUTION

Step 1: Identify the disturbance or stress on the system

Adding NaCl produces Na^+ ions and Cl^- ions as the salt dissolves. Looking at the given equilibrium Cl^- is in the equation and the disturbance is the increase in concentration of the Cl^- ion.

Step 2: Use Le Chatelier's principle to decide how the system will respond

By Le Chatelier's principle, the equilibrium position will shift to reduce the concentration of Cl^- ions.

Step 3: Decide whether the rate of the forward reaction or the rate of the reverse reaction is increased and state the resulting shift in equilibrium

The reverse reaction uses Cl^- ions and hence the rate of the reverse reaction will increase. The reverse reaction is favoured and the equilibrium will shift to the left.

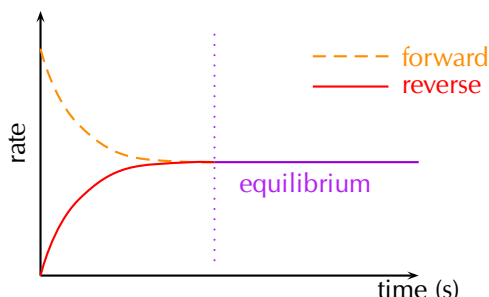
Step 4: What would the colour change be due to this equilibrium shift?

The solution will appear more blue as more blue CoCl_4^{2-} ions are formed.

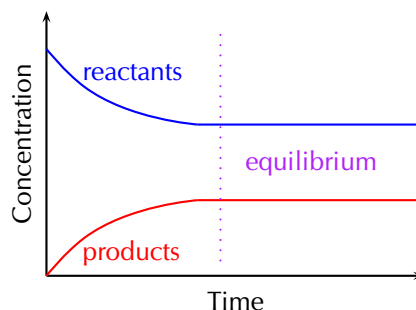
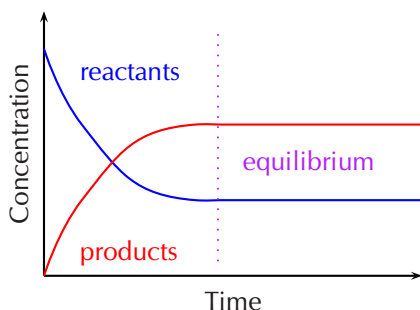
Graphs and Le Chatelier's principle

Graphs can be used to represent data about equilibrium reactions. The following are some points to keep in mind when presented with a graph.

1. Identify the type of graph by looking at the label on the y-axis. You will find either:
 - a) a rate-time graph
 - b) a mole-time or concentration-time graph
2. For rate-time graphs, when the rate for the forward reaction and the rate for the reverse reaction are equal, the system is in equilibrium.



3. For concentration-time graphs or mole-time graphs equilibrium occurs where the concentration or number of moles of the reactants and products are **constant**. These values need not be equal to one another.



4. When calculating K_c make sure you only take values from the sections of the graph where the y-value is constant. K_c can only be calculated when the system is in equilibrium.

Rate-time graphs

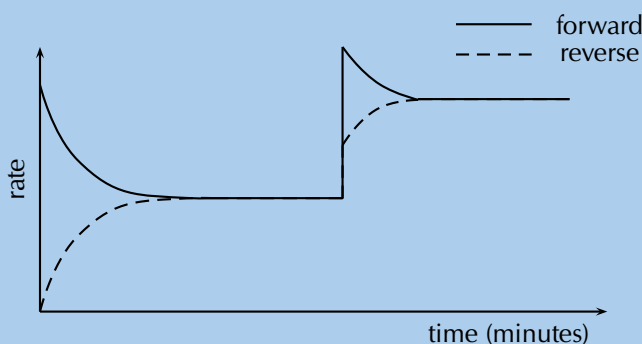
- **A change in concentration** of a substance would favour the reaction that decreases the amount of that substance. This will appear as a sharp increase in the rate of either the forward or reverse reaction and a sharp decrease in the rate of the other reaction.
The increased rate will then gradually decrease and the decreased rate will gradually increase until they are equal again.
- **A change in pressure** of the reaction would cause a sharp increase or decrease in all the reactants and products. This will have the same effect as a change in concentration, although the increase or decrease would be more gradual.
- **A change in temperature** of a substance would affect both rates in the same direction (either both increase or both decrease). However, the effect will be unequal, with the endothermic reaction favoured by an increase in temperature, and the exothermic reaction favoured by a decrease in temperature.
- **The addition of a catalyst** would favour both the forward and reverse reactions by the same amount.

Worked example 7: Rate-time graphs

QUESTION

For the reaction $2\text{AB}(\text{g}) \rightleftharpoons 2\text{A}(\text{g}) + \text{B}_2(\text{g})$, $\Delta H = 26 \text{ kJ}$
the following graph can be plotted:

What stress has occurred in this system? Label the graph with what is happening at each stage.



SOLUTION

Step 1: Check the axes so that you know what the variables are on this graph

The axes are labelled rate and time. Therefore this is a rate-time graph.

Step 2: Are the rates both affected in the same way or is one rate increased and the other decreased when the stress is applied?

Both rates are affected in the same way (increased) therefore the stress must be a catalyst or a change in temperature.

(A change in pressure or concentration would favour one reaction direction only)

Step 3: Are both rates affected equally?

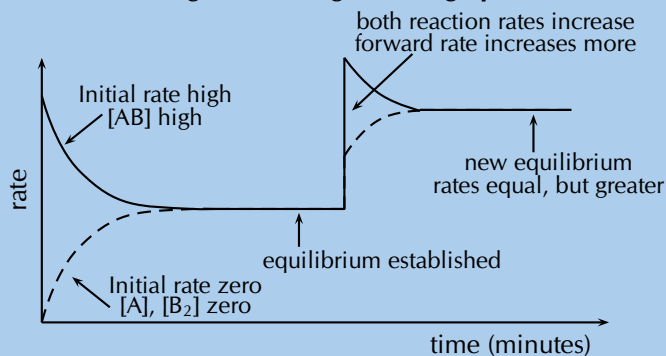
No, the forward rate is increased more than the reverse rate. Therefore the stress must be a change in temperature.

(A catalyst would increase both rates equally)

Step 4: Was the temperature increased or decreased? (Use Le Chatelier's principle)

The forward reaction is endothermic (ΔH is positive). The forward reaction was favoured more than the reverse reaction. An increase in temperature will favour the reaction that cools the reaction vessel (the endothermic reaction). Therefore the stress must have been an increase in temperature.

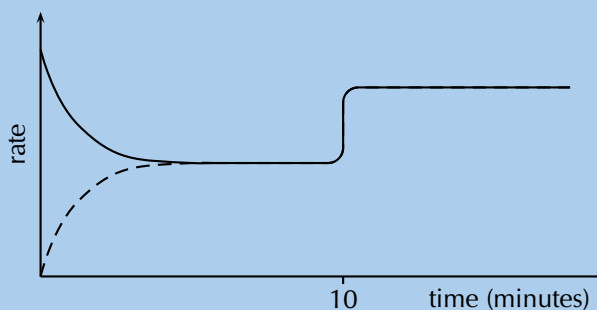
Step 5: Label what is occurring at each stage on the graph



Worked example 8: Rate-time graphs

QUESTION

What is responsible for the change at $t = 10$ minutes in the graph below?



SOLUTION

Step 1: Check the axes so that you know what the variables are on this graph

The axes are labelled rate and time. Therefore this is a rate-time graph.

Step 2: Are the rates both affected in the same way or is one rate increased and the other decreased when the stress is applied?

Both rates are affected in the same way (increased) therefore the stress must be a catalyst or a change in temperature.

(A change in pressure or concentration would favour one reaction direction only)

Step 3: Are both rates affected equally?

Yes, both rates are increased by the same amount.

Step 4: What was responsible for the change at $t = 10$ minutes?

The addition of a catalyst (a change in temperature would affect both rates, but unequally).

Concentration-time and mole-time graphs

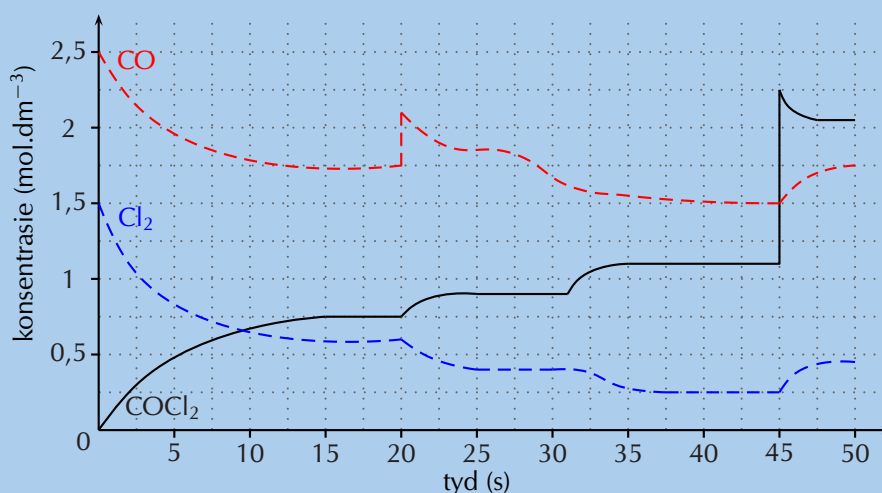
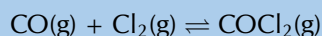
- **A change in concentration** of a substance would appear as a sharp increase or decrease in the concentration or number of moles of that substance and a gradual change in the other substances.
- **A change in temperature** would affect both the forward and reverse reactions. However, one reaction would be affected more than the other. So the reactants and products would be affected gradually, in the opposite direction (one increased, the other decreased).

- A **change in pressure** of the reaction would cause a sharp increase or decrease in all the reactants and products. For an increase in pressure, if the forward reaction is then favoured the reactant concentrations will decrease, and if the reverse reaction is then favoured the product concentrations will decrease.
- **The addition of a catalyst** would increase both the forward and reverse reaction rates, meaning the equilibrium is reached faster. So if the reaction is already at equilibrium there will be no effect on a concentration-time or moles-time graph.

Worked example 9: Concentration-time graphs

QUESTION

Consider the following chemical equilibrium and graph and answer the questions that follow.



1. How much time was necessary for the system to reach equilibrium for the first time?
2. How do the rates of the forward and reverse reactions compare at the following times:
 - $t = 5 \text{ s}$
 - $t = 17 \text{ s}$
 - $t = 23 \text{ s}$
3. Determine the equilibrium constants for the system at $t = 17 \text{ s}$, 25 s , and 45 s .
4. What happens at $t = 20 \text{ s}$? Explain your answer by referring to Le Chatelier's principle.
5. What effect does the stress at $t = 20 \text{ s}$ have on K_c ?

SOLUTION

Step 1: Check the axes so that you know what the variables are on this graph

The axes are labelled concentration and time. Therefore this is a concentration-time graph.

Step 2: How much time was necessary for the system to reach equilibrium?

The concentration of all three compounds becomes constant at $t = 15 \text{ s}$. This means that the reaction has reached equilibrium.

Step 3: How can you determine which rate is faster from the concentration graphs?

If the concentrations of the reactants (CO and Cl₂) are steadily decreasing or the concentration of the product (COCl₂) is increasing then the forward reaction is faster than the reverse reaction.

Similarly, if the concentrations of the reactants (CO and Cl₂) are steadily increasing or the concentration of the product (COCl₂) is decreasing then the reverse reaction is faster than the forward reaction.

Step 4: Compare the forward and reverse reaction rates at t = 5 s, t = 17 s and t = 23 s

At t = 5 s and at t = 23 s the concentrations of the reactants are decreasing and the concentration of the product is increasing. Therefore the rate of the forward reaction is faster than the rate of the reverse reaction.

At t = 17 s the concentrations of the reactants and products are constant (unchanging). Therefore the reaction is in equilibrium and the rate of the forward reaction equals the rate of the reverse reaction.

Step 5: What are the concentrations of CO, Cl₂ and COCl₂ at t = 17 s?

Reading off the graph you can see that:

$$[\text{CO}] = 1,75 \text{ mol.dm}^{-3} \quad [\text{Cl}_2] = 0,6 \text{ mol.dm}^{-3} \quad [\text{COCl}_2] = 0,75 \text{ mol.dm}^{-3}$$

Step 6: Write an expression for the equilibrium constant of this reaction and calculate K_c.

$$K_c = \frac{[\text{COCl}_2]}{[\text{CO}][\text{Cl}_2]} \quad K_c = \frac{0,75}{(1,75)(0,6)} = 0,71$$

Step 7: What happens at t = 20 s?

The concentration of CO increases sharply. The concentrations of the other compounds do not change dramatically. Therefore CO must have been added to the system.

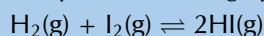
After this addition of CO there is a shift to reduce the amount of CO, that is in the forward direction. Therefore the concentration of the product increases while the concentrations of the reactants decrease.

Step 8: What effect will this have on K_c?

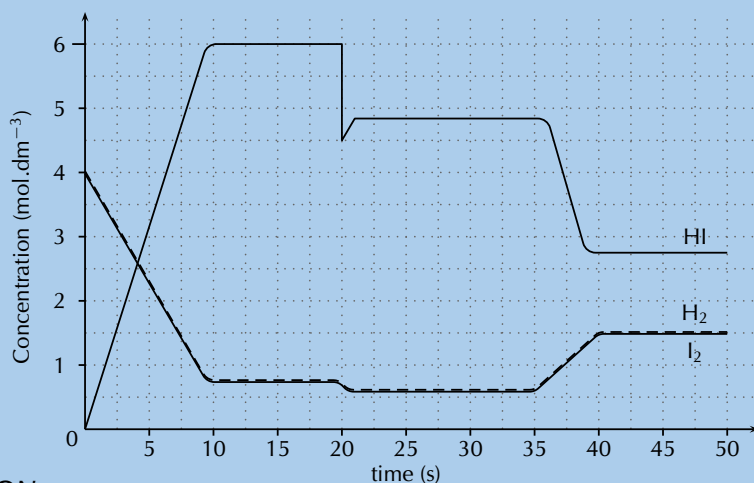
Only a change in temperature affects K_c, therefore this will have no effect on K_c.

Worked example 10: Concentration-time graphs**QUESTION**

Consider the following chemical equilibrium and graph and answer the questions.



1. After how many seconds does the system reach equilibrium?
2. Calculate the value of the equilibrium constant.
3. Explain what happened at t = 20 s.
4. If the change at t = 35 s is due to an increase in temperature, is the reaction exothermic or endothermic? Explain your answer.



SOLUTION

Step 1: Check the axes so that you know what the variables are on this graph

The axes are labelled concentration and time. Therefore this is a concentration-time graph.

Step 2: After how many seconds does the system reach equilibrium?

The concentration of all three compounds becomes constant at $t = 10$ s. This means that the system takes 10 s to reach equilibrium.

Step 3: What are the concentrations of H_2 , I_2 , and HI at $t = 10$ s, 25 s, and 45 s?

Reading off the graph you can see that:

At 10 s: $[H_2]$ and $[I_2] = 0,75 \text{ mol.dm}^{-3}$, $[HI] = 6,0 \text{ mol.dm}^{-3}$

At 25 s: $[H_2]$ and $[I_2] = 0,6 \text{ mol.dm}^{-3}$, $[HI] = 4,8 \text{ mol.dm}^{-3}$

At 45 s: $[H_2]$ and $[I_2] = 1,5 \text{ mol.dm}^{-3}$, $[HI] = 2,75 \text{ mol.dm}^{-3}$

Step 4: Write an expression for the equilibrium constant of this reaction and calculate K_c .

$$K_c = \frac{[HI]^2}{[H_2][I_2]}$$

$$\text{At } t = 10 \text{ s: } K_c = \frac{6,0^2}{(0,75)(0,75)} = 64,0$$

$$\text{At } t = 25 \text{ s: } K_c = \frac{4,8^2}{(0,6)(0,6)} = 64,0$$

$$\text{At } t = 45 \text{ s: } K_c = \frac{2,75^2}{(1,5)(1,5)} = 3,4$$

The different K_c at 45 s means that the event at $t = 35$ s must be a change in temperature.

Step 5: Explain what happened at $t = 20$ s?

The concentration of HI decreases sharply, as a result there is a slight decrease in the concentration of H_2 and I_2 . Therefore HI must have been removed from the system.

After this there is a shift to increase the amount of HI , that is in the forward direction.

Step 6: If the change at $t = 35$ s is due to an increase in temperature, is the reaction exothermic or endothermic? Explain your answer.

An increase in temperature caused the concentration of the product to decrease and the concentrations of the reactants to increase. This means that the reverse reaction has been favoured.

An increase in temperature will favour the reaction that takes heat in and cools the reaction vessel (endothermic). Therefore the reverse reaction must be endothermic and the forward reaction must be exothermic. The reaction is exothermic.

The following rules will help in predicting the changes that take place in equilibrium reactions:

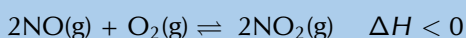
- If the concentration of a **reactant** (on the left) is **increased**, then some of it must be used to form the products (on the right) for equilibrium to be maintained. The equilibrium position will shift to the **right**. K_c is unchanged.
- If the concentration of a **reactant** (on the left) is **decreased**, then some of the products (on the right) must be used to form reactants for equilibrium to be maintained. The equilibrium position will shift to the **left**. K_c is unchanged.
- If the **forward reaction** is **endothermic**, then an **increase** in temperature will favour this reaction and the product **yield** and K_c **will increase**. A **decrease** in temperature will **decrease product yield** and likewise **decrease K_c** .
- If the **forward reaction** is **exothermic**, then an **decrease** in temperature will favour this reaction and the product **yield** and K_c **will increase**. An **increase** in temperature will **decrease product yield** and likewise **decrease K_c** .
- **Increasing** the pressure favours the side of the equilibrium with the **least number of gas molecules**. This is shown in the balanced chemical equation. This rule applies in reactions with one or more gaseous reactants or products. K_c is unchanged.
- **Decreasing** the pressure favours the side of the equilibrium with **more gas molecules**. This rule applies in reactions with one or more gaseous reactants or products. K_c is unchanged.
- A **catalyst** does not affect the equilibrium position of a reaction. It only influences the *rate of the reaction*, in other words, how quickly equilibrium is reached.

The following simulation will help you to understand these concepts.

🕒 See simulation: 27VR at www.everythingscience.co.za

Worked example 11: Chemical equilibrium

QUESTION



How will:

1. the reverse reaction
2. the equilibrium position be affected by:
 - a) A decrease in temperature?
 - b) The addition of a catalyst?
 - c) The addition of more NO_2 gas?

SOLUTION

1. The reverse reaction:

- a) The forward reaction is exothermic ($\Delta H < 0$) so the reverse reaction must be endothermic. A decrease in temperature will cause the equilibrium to shift to favour the exothermic reaction. Therefore the reverse reaction rate will decrease sharply, and then gradually increase until equilibrium is re-established.
- b) The addition of a catalyst will speed up both the forward and reverse reactions.
- c) NO_2 is a product. Therefore the addition of more NO_2 will increase the

rate of the formation of reactants. The rate of the reverse reaction will therefore increase sharply, and then gradually decrease until equilibrium is re-established.

2. The equilibrium position:

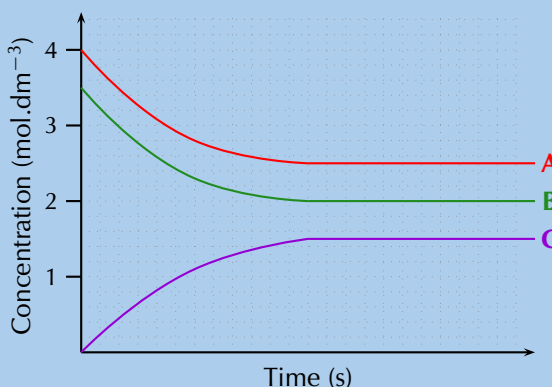
- A decrease in temperature will favour the exothermic reaction and the forward reaction is exothermic. Therefore the equilibrium position will shift to the right.
- The addition of a catalyst will have no effect on the equilibrium position as both the forward and reverse reactions rates would be increased equally.
- The addition of more NO_2 will favour the formation of the reactants and so the equilibrium will shift to the left.

Worked example 12: Graphs of equilibrium

QUESTION

Study the graph and answer the questions that follow:

- Does the equilibrium favour the reactants or the products?
- Determine the value of K_c if the coefficients of the balanced equation all equal 1.
- The forward reaction has $\Delta H > 0$. What effect will an increase in temperature have on $[A]$, $[B]$ and $[C]$?

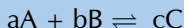


SOLUTION

Step 1: Determine which compounds are the reactants and which are the products

The concentration of reactants decreases from the start of the reaction to equilibrium. Therefore A and B are reactants.

The concentration of products increases from the start of the reaction to equilibrium. Therefore C is a product.



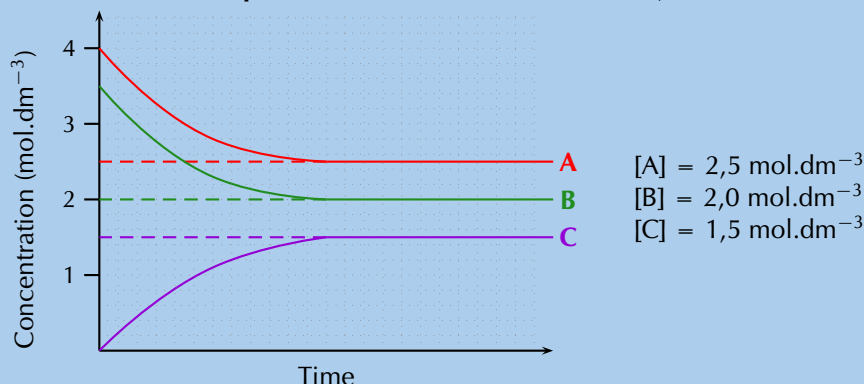
We are told that all coefficients in the balanced equation equal 1. Therefore the general equation is: $A + B \rightleftharpoons C$

Step 2: Do reactants or products have higher concentrations at equilibrium?

A and B (the reactants) have higher concentrations at equilibrium.

Step 3: Does the equilibrium favour the reactants or products?

The reactants have higher concentrations than the products, therefore the equilibrium must favour the reactants.

Step 4: Determine the equilibrium concentration values of A, B and C

Step 5: Calculate K_c

$$K_c = \frac{[C]}{[A][B]} = \frac{1,5}{(2,5)(2,0)} = 0,3$$

Step 6: Determine which reaction is exothermic and which is endothermic

The forward reaction has $\Delta H > 0$. This means that the forward reaction is endothermic. The reverse reaction must therefore be exothermic.

Step 7: Which reaction is favoured by an increase in temperature?

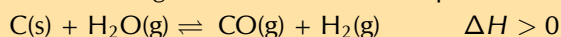
The endothermic reaction would be favoured by an increase in temperature (to lower the temperature). This is the forward reaction.

Step 8: How would [A], [B] and [C] change?

The forward reaction is favoured, therefore the equilibrium would shift to the right. This means that the reactant concentrations ([A] and [B]) would decrease and the product concentration ([C]) would increase.

Exercise 8 – 3: Equilibrium

1. The following reaction has reached equilibrium in a closed container:



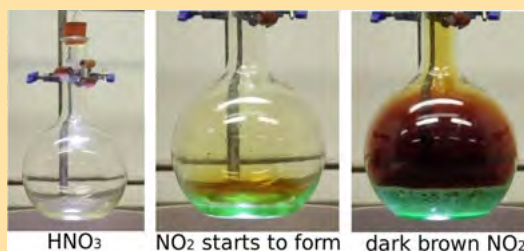
The pressure of the system is then decreased. How will the concentration of the H₂ (g) and the value of K_c be affected when the new equilibrium is established?

	Hydrogen concentration	K_c
(a)	increases	increases
(b)	increases	stays the same
(c)	stays the same	stays the same
(d)	decreases	stays the same

Assume that the temperature of the system remains unchanged.

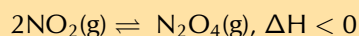
(IEB Paper 2, 2004)

2. During a classroom experiment copper metal reacts with concentrated nitric acid to produce NO₂ gas.

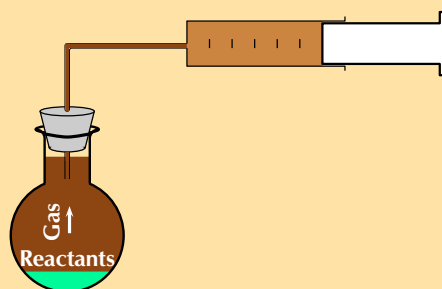


The NO₂ is collected in a gas syringe.

When enough gas has collected in the syringe, the delivery tube is clamped so that no gas can escape. The brown NO_2 gas collected reaches an equilibrium with colourless N_2O_4 gas as represented by the following equation:



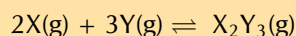
Once this equilibrium has been established, there are 0,01 moles of NO_2 gas and 0,03 moles of N_2O_4 gas present in the syringe.



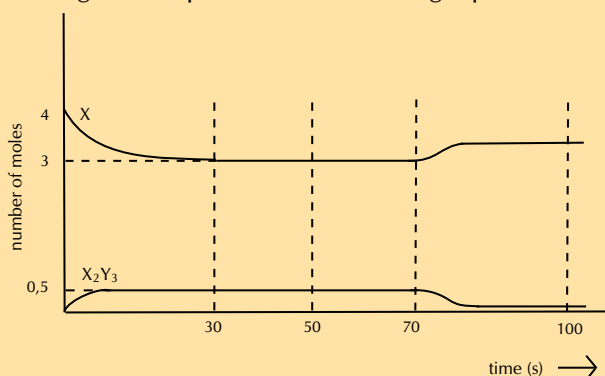
- A learner, noticing that the colour of the gas mixture in the syringe is no longer changing, comments that all chemical reactions in the syringe must have stopped. Is this assumption correct? Explain.
- The gas in the syringe is cooled. The volume of the gas is kept constant during the cooling process. Will the gas be lighter or darker at the lower temperature? Explain your answer.
- The volume of the syringe is now reduced (at constant temperature) to 75 cm^3 by pushing the plunger in and holding it in the new position. There are 0,032 moles of N_2O_4 gas present once the equilibrium has been re-established at the reduced volume (75 cm^3). Calculate the value of the equilibrium constant for this equilibrium.

(IEB Paper 2, 2004)

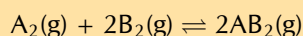
- Gases X and Y are pumped into a 2 dm^3 container. When the container is sealed, 4 moles of gas X and 4 moles of gas Y are present. The following equilibrium is established:



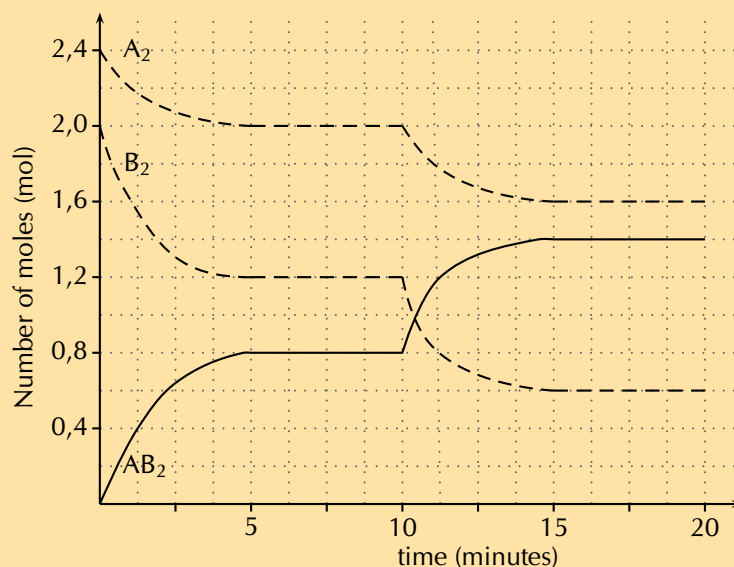
The graph shows the number of moles of gas X and gas X_2Y_3 that are present from the time the container is sealed.



- How many moles of gas X_2Y_3 are formed by the time the reaction reaches equilibrium at 30 seconds?
 - Calculate the value of the equilibrium constant at $t = 50 \text{ s}$.
 - At 70 s the temperature is increased. Is the forward reaction endothermic or exothermic? Explain in terms of Le Chatelier's Principle.
 - How will this increase in temperature affect the value of the equilibrium constant?
- Consider the following hypothetical reaction that takes place in a **closed 2 dm^3** flask at 298 K.



The graph represents the change in the number of moles of each gas in the flask over a period of 20 minutes.



- State how long (in minutes) it took for the reaction to reach equilibrium for the first time.
- Write down an expression for the equilibrium constant, K_c , for this particular reaction.
- Calculate the concentration of each of the reactants and the product using figures from the graph between 5 minutes and 10 minutes and hence calculate the equilibrium constant K_c for this reaction at 298 K
- State what a low value of K_c indicates about the yield of product for a reaction.
- Why is it not possible to calculate K_c using figures from the graph during the first 5 minutes
- State Le Chatelier's principle.
- At 10 minutes the temperature of the flask was increased. Using Le Chatelier's principle, determine if the production of AB₂ is exothermic or endothermic?
- How would the equilibrium **constant** be affected by each of the following changes (state either increase, decrease or no effect):
 - Increasing the pressure of the flask with no change to temperature.
 - Adding a catalyst to the flask.
 - Increasing the temperature of the flask.

5. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27VS 2. 27VT 3. 27VV 4. 27VW



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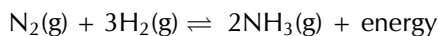
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Industrial applications

ESCNX

In industrial processes, it is important to get the product as quickly and as efficiently as possible. The less expensive the process the better.

The **Haber process** is a good example of an industrial process which uses the equilibrium principles that have been discussed. The equation for the process is as follows:



Since the forward reaction is exothermic, to produce a lot of product and favour the forward reaction the system needs to be colder. However, cooling a system slows down all chemical reactions and so the system can't be too cold. This process is carried out at a much higher temperature to ensure the speed of production.

Because high temperature favours the reverse reaction, the NH_3 product is actually removed as it is made (product concentration decreased) to prevent ammonia being used in the reverse reaction. The decrease of product concentration favours the forward reaction.

High pressure is also used to ensure faster reaction time and to favour the production of NH_3 . The forward reaction is favoured by higher pressures because there are 2 gas molecules of product for every 4 gas molecules of reactant.

Refer to Chapter 14 for more information on the Haber process and other industrial applications.

Exercise 8 – 4: Applying equilibrium principles

- Look at the values of K_c calculated for the Haber process reaction at different temperatures, and then answer the questions that follow:

$T (^{\circ}\text{C})$	K_c
25	$6,4 \times 10^2$
200	$4,4 \times 10^{-1}$
300	$4,3 \times 10^{-3}$
400	$1,6 \times 10^{-4}$
500	$1,5 \times 10^{-5}$

- What happens to the value of K_c as the temperature increases?
- Which reaction is being favoured when the temperature is 300°C ?
- According to this table, which temperature would be best if you wanted to produce as much ammonia as possible? Explain.

- More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27VX



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8.4 Chapter Summary

ESCNY

- A reaction is **reversible** when reactants can react to form products, and products can react to form the reactants again.
- A reaction is in **chemical equilibrium** when the rate of the forward reaction equals the rate of the reverse reaction.
- In an **open** system energy and matter can enter and leave the system. In a **closed** system energy can enter and leave the system, but matter cannot.
- The **equilibrium constant** (K_c) relates to the concentration of the reactants and products at equilibrium, and can be calculated using the following expression:

$$K_c = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b} \text{ where } a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$$

A and B are reactants, C and D are products and a, b, c, and d are the coefficients of the respective reactants and products.

- A **high K_c value** means that the concentration of products at equilibrium is high

and the reaction has a high yield of the products. A **low K_c value** means that the concentration of products at equilibrium is low and the reaction has a low yield of the products.

- **Le Chatelier's Principle** states that if an external stress (change in pressure, temperature or concentration) is applied to a system in chemical equilibrium, the equilibrium will change in such a way as to reduce the effect of the stress.
- Although a change in temperature, concentration or total pressure will affect the equilibrium position, only **temperature** will affect the equilibrium constant K_c .
- The principles of equilibrium are very important in **industrial applications** such as the Haber process, so that productivity can be maximised.

Exercise 8 – 5:

1. Explain the following concepts:

- a) Chemical equilibrium
- b) A closed system
- c) A reversible reaction

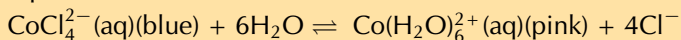
2. The following equilibrium constant expression is given $K_c = \frac{[\text{H}_2\text{O}]^4[\text{CO}_2]^3}{[\text{C}_3\text{H}_8][\text{O}_2]^5}$ for a particular reaction:

For which one of the following reactions is the above expression of K_c correct?

- a) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{H}_2\text{O}(\text{g}) + 3\text{CO}_2(\text{g})$
- b) $4\text{H}_2\text{O}(\text{g}) + 3\text{CO}_2(\text{g}) \rightleftharpoons \text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g})$
- c) $2\text{C}_3\text{H}_8(\text{g}) + 7\text{O}_2(\text{g}) \rightleftharpoons 6\text{CO}(\text{g}) + 8\text{H}_2\text{O}(\text{g})$
- d) $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightleftharpoons 4\text{H}_2\text{O}(\text{l}) + 3\text{CO}_2(\text{g})$

(IEB Paper 2, 2001)

3. Cobalt chloride crystals are dissolved in a beaker containing ethanol and then a few drops of water are added. After a period of time, the reaction reaches equilibrium as follows:



The solution, which is now purple, is poured into three test tubes. State, in each case, what colour changes, if any, will be observed if the following are added in turn to each test tube:

- a) 1 cm³ of distilled water
- b) A few crystals of sodium chloride

4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27VY 2. 27VZ 3. 27W2



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